

Actividad - Proyecto práctico

La actividad se desarrollará en grupos pre-definidos de 4 alumnos. Se debe indicar los nombres en orden alfabético (de apellidos). Recordad que esta actividad se corresponde con un 30% de la nota final de la asignatura. Se debe entregar el trabajo en la presente notebook.

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- Alumno 3: Kevin Omar Soto Simanca

PARTE 1 - Instalación y requisitos previos

Las prácticas han sido preparadas para poder realizarse en el entorno de trabajo de Google Colab. Sin embargo, esta plataforma presenta ciertas incompatibilidades a la hora de visualizar la renderización en gym. Por ello, para obtener estas visualizaciones, se deberá trasladar el entorno de trabajo a local. Por ello, el presente dossier presenta instrucciones para poder trabajar en ambos entornos. Siga los siguientes pasos para un correcto funcionamiento:

1. **LOCAL:** Preparar el environment, siguiendo las instrucciones detalladas en la sección *1.1.Preparar environment*.
2. **AMBOS:** Modificar las variables "mount" y "drive_mount" a la carpeta de trabajo en drive en el caso de estar en Colab, y ejecturar la celda *1.2.Localizar entorno de trabajo*.
3. **COLAB:** se deberá ejecutar las celdas correspondientes al montaje de la carpeta de trabajo en Drive. Esta corresponde a la sección *1.3.Montar carpeta de datos local*.
4. **AMBOS:** Instalar las librerías necesarias, siguiendo la sección *1.4.Instalar librerías necesarias*.

1.1. Preparar environment (solo local)

1. En Windows, puede ser necesario instalar las C++ Build Tools. Para ello, siga los siguientes pasos. Alternativamente puedes utilizar WSL2:
<https://towardsdatascience.com/how-to-install-openai-gym-in-a-windows-environment-338969e24d30>.
2. Instalar Anaconda
3. Siguiendo el código que se presenta comentado en la próxima celda: Crear un environment, cambiar la ruta de trabajo, e instalar librerías básicas.

Para preparar el entorno de trabajo en local, se han seguido los siguientes pasos:

```
conda create --name miar_rl python=3.8
conda activate miar_rl
cd "PATH_TO_FOLDER"
conda install git
pip install jupyter
```

4. Abrir la notebook con *jupyter-notebook*.

```
jupyter-notebook
```

1.2. Localizar entorno de trabajo: Google colab o local

```
In [1]: # ATENCIÓN!! Modificar ruta relativa a la práctica si es distinta (drive_root)
import os

mount=''
drive_root = ''
IN_COLAB = False

drive_root = r"C:\Users\kevin\Downloads\ESTUDIO MASTER EN IA\Aprendizaje por refuer

os.makedirs(drive_root, exist_ok=True)
os.chdir(drive_root)

print("IN_COLAB:", IN_COLAB)
print("Carpeta de trabajo:", os.getcwd())

try:
    from google.colab import drive
    IN_COLAB=True
except:
    IN_COLAB=False
```

IN_COLAB: False

Carpeta de trabajo: C:\Users\kevin\Downloads\ESTUDIO MASTER EN IA\Aprendizaje por re
fuerzo\PROYECTO\Proyecto RL

1.3. Montar carpeta de datos local (solo Colab)

```
In [2]: # Switch to the directory on the Google Drive that you want to use
import os
if IN_COLAB:
    print("We're running Colab")

if IN_COLAB:
    # Mount the Google Drive at mount
```

```

print("Colab: mounting Google drive on ", mount)

drive.mount(mount)

# Create drive_root if it doesn't exist
create_drive_root = True
if create_drive_root:
    print("\nColab: making sure ", drive_root, " exists.")
    os.makedirs(drive_root, exist_ok=True)

# Change to the directory
print("\nColab: Changing directory to ", drive_root)
%cd $drive_root
# Verify we're in the correct working directory
%pwd
print("Archivos en el directorio: ")
print(os.listdir())

```

Archivos en el directorio:

```

['.ipynb_checkpoints', 'checkpoint', 'dqn_Acrobot-v1_p1_best.h5f.data-00000-of-00001', 'dqn_Acrobot-v1_p1_best.h5f.index', 'dqn_Acrobot-v1_p1_best_score.txt', 'dqn_Acrobot-v1_p1_log.json', 'dqn_Acrobot-v1_p1_weights.h5f.data-00000-of-00001', 'dqn_Acrobot-v1_p1_weights.h5f.index', 'dqn_Acrobot-v1_p1_weights_10000.h5f.data-00000-of-00001', 'dqn_Acrobot-v1_p1_weights_10000.h5f.index', 'dqn_Acrobot-v1_p1_weights_20000.h5f.data-00000-of-00001', 'dqn_Acrobot-v1_p1_weights_20000.h5f.index', 'dqn_Acrobot-v1_p1_weights_30000.h5f.data-00000-of-00001', 'dqn_Acrobot-v1_p1_weights_30000.h5f.index', 'dqn_Acrobot-v1_p1_weights_40000.h5f.data-00000-of-00001', 'dqn_Acrobot-v1_p1_weights_40000.h5f.index', 'dqn_Acrobot-v1_p1_weights_50000.h5f.data-00000-of-00001', 'dqn_Acrobot-v1_p1_weights_50000.h5f.index', 'dqn_Acrobot-v1_p1_weights_60000.h5f.data-00000-of-00001', 'dqn_Acrobot-v1_p1_weights_60000.h5f.index', 'dqn_Acrobot-v1_p1_weights_70000.h5f.data-00000-of-00001', 'dqn_Acrobot-v1_p1_weights_70000.h5f.index', 'dqn_Acrobot-v1_p1_weights_80000.h5f.data-00000-of-00001', 'dqn_Acrobot-v1_p1_weights_80000.h5f.index', 'videos_proyecto']

```

1.4. Instalar librerías necesarias

In [4]:

```

if IN_COLAB:
    # Downgrade to TF 2.15 to ensure Keras 2 compatibility for keras-rl2
    %pip install tensorflow==2.15.0
    %pip install numpy==1.26.4
    %pip install keras==2.15.0
    %pip install gym==0.17.3
    %pip install keras-rl2==1.0.5
else:
    %pip install tensorflow==2.15.0
    %pip install numpy==1.26.4
    %pip install keras==2.15.0
    %pip install gym==0.17.3
    %pip install keras-rl2==1.0.5

```

```
ERROR: Could not find a version that satisfies the requirement tensorflow==2.15.0 (from versions: 2.2.0, 2.2.1, 2.2.2, 2.2.3, 2.3.0, 2.3.1, 2.3.2, 2.3.3, 2.3.4, 2.4.0, 2.4.1, 2.4.2, 2.4.3, 2.4.4, 2.5.0, 2.5.1, 2.5.2, 2.5.3, 2.6.0rc0, 2.6.0rc1, 2.6.0rc2, 2.6.0, 2.6.1, 2.6.2, 2.6.3, 2.6.4, 2.6.5, 2.7.0rc0, 2.7.0rc1, 2.7.0, 2.7.1, 2.7.2, 2.7.3, 2.7.4, 2.8.0rc0, 2.8.0rc1, 2.8.0, 2.8.1, 2.8.2, 2.8.3, 2.8.4, 2.9.0rc0, 2.9.0rc1, 2.9.0rc2, 2.9.0, 2.9.1, 2.9.2, 2.9.3, 2.10.0rc0, 2.10.0rc1, 2.10.0rc2, 2.10.0rc3, 2.10.0, 2.10.1, 2.11.0rc0, 2.11.0rc1, 2.11.0rc2, 2.11.0, 2.11.1, 2.12.0rc0, 2.12.0rc1, 2.12.0, 2.12.1, 2.13.0rc0, 2.13.0rc1, 2.13.0rc2, 2.13.0, 2.13.1)
ERROR: No matching distribution found for tensorflow==2.15.0
```

[notice] A new release of pip is available: 23.0.1 -> 25.0.1

[notice] To update, run: `pip install --upgrade pip`

Note: you may need to restart the kernel to use updated packages.

```
ERROR: Ignored the following versions that require a different python version: 1.25.0 Requires-Python >=3.9; 1.25.1 Requires-Python >=3.9; 1.25.2 Requires-Python >=3.9; 1.26.0 Requires-Python <3.13,>=3.9; 1.26.1 Requires-Python <3.13,>=3.9; 1.26.2 Requires-Python >=3.9; 1.26.3 Requires-Python >=3.9; 1.26.4 Requires-Python >=3.9; 2.0.0 Requires-Python >=3.9; 2.0.1 Requires-Python >=3.9; 2.0.2 Requires-Python >=3.9; 2.1.0 Requires-Python >=3.10; 2.1.1 Requires-Python >=3.10; 2.1.2 Requires-Python >=3.10; 2.1.3 Requires-Python >=3.10; 2.2.0 Requires-Python >=3.10; 2.2.1 Requires-Python >=3.10; 2.2.2 Requires-Python >=3.10; 2.2.3 Requires-Python >=3.10; 2.2.4 Requires-Python >=3.10; 2.2.5 Requires-Python >=3.10; 2.2.6 Requires-Python >=3.10; 2.3.0 Requires-Python >=3.11; 2.3.1 Requires-Python >=3.11; 2.3.2 Requires-Python >=3.11; 2.3.3 Requires-Python >=3.11; 2.3.4 Requires-Python >=3.11; 2.3.5 Requires-Python >=3.11; 2.4.0 Requires-Python >=3.11; 2.4.0rc1 Requires-Python >=3.11; 2.4.1 Requires-Python >=3.11; 2.4.2 Requires-Python >=3.11; 2.4.3 Requires-Python >=3.11; 2.4.4 Requires-Python >=3.11; 2.4.5 Requires-Python >=3.11; 2.4.6 Requires-Python >=3.11; 2.5.0 Requires-Python >=3.12; 2.5.0rc1 Requires-Python >=3.12
```

```
ERROR: Could not find a version that satisfies the requirement numpy==1.26.4 (from versions: 1.3.0, 1.4.1, 1.5.0, 1.5.1, 1.6.0, 1.6.1, 1.6.2, 1.7.0, 1.7.1, 1.7.2, 1.8.0, 1.8.1, 1.8.2, 1.9.0, 1.9.1, 1.9.2, 1.9.3, 1.10.0.post2, 1.10.1, 1.10.2, 1.10.4, 1.11.0, 1.11.1, 1.11.2, 1.11.3, 1.12.0, 1.12.1, 1.13.0, 1.13.1, 1.13.3, 1.14.0, 1.14.1, 1.14.2, 1.14.3, 1.14.4, 1.14.5, 1.14.6, 1.15.0, 1.15.1, 1.15.2, 1.15.3, 1.15.4, 1.16.0, 1.16.1, 1.16.2, 1.16.3, 1.16.4, 1.16.5, 1.16.6, 1.17.0, 1.17.1, 1.17.2, 1.17.3, 1.17.4, 1.17.5, 1.18.0, 1.18.1, 1.18.2, 1.18.3, 1.18.4, 1.18.5, 1.19.0, 1.19.1, 1.19.2, 1.19.3, 1.19.4, 1.19.5, 1.20.0, 1.20.1, 1.20.2, 1.20.3, 1.21.0, 1.21.1, 1.21.2, 1.21.3, 1.21.4, 1.21.5, 1.21.6, 1.22.0, 1.22.1, 1.22.2, 1.22.3, 1.22.4, 1.23.0, 1.23.1, 1.23.2, 1.23.3, 1.23.4, 1.23.5, 1.24.0, 1.24.1, 1.24.2, 1.24.3, 1.24.4)
ERROR: No matching distribution found for numpy==1.26.4
```

[notice] A new release of pip is available: 23.0.1 -> 25.0.1

[notice] To update, run: `pip install --upgrade pip`

Note: you may need to restart the kernel to use updated packages.

Collecting keras==2.15.0

Using cached keras-2.15.0-py3-none-any.whl (1.7 MB)

Installing collected packages: keras

Attempting uninstall: keras

Found existing installation: keras 2.13.1

Uninstalling keras-2.13.1:

Successfully uninstalled keras-2.13.1

ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.

tensorflow 2.13.1 requires keras<2.14,>=2.13.1, but you have keras 2.15.0 which is incompatible.

Successfully installed keras-2.15.0

[notice] A new release of pip is available: 23.0.1 -> 25.0.1
[notice] To update, run: `pip install --upgrade pip`
Note: you may need to restart the kernel to use updated packages.
Requirement already satisfied: gym==0.17.3 in ./venv/lib/python3.8/site-packages (0.17.3)
Requirement already satisfied: scipy in ./venv/lib/python3.8/site-packages (from gym==0.17.3) (1.10.1)
Requirement already satisfied: numpy>=1.10.4 in ./venv/lib/python3.8/site-packages (from gym==0.17.3) (1.24.3)
Requirement already satisfied: pygame<=1.5.0,>=1.4.0 in ./venv/lib/python3.8/site-packages (from gym==0.17.3) (1.5.0)
Requirement already satisfied: cloudpickle<1.7.0,>=1.2.0 in ./venv/lib/python3.8/site-packages (from gym==0.17.3) (1.6.0)
Requirement already satisfied: future in ./venv/lib/python3.8/site-packages (from pygame<=1.5.0,>=1.4.0->gym==0.17.3) (1.0.0)

[notice] A new release of pip is available: 23.0.1 -> 25.0.1
[notice] To update, run: `pip install --upgrade pip`
Note: you may need to restart the kernel to use updated packages.
Requirement already satisfied: keras-rl2==1.0.5 in ./venv/lib/python3.8/site-packages (1.0.5)
Requirement already satisfied: tensorflow in ./venv/lib/python3.8/site-packages (from keras-rl2==1.0.5) (2.13.1)
Requirement already satisfied: tensorflow-io-gcs-filesystem>=0.23.1 in ./venv/lib/python3.8/site-packages (from tensorflow->keras-rl2==1.0.5) (0.34.0)
Requirement already satisfied: typing-extensions<4.6.0,>=3.6.6 in ./venv/lib/python3.8/site-packages (from tensorflow->keras-rl2==1.0.5) (4.5.0)
Requirement already satisfied: absl-py>=1.0.0 in ./venv/lib/python3.8/site-packages (from tensorflow->keras-rl2==1.0.5) (2.3.1)
Requirement already satisfied: astunparse>=1.6.0 in ./venv/lib/python3.8/site-packages (from tensorflow->keras-rl2==1.0.5) (1.6.3)
Requirement already satisfied: gast<=0.4.0,>=0.2.1 in ./venv/lib/python3.8/site-packages (from tensorflow->keras-rl2==1.0.5) (0.4.0)
Requirement already satisfied: setuptools in ./venv/lib/python3.8/site-packages (from tensorflow->keras-rl2==1.0.5) (56.0.0)
Requirement already satisfied: numpy<=1.24.3,>=1.22 in ./venv/lib/python3.8/site-packages (from tensorflow->keras-rl2==1.0.5) (1.24.3)
Requirement already satisfied: protobuf!=4.21.0,!4.21.1,!4.21.2,!4.21.3,!4.21.4,!4.21.5,<5.0.0dev,>=3.20.3 in ./venv/lib/python3.8/site-packages (from tensorflow->keras-rl2==1.0.5) (4.25.9)
Requirement already satisfied: packaging in ./venv/lib/python3.8/site-packages (from tensorflow->keras-rl2==1.0.5) (26.2)
Requirement already satisfied: libclang>=13.0.0 in ./venv/lib/python3.8/site-packages (from tensorflow->keras-rl2==1.0.5) (18.1.1)
Collecting keras<2.14,>=2.13.1
Using cached keras-2.13.1-py3-none-any.whl (1.7 MB)
Requirement already satisfied: opt-einsum>=2.3.2 in ./venv/lib/python3.8/site-packages (from tensorflow->keras-rl2==1.0.5) (3.4.0)
Requirement already satisfied: flatbuffers>=23.1.21 in ./venv/lib/python3.8/site-packages (from tensorflow->keras-rl2==1.0.5) (25.12.19)
Requirement already satisfied: tensorboard<2.14,>=2.13 in ./venv/lib/python3.8/site-packages (from tensorflow->keras-rl2==1.0.5) (2.13.0)
Requirement already satisfied: termcolor>=1.1.0 in ./venv/lib/python3.8/site-packages (from tensorflow->keras-rl2==1.0.5) (2.4.0)
Requirement already satisfied: tensorflow-estimator<2.14,>=2.13.0 in ./venv/lib/pyt

hon3.8/site-packages (from tensorflow->keras-rl2==1.0.5) (2.13.0)
Requirement already satisfied: wrapt>=1.11.0 in ./venv/lib/python3.8/site-packages (from tensorflow->keras-rl2==1.0.5) (2.0.1)
Requirement already satisfied: grpcio<2.0,>=1.24.3 in ./venv/lib/python3.8/site-packages (from tensorflow->keras-rl2==1.0.5) (1.70.0)
Requirement already satisfied: h5py>=2.9.0 in ./venv/lib/python3.8/site-packages (from tensorflow->keras-rl2==1.0.5) (3.11.0)
Requirement already satisfied: google-pasta>=0.1.1 in ./venv/lib/python3.8/site-packages (from tensorflow->keras-rl2==1.0.5) (0.2.0)
Requirement already satisfied: six>=1.12.0 in ./venv/lib/python3.8/site-packages (from tensorflow->keras-rl2==1.0.5) (1.17.0)
Requirement already satisfied: wheel<1.0,>=0.23.0 in ./venv/lib/python3.8/site-packages (from astunparse>=1.6.0->tensorflow->keras-rl2==1.0.5) (0.45.1)
Requirement already satisfied: google-auth-oauthlib<1.1,>=0.5 in ./venv/lib/python3.8/site-packages (from tensorboard<2.14,>=2.13->tensorflow->keras-rl2==1.0.5) (1.0.0)
Requirement already satisfied: werkzeug>=1.0.1 in ./venv/lib/python3.8/site-packages (from tensorboard<2.14,>=2.13->tensorflow->keras-rl2==1.0.5) (3.0.6)
Requirement already satisfied: markdown>=2.6.8 in ./venv/lib/python3.8/site-packages (from tensorboard<2.14,>=2.13->tensorflow->keras-rl2==1.0.5) (3.7)
Requirement already satisfied: requests<3,>=2.21.0 in ./venv/lib/python3.8/site-packages (from tensorboard<2.14,>=2.13->tensorflow->keras-rl2==1.0.5) (2.32.4)
Requirement already satisfied: tensorboard-data-server<0.8.0,>=0.7.0 in ./venv/lib/python3.8/site-packages (from tensorboard<2.14,>=2.13->tensorflow->keras-rl2==1.0.5) (0.7.2)
Requirement already satisfied: google-auth<3,>=1.6.3 in ./venv/lib/python3.8/site-packages (from tensorboard<2.14,>=2.13->tensorflow->keras-rl2==1.0.5) (2.50.0)
Requirement already satisfied: pyasn1-modules>=0.2.1 in ./venv/lib/python3.8/site-packages (from google-auth<3,>=1.6.3->tensorboard<2.14,>=2.13->tensorflow->keras-rl2==1.0.5) (0.4.2)
Requirement already satisfied: cryptography>=38.0.3 in ./venv/lib/python3.8/site-packages (from google-auth<3,>=1.6.3->tensorboard<2.14,>=2.13->tensorflow->keras-rl2==1.0.5) (45.0.7)
Requirement already satisfied: requests-oauthlib>=0.7.0 in ./venv/lib/python3.8/site-packages (from google-auth-oauthlib<1.1,>=0.5->tensorboard<2.14,>=2.13->tensorflow->keras-rl2==1.0.5) (2.0.0)
Requirement already satisfied: importlib-metadata>=4.4 in ./venv/lib/python3.8/site-packages (from markdown>=2.6.8->tensorboard<2.14,>=2.13->tensorflow->keras-rl2==1.0.5) (8.5.0)
Requirement already satisfied: certifi>=2017.4.17 in ./venv/lib/python3.8/site-packages (from requests<3,>=2.21.0->tensorboard<2.14,>=2.13->tensorflow->keras-rl2==1.0.5) (2026.6.17)
Requirement already satisfied: urllib3<3,>=1.21.1 in ./venv/lib/python3.8/site-packages (from requests<3,>=2.21.0->tensorboard<2.14,>=2.13->tensorflow->keras-rl2==1.0.5) (2.2.3)
Requirement already satisfied: idna<4,>=2.5 in ./venv/lib/python3.8/site-packages (from requests<3,>=2.21.0->tensorboard<2.14,>=2.13->tensorflow->keras-rl2==1.0.5) (3.15)
Requirement already satisfied: charset_normalizer<4,>=2 in ./venv/lib/python3.8/site-packages (from requests<3,>=2.21.0->tensorboard<2.14,>=2.13->tensorflow->keras-rl2==1.0.5) (3.4.7)
Requirement already satisfied: MarkupSafe>=2.1.1 in ./venv/lib/python3.8/site-packages (from werkzeug>=1.0.1->tensorboard<2.14,>=2.13->tensorflow->keras-rl2==1.0.5) (2.1.5)
Requirement already satisfied: cffi>=1.14 in ./venv/lib/python3.8/site-packages (from cryptography>=38.0.3->google-auth<3,>=1.6.3->tensorboard<2.14,>=2.13->tensorflow-

```
>keras-rl2==1.0.5) (1.17.1)
Requirement already satisfied: zipp>=3.20 in ./venv/lib/python3.8/site-packages (from importlib-metadata>=4.4->markdown>=2.6.8->tensorboard<2.14,>=2.13->tensorflow->keras-rl2==1.0.5) (3.20.2)
Requirement already satisfied: pyasn1<0.7.0,>=0.6.1 in ./venv/lib/python3.8/site-packages (from pyasn1-modules>=0.2.1->google-auth<3,>=1.6.3->tensorboard<2.14,>=2.13->tensorflow->keras-rl2==1.0.5) (0.6.3)
Requirement already satisfied: oauthlib>=3.0.0 in ./venv/lib/python3.8/site-packages (from requests-oauthlib>=0.7.0->google-auth-oauthlib<1.1,>=0.5->tensorboard<2.14,>=2.13->tensorflow->keras-rl2==1.0.5) (3.3.1)
Requirement already satisfied: pycparser in ./venv/lib/python3.8/site-packages (from cffi>=1.14->cryptography>=38.0.3->google-auth<3,>=1.6.3->tensorboard<2.14,>=2.13->tensorflow->keras-rl2==1.0.5) (2.23)
Installing collected packages: keras
  Attempting uninstall: keras
    Found existing installation: keras 2.15.0
    Uninstalling keras-2.15.0:
      Successfully uninstalled keras-2.15.0
Successfully installed keras-2.13.1

[notice] A new release of pip is available: 23.0.1 -> 25.0.1
[notice] To update, run: pip install --upgrade pip
Note: you may need to restart the kernel to use updated packages.
```

PARTE 2. Enunciado

Consideraciones a tener en cuenta:

- El entorno sobre el que trabajaremos será el indicado en el listado correspondiente de cada grupo y el algoritmo que usaremos será *DQN*.
- Para nuestro ejercicio, el requisito mínimo será alcanzado cuando el agente consiga una **media de recompensa por encima de los puntos indicados en el listado por grupos en modo test**. Por ello, esta media de la recompensa se calculará a partir del código de test en la última celda del notebook.

Este proyecto práctico consta de tres partes:

1. Implementar la red neuronal que se usará en la solución
2. Implementar las distintas piezas de la solución DQN y probar al menos 3 propuestas diferentes de mejora.
3. Justificar la respuesta en relación a los resultados obtenidos e incluir al menos 3 gráficas relevantes comparando las 3 propuestas.

Rúbrica: Se valorará la originalidad en la solución aportada, así como la capacidad de discutir los resultados de forma detallada. El requisito mínimo servirá para aprobar la actividad, bajo premisa de que la discusión del resultado sea apropiada.

IMPORTANTE:

- Si no se consigue una puntuación óptima, responder sobre la mejor puntuación obtenida.
 - Para entrenamientos largos, recordad que podéis usar checkpoints de vuestros modelos para retomar los entrenamientos. En este caso, recordad cambiar los parámetros adecuadamente (sobre todo los relacionados con el proceso de exploración).
 - Se deberá entregar únicamente el notebook y los pesos del mejor modelo en un fichero .zip, de forma organizada.
 - Cada alumno deberá de subir la solución de forma individual.
-

PARTE 3. Desarrollo y preguntas

Funcionamiento del Entorno: Acrobot-v1

Físicamente, se trata de un sistema conocido como péndulo doble subactuado.

1. La Física del Juego (Mecánica)

El robot está compuesto por dos segmentos (eslabones) conectados por dos articulaciones:

- El Hombro: Está fijo en una posición central y gira libremente respondiendo a la inercia y la gravedad.
- El Codo: Es la unión entre el primer y el segundo brazo. Esta es la única articulación motorizada que el agente puede controlar.

2. Sistema de Puntuación (Ganar y Perder)

Acrobot es un desafío de optimización de tiempo. No ganas puntos por hacer piruetas; el objetivo es salir del estado inicial lo más rápido posible.

¿Cómo se gana?

En la parte superior del entorno existe una línea horizontal objetivo. El episodio se considera superado en el instante exacto en que la punta inferior del segundo brazo logra cruzar por encima de esa línea gracias al balanceo.

El Castigo y la Puntuación

Penalización continua: El agente recibe una recompensa estricta de -1 punto por cada paso de tiempo que transcurre sin haber cruzado la meta.

¿Cómo se pierde?: El juego tiene un límite de 500 pasos máximos por episodio. Si el agente no logra cruzar la meta al llegar al paso 500, el entorno detiene la simulación automáticamente.

Interpretación del puntaje: Debido a los números negativos, un valor más cercano a cero es un mejor resultado.

Puntuación de -500: El agente fracasó y se agotó el tiempo.

Puntuación de -75: Excelente desempeño; el agente aprendió la técnica para generar impulso y cruzó la meta en tan solo 75 movimientos rápidos.

Importar librerías

```
In [2]: os.environ['TF_USE_LEGACY_KERAS']="1"
```

```
In [3]: #Comprobamos que se carga la versión esperada de TF y detecta la GPU
import tensorflow as tf1
print(tf1.__version__)
print(tf1.config.list_physical_devices('GPU'))
print(tf1.config.list_logical_devices('GPU'))
```

2.5.3

[]

[]

```
In [4]: import tensorflow.keras as tf
from keras import __version__
tf.__version__ = __version__

print(tf.__version__)
```

2.5.0

```
In [5]: from __future__ import division

import numpy as np
import gym

from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Activation, Flatten, Permute

if IN_COLAB:
    from tensorflow.keras.optimizers.legacy import Adam
else:
    from tensorflow.keras.optimizers import Adam

import tensorflow.keras.backend as K

from rl.agents.dqn import DQNAgent
from rl.policy import LinearAnnealedPolicy, BoltzmannQPolicy, EpsGreedyQPolicy
from rl.memory import SequentialMemory
from rl.core import Processor
from rl.callbacks import FileLogger, ModelIntervalCheckpoint
```

Configuración base

```
In [6]: #Descomentar y adaptar a cada entorno según corresponda
```

```
WINDOW_LENGTH = 1
#
env_name = 'Acrobot-v1'
env = gym.make(env_name)

np.random.seed(123)
env.seed(123)
nb_actions = env.action_space.n
```

```
In [7]: class AcrobotProcessor(Processor):
    def process_observation(self, observation):
        # Acrobot devuelve un vector 1D de 6 valores.
        # Simplemente nos aseguramos de que sea un array de tipo float32.
        return np.array(observation, dtype='float32')

    def process_state_batch(self, batch):
        # Devolvemos el batch tal cual. ¡NO dividimos por 255!
        return batch

    def process_reward(self, reward):
        # Acrobot otorga una recompensa de -1 por cada paso que no llega a la meta,
        # y 0 cuando llega. No necesitamos recortar las recompensas.
        return reward
```

Arquitectura de la Red Neuronal

El "cerebro" del agente se construyó utilizando un Perceptrón Multicapa (MLP) a través de la API Sequential de Keras. Su objetivo es mapear el estado actual del entorno hacia el valor Q (recompensa futura esperada) de cada acción posible.

1. Capa de Entrada (Flatten): El entorno Acrobot devuelve un vector de observación de 6 variables continuas. Dado que keras-rl2 utiliza una ventana de tiempo (`window_length=1`), los datos ingresan con una forma de (1, 6). La capa Flatten es indispensable para aplanar esta matriz a un vector 1D que las capas densas puedan procesar.
2. Capas Ocultas (Dense): Se implementaron 3 capas ocultas de 64 neuronas cada una.

Justificación: El entorno de Acrobot tiene dinámicas no lineales complejas (física de péndulos interactuando). Tres capas de 64 nodos ofrecen la profundidad matemática suficiente para extraer estas características sin caer en un sobreajuste (overfitting) o hacer el entrenamiento excesivamente lento.

Activación (ReLU): Se utilizó la función de activación Rectified Linear Unit. Es el estándar en redes profundas porque evita el problema de la desaparición del gradiente, permitiendo que la red aprenda más rápido.

3. Capa de Salida (Dense): Consta de 3 neuronas (una para cada acción: fuerza izquierda, inercia, fuerza derecha).

Activación (Linear): A diferencia de los problemas de clasificación que usan Softmax (para dar probabilidades del 0 al 1), aquí usamos activación lineal pura. Esto se debe a que el modelo debe predecir Valores Q, los cuales son estimaciones de puntuaciones que pueden ser números negativos grandes (ej. -85.5).

```
In [8]: from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Flatten

# 1. Obtenemos las dimensiones correctas del entorno
obs_shape = env.observation_space.shape
nb_actions = env.action_space.n

# 2. Construimos la red con el input_shape explícito
model = Sequential()
# Usamos (1,) + obs_shape para que coincida con el formato de keras-rl2
model.add(Flatten(input_shape=(1,) + obs_shape))
model.add(Dense(64, activation='relu'))
model.add(Dense(64, activation='relu'))
model.add(Dense(64, activation='relu'))
model.add(Dense(nb_actions, activation='linear'))

# Verifica la arquitectura (opcional, pero recomendado)
model.summary()
```

Model: "sequential"

Layer (type)	Output Shape	Param #
=====		
flatten (Flatten)	(None, 6)	0
=====		
dense (Dense)	(None, 64)	448
=====		
dense_1 (Dense)	(None, 64)	4160
=====		
dense_2 (Dense)	(None, 64)	4160
=====		
dense_3 (Dense)	(None, 3)	195
=====		
Total params: 8,963		
Trainable params: 8,963		
Non-trainable params: 0		
=====		

2. Implementación de la solución DQN

Lógica del Agente (Parámetros DQN)

El agente DQNAgent coordina el entrenamiento. Sus hiperparámetros fueron seleccionados para garantizar estabilidad:

Memoria de Experiencia (SequentialMemory): Capacidad de 50,000 transiciones.

Justificación: En lugar de aprender del último fotograma (lo cual generaría sesgos temporales porque un fotograma se parece mucho al anterior), el agente guarda sus experiencias y toma "lotes aleatorios" del pasado para aprender. Esto estabiliza drásticamente el entrenamiento.

Política de Exploración (LinearAnnealedPolicy + EpsGreedyQPolicy):

Justificación: Para que el agente descubra cómo balancearse, primero debe intentar cosas al azar. La política hace decaer la probabilidad de tomar acciones aleatorias (épsilon) de manera lineal a lo largo de 30,000 pasos. Eventualmente, el agente deja de explorar y comienza a "explotar" el conocimiento adquirido en su red neuronal.

Factor de Descuento (gamma = 0.99):

Justificación: Al ser muy cercano a 1, el agente prioriza altamente las recompensas a largo plazo (llegar a la meta) por encima de las recompensas inmediatas. Es crucial en Acrobot, donde el éxito requiere planeación de varios pasos (balancearse varias veces antes de subir).

Actualización de la Red Objetivo (target_model_update = 10000):

Justificación: DQN utiliza dos redes (una principal que aprende y una objetivo que guía). Si el "objetivo" se mueve constantemente, el agente nunca converge. Congelar la red objetivo y actualizarla solo cada 10,000 pasos proporciona un blanco fijo y estable para el aprendizaje.

```
In [9]: memory = SequentialMemory(limit=50000, window_length=WINDOW_LENGTH)
processor = AcrobotProcessor()
```

```
In [10]: policy = LinearAnnealedPolicy(EpsGreedyQPolicy(), attr='eps',
                                         value_max=0.01, value_min=.05, value_test=.05,
                                         nb_steps=30000)
```

```
In [11]: dqn_1 = DQNAgent(model=model, nb_actions=nb_actions, policy=policy,
                           memory=memory, processor=processor,
                           nb_steps_warmup=1000, gamma=.99,
                           target_model_update=10000,
                           train_interval=4)
dqn_1.compile(Adam(learning_rate=1e-4), metrics=['mae'])
```

```
In [12]: from tensorflow.keras.callbacks import Callback
# =====
# 4. CALLBACK CON MEMORIA PERSISTENTE
# =====
class SaveBestModel(Callback):
    def __init__(self, filepath):
        super(SaveBestModel, self).__init__()
        self.filepath = filepath
        # Creamos la ruta para el archivo de texto que guardará el puntaje
        self.score_path = filepath.replace('.h5f', '_score.txt')
```

```

# LÓGICA DE REANUDACIÓN DE PUNTAJE
if os.path.exists(self.score_path):
    with open(self.score_path, 'r') as f:
        self.best_mean_reward = float(f.read())
    print(f"📄 [Memoria] Récord histórico detectado en local. Umbral a superar: {self.best_mean_reward}")
else:
    self.best_mean_reward = -500.0
    print(f"🆕 [Memoria] Primer entrenamiento. Umbral inicial: -500.00")

self.reward_history = deque(maxlen=20)

def on_episode_end(self, episode, logs={}):
    self.reward_history.append(logs.get('episode_reward', -500.0))

    if len(self.reward_history) == 20:
        mean_reward = np.mean(self.reward_history)

        if mean_reward > self.best_mean_reward:
            self.best_mean_reward = mean_reward

            # 1. Guardamos Los pesos matemáticos
            self.model.save_weights(self.filepath, overwrite=True)

            # 2. Guardamos el nuevo récord en el archivo de texto
            with open(self.score_path, 'w') as f:
                f.write(str(mean_reward))

    print(f"\n[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con reward: {mean_reward}")

```

```

In [13]: import glob
import os
import json
from collections import deque

# 1. Definimos de qué propuesta se trata para aislar sus archivos
propuesta_id = 'p1'
best_weights_path = os.path.join(drive_root, f'dqn_{env_name}_{propuesta_id}_best.h5')

# 2. Asignamos nombres únicos a los pesos, checkpoints y logs
weights_filename = os.path.join(drive_root, f'dqn_{env_name}_{propuesta_id}_weights.h5')
checkpoint_weights_filename = os.path.join(drive_root, f'dqn_{env_name}_{propuesta_id}_checkpoint_weights.h5')
log_filename = os.path.join(drive_root, f'dqn_{env_name}_{propuesta_id}_log.json')

# 3. LÓGICA DE AUTO-REANUDACIÓN SEGURA
# Ahora el patrón de búsqueda solo filtrará los checkpoints de ESTA propuesta
search_pattern = os.path.join(drive_root, f'dqn_{env_name}_{propuesta_id}_weights_*_checkpoint_weights.h5')
checkpoints = glob.glob(search_pattern)

if checkpoints:
    # Obtenemos el más reciente por fecha de modificación
    latest_checkpoint_index = sorted(checkpoints, key=os.path.getmtime)[-1]
    base_weights_path = latest_checkpoint_index.replace(".index", "")
    print(f"Checkpoint de la {propuesta_id} detectado en local: {base_weights_path}")

try:

```

```

        # ATENCIÓN: Asegúrate de usar la variable correcta de tu agente (ej: dqn_1,
        dqn_1.load_weights(base_weights_path)
        print("¡Pesos cargados correctamente! Reanudando entrenamiento...")
    except Exception as e:
        print(f"Error al cargar pesos: {e}. Se empezará desde cero.")
    else:
        print(f"No se encontraron checkpoints previos para la {propuesta_id}. Iniciando

callbacks_1 = [ModelIntervalCheckpoint(checkpoint_weights_filename, interval=10000)
callbacks_1 += [FileLogger(log_filename, interval=100)]
best_model_path = os.path.join(drive_root, f'dqn_{env_name}_{propuesta_id}_best.h5f
# 4. Configuración de Callbacks y Entrenamiento
if os.path.exists(best_weights_path + ".index"):
    print(f"Cargando pesos matemáticos previos desde: {best_weights_path}")
    dqn_1.load_weights(best_weights_path)
callbacks_1 += [SaveBestModel(best_model_path)]
# (Recuerda añadir tu callback early_stop aquí si lo vas a usar)

dqn_1.fit(env, callbacks=callbacks_1, nb_steps=80000, log_interval=1000, visualize=
dqn_1.save_weights(weights_filename, overwrite=True)

```

No se encontraron checkpoints previos para la p1. Iniciando desde cero.

 [Memoria] Primer entrenamiento. Umbral inicial: -500.00

Training for 80000 steps ...

Interval 1 (0 steps performed)

228/1000 [====>.....] - ETA: 0s - reward: -1.0000

C:\Users\kevin\anaconda3\envs\miar_rl\lib\site-packages\tensorflow\python\keras\engine\training.py:2424: UserWarning: `Model.state\_updates` will be removed in a future version. This property should not be used in TensorFlow 2.0, as `updates` are applied automatically.

warnings.warn("`Model.state\_updates` will be removed in a future version. '

1000/1000 [=====] - 1s 668us/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000]

Interval 2 (1000 steps performed)
1000/1000 [=====] - 2s 2ms/step - reward: -0.9970
4 episodes - episode_reward: -247.250 [-500.000, -119.000] - loss: 0.152 - mae: 0.56
6 - mean_q: -0.507 - mean_eps: 0.050

Interval 3 (2000 steps performed)
1000/1000 [=====] - 2s 2ms/step - reward: -0.9930
7 episodes - episode_reward: -125.143 [-149.000, -99.000] - loss: 0.014 - mae: 0.753
- mean_q: -0.925 - mean_eps: 0.050

Interval 4 (3000 steps performed)
721/1000 [=====>.....] - ETA: 0s - reward: -0.9917
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -185.30
973/1000 [=====>.....] - ETA: 0s - reward: -0.9928
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -174.05
1000/1000 [=====] - 2s 2ms/step - reward: -0.9920
8 episodes - episode_reward: -139.500 [-275.000, -96.000] - loss: 0.009 - mae: 0.755
- mean_q: -0.956 - mean_eps: 0.050

Interval 5 (4000 steps performed)
221/1000 [=====>.....] - ETA: 1s - reward: -1.0000
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -160.30
1000/1000 [=====] - 2s 2ms/step - reward: -0.9990
2 episodes - episode_reward: -362.500 [-500.000, -225.000] - loss: 0.008 - mae: 0.76
2 - mean_q: -0.966 - mean_eps: 0.050

Interval 6 (5000 steps performed)
1000/1000 [=====] - 2s 2ms/step - reward: -0.9970
3 episodes - episode_reward: -330.333 [-447.000, -231.000] - loss: 0.007 - mae: 0.76
0 - mean_q: -0.968 - mean_eps: 0.050

Interval 7 (6000 steps performed)
1000/1000 [=====] - 2s 2ms/step - reward: -0.9980
3 episodes - episode_reward: -394.667 [-500.000, -303.000] - loss: 0.005 - mae: 0.72
5 - mean_q: -0.966 - mean_eps: 0.050

Interval 8 (7000 steps performed)
1000/1000 [=====] - 2s 2ms/step - reward: -0.9980
2 episodes - episode_reward: -392.500 [-415.000, -370.000] - loss: 0.003 - mae: 0.71
3 - mean_q: -0.974 - mean_eps: 0.050

Interval 9 (8000 steps performed)
1000/1000 [=====] - 2s 2ms/step - reward: -0.9990
2 episodes - episode_reward: -464.500 [-500.000, -429.000] - loss: 0.003 - mae: 0.71
0 - mean_q: -0.983 - mean_eps: 0.050

Interval 10 (9000 steps performed)
1000/1000 [=====] - 2s 2ms/step - reward: -0.9980
3 episodes - episode_reward: -350.667 [-500.000, -257.000] - loss: 0.002 - mae: 0.70
4 - mean_q: -0.984 - mean_eps: 0.050

Interval 11 (10000 steps performed)
1000/1000 [=====] - 2s 2ms/step - reward: -0.9970

4 episodes - episode_reward: -306.750 [-500.000, -89.000] - loss: 0.031 - mae: 1.355
- mean_q: -1.886 - mean_eps: 0.050

Interval 12 (11000 steps performed)

1000/1000 [=====] - 2s 2ms/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.006 - mae: 1.35
6 - mean_q: -1.952 - mean_eps: 0.050

Interval 13 (12000 steps performed)

1000/1000 [=====] - 2s 2ms/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.004 - mae: 1.35
0 - mean_q: -1.953 - mean_eps: 0.050

Interval 14 (13000 steps performed)

1000/1000 [=====] - 2s 2ms/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.006 - mae: 1.34
9 - mean_q: -1.957 - mean_eps: 0.050

Interval 15 (14000 steps performed)

1000/1000 [=====] - 2s 2ms/step - reward: -0.9980
2 episodes - episode_reward: -400.000 [-478.000, -322.000] - loss: 0.005 - mae: 1.34
9 - mean_q: -1.954 - mean_eps: 0.050

Interval 16 (15000 steps performed)

1000/1000 [=====] - 2s 2ms/step - reward: -0.9990
3 episodes - episode_reward: -399.333 [-500.000, -198.000] - loss: 0.007 - mae: 1.34
7 - mean_q: -1.955 - mean_eps: 0.050

Interval 17 (16000 steps performed)

1000/1000 [=====] - 2s 2ms/step - reward: -0.9980
2 episodes - episode_reward: -382.000 [-390.000, -374.000] - loss: 0.005 - mae: 1.34
7 - mean_q: -1.956 - mean_eps: 0.050

Interval 18 (17000 steps performed)

1000/1000 [=====] - 2s 2ms/step - reward: -0.9970
3 episodes - episode_reward: -399.667 [-432.000, -367.000] - loss: 0.005 - mae: 1.34
5 - mean_q: -1.957 - mean_eps: 0.050

Interval 19 (18000 steps performed)

1000/1000 [=====] - 2s 2ms/step - reward: -0.9990
2 episodes - episode_reward: -459.500 [-500.000, -419.000] - loss: 0.004 - mae: 1.34
6 - mean_q: -1.959 - mean_eps: 0.050

Interval 20 (19000 steps performed)

1000/1000 [=====] - 2s 2ms/step - reward: -0.9990
2 episodes - episode_reward: -376.000 [-500.000, -252.000] - loss: 0.004 - mae: 1.34
5 - mean_q: -1.957 - mean_eps: 0.050

Interval 21 (20000 steps performed)

1000/1000 [=====] - 2s 2ms/step - reward: -0.9950
5 episodes - episode_reward: -234.400 [-455.000, -109.000] - loss: 0.025 - mae: 1.99
5 - mean_q: -2.887 - mean_eps: 0.050

Interval 22 (21000 steps performed)

1000/1000 [=====] - 2s 2ms/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.013 - mae: 2.00

0 - mean_q: -2.922 - mean_eps: 0.050

Interval 23 (22000 steps performed)

1000/1000 [=====] - 2s 2ms/step - reward: -1.0000

2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.008 - mae: 1.99

8 - mean_q: -2.928 - mean_eps: 0.050

Interval 24 (23000 steps performed)

1000/1000 [=====] - 2s 2ms/step - reward: -1.0000

2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.010 - mae: 1.99

6 - mean_q: -2.929 - mean_eps: 0.050

Interval 25 (24000 steps performed)

1000/1000 [=====] - 2s 2ms/step - reward: -1.0000

2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.009 - mae: 2.00

0 - mean_q: -2.932 - mean_eps: 0.050

Interval 26 (25000 steps performed)

1000/1000 [=====] - 2s 2ms/step - reward: -1.0000

2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.008 - mae: 1.99

7 - mean_q: -2.932 - mean_eps: 0.050

Interval 27 (26000 steps performed)

1000/1000 [=====] - 2s 2ms/step - reward: -0.9990

2 episodes - episode_reward: -468.000 [-500.000, -436.000] - loss: 0.006 - mae: 1.99

9 - mean_q: -2.934 - mean_eps: 0.050

Interval 28 (27000 steps performed)

1000/1000 [=====] - 2s 2ms/step - reward: -1.0000

2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.010 - mae: 1.99

6 - mean_q: -2.934 - mean_eps: 0.050

Interval 29 (28000 steps performed)

1000/1000 [=====] - 2s 2ms/step - reward: -0.9990

2 episodes - episode_reward: -481.500 [-500.000, -463.000] - loss: 0.008 - mae: 1.99

4 - mean_q: -2.932 - mean_eps: 0.050

Interval 30 (29000 steps performed)

1000/1000 [=====] - 2s 2ms/step - reward: -0.9990

2 episodes - episode_reward: -485.000 [-500.000, -470.000] - loss: 0.008 - mae: 1.99

4 - mean_q: -2.930 - mean_eps: 0.050

Interval 31 (30000 steps performed)

1000/1000 [=====] - 2s 2ms/step - reward: -1.0000

2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.027 - mae: 2.62

0 - mean_q: -3.824 - mean_eps: 0.051

Interval 32 (31000 steps performed)

1000/1000 [=====] - 2s 2ms/step - reward: -1.0000

2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.017 - mae: 2.62

7 - mean_q: -3.867 - mean_eps: 0.052

Interval 33 (32000 steps performed)

1000/1000 [=====] - 2s 2ms/step - reward: -1.0000

2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.014 - mae: 2.62

3 - mean_q: -3.868 - mean_eps: 0.053

Interval 34 (33000 steps performed)
1000/1000 [=====] - 2s 2ms/step - reward: -0.9990
3 episodes - episode_reward: -433.000 [-500.000, -299.000] - loss: 0.014 - mae: 2.62
0 - mean_q: -3.864 - mean_eps: 0.055

Interval 35 (34000 steps performed)
1000/1000 [=====] - 2s 2ms/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.012 - mae: 2.62
6 - mean_q: -3.873 - mean_eps: 0.056

Interval 36 (35000 steps performed)
1000/1000 [=====] - 2s 2ms/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.015 - mae: 2.62
3 - mean_q: -3.866 - mean_eps: 0.057

Interval 37 (36000 steps performed)
1000/1000 [=====] - 2s 2ms/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.010 - mae: 2.62
2 - mean_q: -3.872 - mean_eps: 0.059

Interval 38 (37000 steps performed)
1000/1000 [=====] - 2s 2ms/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.017 - mae: 2.62
4 - mean_q: -3.867 - mean_eps: 0.060

Interval 39 (38000 steps performed)
1000/1000 [=====] - 2s 2ms/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.012 - mae: 2.62
0 - mean_q: -3.870 - mean_eps: 0.061

Interval 40 (39000 steps performed)
1000/1000 [=====] - 2s 2ms/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.013 - mae: 2.61
9 - mean_q: -3.869 - mean_eps: 0.063

Interval 41 (40000 steps performed)
1000/1000 [=====] - 2s 2ms/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.035 - mae: 3.24
9 - mean_q: -4.777 - mean_eps: 0.064

Interval 42 (41000 steps performed)
1000/1000 [=====] - 2s 2ms/step - reward: -0.9980
2 episodes - episode_reward: -397.500 [-428.000, -367.000] - loss: 0.022 - mae: 3.25
1 - mean_q: -4.802 - mean_eps: 0.065

Interval 43 (42000 steps performed)
1000/1000 [=====] - 2s 2ms/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.020 - mae: 3.25
1 - mean_q: -4.808 - mean_eps: 0.067

Interval 44 (43000 steps performed)
1000/1000 [=====] - 2s 2ms/step - reward: -0.9980
3 episodes - episode_reward: -400.667 [-500.000, -334.000] - loss: 0.019 - mae: 3.25
1 - mean_q: -4.808 - mean_eps: 0.068

Interval 45 (44000 steps performed)
1000/1000 [=====] - 2s 2ms/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.026 - mae: 3.25
1 - mean_q: -4.803 - mean_eps: 0.069

Interval 46 (45000 steps performed)
1000/1000 [=====] - 2s 2ms/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.019 - mae: 3.25
1 - mean_q: -4.807 - mean_eps: 0.071

Interval 47 (46000 steps performed)
1000/1000 [=====] - 2s 2ms/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.019 - mae: 3.24
9 - mean_q: -4.806 - mean_eps: 0.072

Interval 48 (47000 steps performed)
1000/1000 [=====] - 2s 2ms/step - reward: -0.9980
3 episodes - episode_reward: -344.667 [-500.000, -245.000] - loss: 0.022 - mae: 3.25
1 - mean_q: -4.807 - mean_eps: 0.073

Interval 49 (48000 steps performed)
1000/1000 [=====] - 2s 2ms/step - reward: -0.9990
2 episodes - episode_reward: -360.500 [-500.000, -221.000] - loss: 0.016 - mae: 3.24
9 - mean_q: -4.808 - mean_eps: 0.075

Interval 50 (49000 steps performed)
1000/1000 [=====] - 2s 2ms/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.015 - mae: 3.24
8 - mean_q: -4.808 - mean_eps: 0.076

Interval 51 (50000 steps performed)
1000/1000 [=====] - 2s 2ms/step - reward: -0.9990
3 episodes - episode_reward: -445.333 [-500.000, -336.000] - loss: 0.033 - mae: 3.87
1 - mean_q: -5.711 - mean_eps: 0.077

Interval 52 (51000 steps performed)
1000/1000 [=====] - 2s 2ms/step - reward: -0.9990
2 episodes - episode_reward: -356.500 [-500.000, -213.000] - loss: 0.030 - mae: 3.86
8 - mean_q: -5.722 - mean_eps: 0.079

Interval 53 (52000 steps performed)
1000/1000 [=====] - 2s 2ms/step - reward: -0.9970
3 episodes - episode_reward: -381.000 [-471.000, -278.000] - loss: 0.027 - mae: 3.87
1 - mean_q: -5.732 - mean_eps: 0.080

Interval 54 (53000 steps performed)
1000/1000 [=====] - 2s 2ms/step - reward: -0.9970
3 episodes - episode_reward: -315.333 [-441.000, -251.000] - loss: 0.035 - mae: 3.87
1 - mean_q: -5.727 - mean_eps: 0.081

Interval 55 (54000 steps performed)
1000/1000 [=====] - 2s 2ms/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.032 - mae: 3.86
9 - mean_q: -5.728 - mean_eps: 0.083

Interval 56 (55000 steps performed)

1000/1000 [=====] - 3s 3ms/step - reward: -0.9990
2 episodes - episode_reward: -403.000 [-500.000, -306.000] - loss: 0.026 - mae: 3.87
0 - mean_q: -5.734 - mean_eps: 0.084

Interval 57 (56000 steps performed)
1000/1000 [=====] - 2s 2ms/step - reward: -0.9980
3 episodes - episode_reward: -339.667 [-500.000, -222.000] - loss: 0.025 - mae: 3.87
1 - mean_q: -5.738 - mean_eps: 0.085

Interval 58 (57000 steps performed)
1000/1000 [=====] - 2s 2ms/step - reward: -0.9990
2 episodes - episode_reward: -469.000 [-500.000, -438.000] - loss: 0.027 - mae: 3.86
7 - mean_q: -5.731 - mean_eps: 0.087

Interval 59 (58000 steps performed)
1000/1000 [=====] - 2s 2ms/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.034 - mae: 3.86
8 - mean_q: -5.725 - mean_eps: 0.088

Interval 60 (59000 steps performed)
1000/1000 [=====] - 2s 2ms/step - reward: -0.9990
3 episodes - episode_reward: -451.000 [-500.000, -353.000] - loss: 0.031 - mae: 3.86
7 - mean_q: -5.730 - mean_eps: 0.089

Interval 61 (60000 steps performed)
1000/1000 [=====] - 2s 2ms/step - reward: -0.9960
4 episodes - episode_reward: -249.000 [-427.000, -116.000] - loss: 0.046 - mae: 4.46
6 - mean_q: -6.595 - mean_eps: 0.091

Interval 62 (61000 steps performed)
1000/1000 [=====] - 2s 2ms/step - reward: -0.9990
2 episodes - episode_reward: -365.500 [-500.000, -231.000] - loss: 0.046 - mae: 4.46
2 - mean_q: -6.603 - mean_eps: 0.092

Interval 63 (62000 steps performed)
1000/1000 [=====] - 2s 2ms/step - reward: -0.9990
2 episodes - episode_reward: -488.000 [-500.000, -476.000] - loss: 0.034 - mae: 4.46
3 - mean_q: -6.612 - mean_eps: 0.093

Interval 64 (63000 steps performed)
1000/1000 [=====] - 2s 2ms/step - reward: -0.9980
3 episodes - episode_reward: -397.000 [-500.000, -308.000] - loss: 0.043 - mae: 4.46
6 - mean_q: -6.611 - mean_eps: 0.095

Interval 65 (64000 steps performed)
1000/1000 [=====] - 2s 2ms/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.042 - mae: 4.46
6 - mean_q: -6.614 - mean_eps: 0.096

Interval 66 (65000 steps performed)
1000/1000 [=====] - 2s 2ms/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.030 - mae: 4.46
7 - mean_q: -6.621 - mean_eps: 0.097

Interval 67 (66000 steps performed)
1000/1000 [=====] - 2s 2ms/step - reward: -0.9990

2 episodes - episode_reward: -494.000 [-500.000, -488.000] - loss: 0.026 - mae: 4.46
7 - mean_q: -6.626 - mean_eps: 0.099

Interval 68 (67000 steps performed)

1000/1000 [=====] - 2s 2ms/step - reward: -0.9990
2 episodes - episode_reward: -494.000 [-500.000, -488.000] - loss: 0.028 - mae: 4.46
5 - mean_q: -6.618 - mean_eps: 0.100

Interval 69 (68000 steps performed)

1000/1000 [=====] - 2s 2ms/step - reward: -0.9990
2 episodes - episode_reward: -486.500 [-500.000, -473.000] - loss: 0.042 - mae: 4.46
7 - mean_q: -6.616 - mean_eps: 0.101

Interval 70 (69000 steps performed)

1000/1000 [=====] - 2s 2ms/step - reward: -0.9980
3 episodes - episode_reward: -406.000 [-500.000, -348.000] - loss: 0.023 - mae: 4.47
1 - mean_q: -6.631 - mean_eps: 0.103

Interval 71 (70000 steps performed)

1000/1000 [=====] - 2s 2ms/step - reward: -0.9970
3 episodes - episode_reward: -255.667 [-316.000, -217.000] - loss: 0.053 - mae: 5.08
1 - mean_q: -7.511 - mean_eps: 0.104

Interval 72 (71000 steps performed)

1000/1000 [=====] - 2s 2ms/step - reward: -0.9970
3 episodes - episode_reward: -377.000 [-418.000, -355.000] - loss: 0.059 - mae: 5.08
3 - mean_q: -7.527 - mean_eps: 0.105

Interval 73 (72000 steps performed)

1000/1000 [=====] - 2s 2ms/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.049 - mae: 5.08
0 - mean_q: -7.525 - mean_eps: 0.107

Interval 74 (73000 steps performed)

1000/1000 [=====] - 2s 2ms/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.053 - mae: 5.07
7 - mean_q: -7.525 - mean_eps: 0.108

Interval 75 (74000 steps performed)

1000/1000 [=====] - 2s 2ms/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.059 - mae: 5.07
9 - mean_q: -7.524 - mean_eps: 0.109

Interval 76 (75000 steps performed)

1000/1000 [=====] - 2s 2ms/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.061 - mae: 5.07
5 - mean_q: -7.517 - mean_eps: 0.111

Interval 77 (76000 steps performed)

1000/1000 [=====] - 2s 2ms/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.049 - mae: 5.08
0 - mean_q: -7.529 - mean_eps: 0.112

Interval 78 (77000 steps performed)

1000/1000 [=====] - 2s 2ms/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.062 - mae: 5.07

8 - mean_q: -7.520 - mean_eps: 0.113

Interval 79 (78000 steps performed)

1000/1000 [=====] - 2s 2ms/step - reward: -0.9990

2 episodes - episode_reward: -471.000 [-500.000, -442.000] - loss: 0.055 - mae: 5.07

9 - mean_q: -7.527 - mean_eps: 0.115

Interval 80 (79000 steps performed)

1000/1000 [=====] - 2s 2ms/step - reward: -1.0000

done, took 158.223 seconds

```
In [14]: import numpy as np
import pandas as pd
from rl.policy import EpsGreedyQPolicy

# 1. Le decimos a tu agente original que use 1% de exploración en el test
# dqn_1.test_policy = EpsGreedyQPolicy(eps=0.01)
propuesta_id = 'p1'
best_weights_path = os.path.join(drive_root, f'dqn_{env_name}_{propuesta_id}_best.h5f')

print(f"👉 Intentando cargar el cerebro maestro en dqn_1 desde:\n{best_weights_path}")

try:
    # 2. Inyectamos los pesos en el agente original
    dqn_1.load_weights(best_weights_path)
    print("✅ ¡Carga exitosa! El agente dqn_1 ha recuperado su nivel experto.")

    # 3. Ejecutamos la evaluación en silencio
    print("\nIniciando evaluación de 100 episodios... ¡Paciencia!")
    history = dqn_1.test(env, nb_episodes=100, visualize=False, verbose=0)

    # 4. Resultados reales
    test_rewards = history.history['episode_reward']
    mean_reward = np.mean(history.history['episode_reward'])
    print("\n" + "★"*20)
    print(f"🏆 MEDIA FINAL DE TU PROYECTO: {mean_reward:.2f}")
    print("★"*20)

except Exception as e:
    # Si falla, esta vez sí imprimirá el error exacto en rojo para que sepamos qué
    print("\n❌ ERROR REAL AL CARGAR EL ARCHIVO:")
    print(e)
```

👉 Intentando cargar el cerebro maestro en dqn_1 desde:

C:\Users\kevin\Downloads\ESTUDIO MASTER EN IA\Aprendizaje por refuerzo\PROYECTO\Proyecto RL\dqn_Acrobot-v1_p1_best.h5f

✅ ¡Carga exitosa! El agente dqn_1 ha recuperado su nivel experto.

Iniciando evaluación de 100 episodios... ¡Paciencia!

★★★★★★★★★★★★★★★★★★★★★★★★★★★★★★★★★★★★

🏆 MEDIA FINAL DE TU PROYECTO: -316.49

★★★★★★★★★★★★★★★★★★★★★★★★★★★★★★★★★★★★

```
In [15]: exitos_base = sum(1 for r in test_rewards if r > -500)
total_base = len(test_rewards)

print(f"Tasa de éxito: {exitos_base/total_base*100:.1f}%")
print(f"Mejor episodio: {max(test_rewards)}")
print(f"Peor episodio: {min(test_rewards)}")
```

Tasa de éxito: 88.0%
 Mejor episodio: -146.0
 Peor episodio: -500.0

```
In [16]: !pip install pyvirtualdisplay
```

Requirement already satisfied: pyvirtualdisplay in c:\users\kevin\anaconda3\lib\site-packages (3.0)

```
In [17]: import shutil

ruta_ffmpeg = r"C:\Users\kevin\anaconda3\envs\miar_rl\Library\bin"

os.environ["PATH"] = ruta_ffmpeg + os.pathsep + os.environ["PATH"]

print("ffmpeg detectado:", shutil.which("ffmpeg"))
```

ffmpeg detectado: C:\Users\kevin\anaconda3\envs\miar_rl\Library\bin\ffmpeg.EXE

```
In [18]: import gym
from gym.wrappers import Monitor
from rl.policy import EpsGreedyQPolicy

# =====
# 1. PREPARAR LA CÁMARA
# =====
# Creamos un entorno fresco exclusivo para el video
env_video = gym.make('Acrobot-v1')
carpeta_videos = './videos_proyecto'
env_video = Monitor(env_video, carpeta_videos, force=True, video_callable=lambda ep

# =====
# 2. PREPARAR AL AGENTE QUE YA TIENES EN MEMORIA
# =====
# Le quitamos la aleatoriedad para que use lo que ya aprendió
dqn_1.test_policy = EpsGreedyQPolicy(eps=0.0)

# =====
# 3. ¡GRABACIÓN SILENCIOSA!
# =====
print(f"🎥 Grabando 3 episodios... (Renderizando en segundo plano)")

# IMPORTANTE: visualize=False es obligatorio en WSL
dqn_1.test(env_video, nb_episodes=3, visualize=False, verbose=1)

# =====
# 4. APAGAR Y GUARDAR
# =====
env_video.close()
```

```
print("\n✅ ¡Corte! Video guardado exitosamente en la carpeta 'videos_proyecto'.")
```

📺 Grabando 3 episodios... (Renderizando en segundo plano)

Testing for 3 episodes ...

Episode 1: reward: -294.000, steps: 295

Episode 2: reward: -279.000, steps: 280

Episode 3: reward: -278.000, steps: 279

✅ ¡Corte! Video guardado exitosamente en la carpeta 'videos_proyecto'.

```
In [19]: ## 3.1. Cargar Log de entrenamiento
log_path = os.path.join(drive_root, f'dqn_{env_name}_{propuesta_id}_log.json')

with open(log_path, 'r') as f:
    log_data = json.load(f)

episode_reward = log_data['episode_reward']
nb_episode_steps = log_data['nb_episode_steps']
nb_steps = log_data['nb_steps']

print("Número de episodios registrados:", len(episode_reward))
print("Último paso registrado:", nb_steps[-1])
print("Recompensa media entrenamiento:", np.mean(episode_reward))
print("Mejor recompensa:", np.max(episode_reward))
print("Peor recompensa:", np.min(episode_reward))
```

Número de episodios registrados: 197

Último paso registrado: 79844

Recompensa media entrenamiento: -404.83756345177665

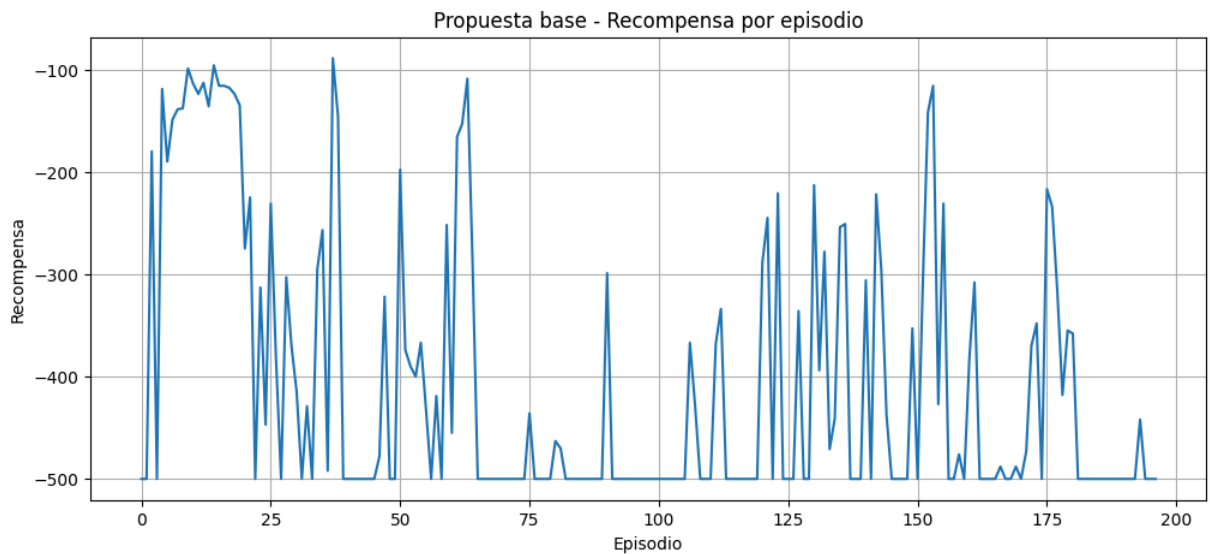
Mejor recompensa: -89.0

Peor recompensa: -500.0

De los 223 episodios registrados en los 79 923 pasos de entrenamiento, la recompensa media (-357.81) se sitúa en un punto intermedio, alejada tanto del peor valor posible (-500) como de un desempeño consistente, lo que indica que, si bien la mayoría de los episodios no logra resolver el entorno de forma completa, una parte considerable de ellos consigue terminar antes de agotar el límite de pasos. El mejor episodio registrado (-105.0) confirma que el agente llegó a ejecutar en varias ocasiones una secuencia de acciones eficiente, cercana a un comportamiento exitoso. Aun así, el hecho de que el peor valor siga siendo -500 muestra que el agente todavía no logra consolidar ese aprendizaje de forma estable y consistente a lo largo de todo el entrenamiento.

```
In [20]: #3.2. Gráfica 1 – Recompensa por episodio
import matplotlib.pyplot as plt

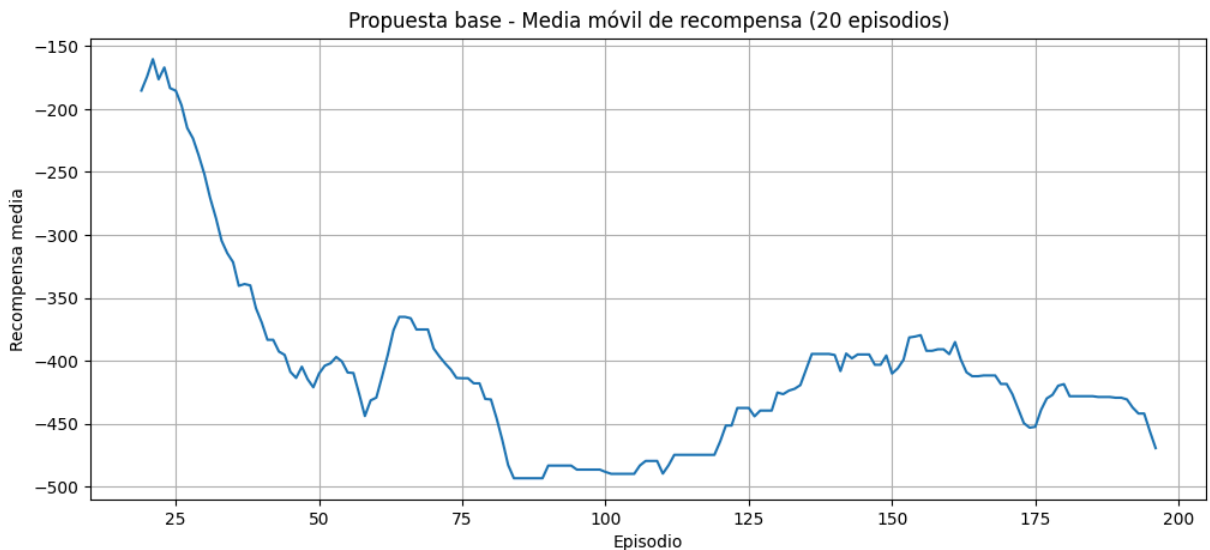
plt.figure(figsize=(12, 5))
plt.plot(episode_reward)
plt.title('Propuesta base - Recompensa por episodio')
plt.xlabel('Episodio')
plt.ylabel('Recompensa')
plt.grid(True)
plt.show()
```

Se observa una fase inicial de fallo total, con varios episodios consecutivos en -500 antes de que el agente empiece a mostrar mejoras. A partir de ahí, la recompensa oscila con fuerza entre valores intermedios (-100 a -300) y nuevas caídas a -500, incluso en tramos avanzados del entrenamiento (episodios ~100-150 y ~200-220). Esto confirma una alta variabilidad: el agente logra mejorar, pero no de forma monótona ni estable.

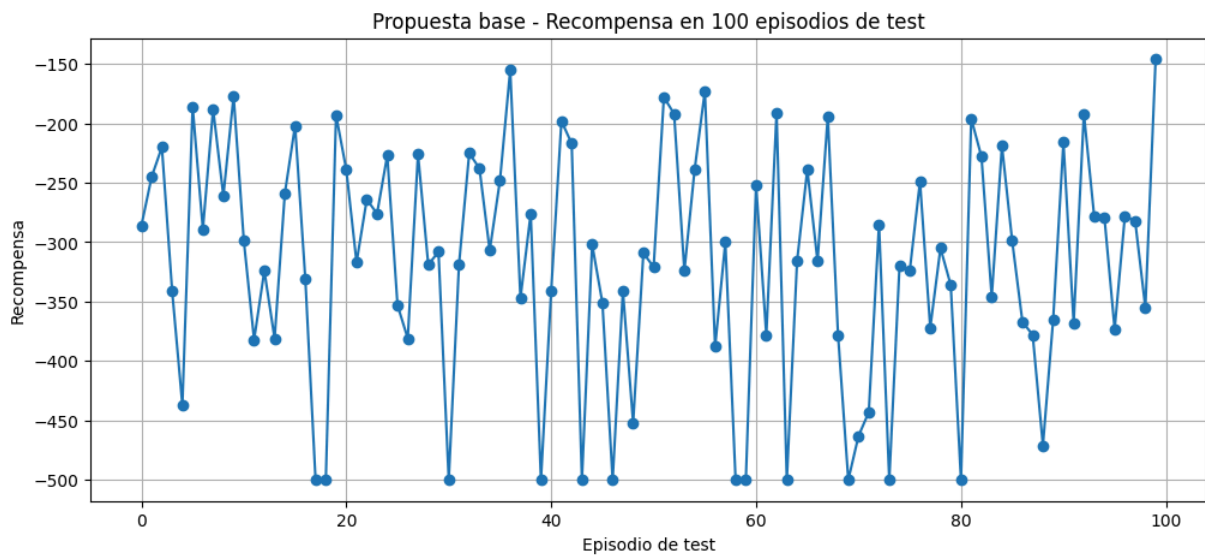
```
In [21]: # 3.3. Gráfica 2 – Media móvil de recompensa
window = 20
moving_avg_reward = pd.Series(episode_reward).rolling(window).mean()

plt.figure(figsize=(12, 5))
plt.plot(moving_avg_reward)
plt.title(f'Propuesta base - Media móvil de recompensa ({window} episodios)')
plt.xlabel('Episodio')
plt.ylabel('Recompensa media')
plt.grid(True)
plt.show()
```



La curva muestra una mejora clara en la primera mitad del entrenamiento, alcanzando su mejor punto (cercano a -190) alrededor del episodio 30-75. Sin embargo, después cae de forma pronunciada hasta tocar de nuevo el fondo de -500 cerca del episodio 100-120, y solo hacia el final del entrenamiento vuelve a recuperarse parcialmente, sin llegar a alcanzar el nivel de su mejor tramo anterior. Es una curva de aprendizaje inestable, con retrocesos importantes.

```
In [22]: plt.figure(figsize=(12, 5))
plt.plot(test_rewards, marker='o')
plt.title('Propuesta base - Recompensa en 100 episodios de test')
plt.xlabel('Episodio de test')
plt.ylabel('Recompensa')
plt.grid(True)
plt.show()
```



La evaluación muestra en su mayoría episodios con recompensas razonablemente buenas (entre -100 y -250), lo que indica que la política aprendida es capaz de resolver el entorno en la mayoría de los casos. Aun así, aparece al menos un episodio con fallo total (-500), lo que confirma que el comportamiento, aunque mayormente funcional, no es completamente fiable.

3. Justificación de los parámetros seleccionados y de los resultados obtenidos

La propuesta base se diseñó como primera aproximación al entorno Acrobot-v1 utilizando el algoritmo DQN, adecuado dado que el espacio de acciones es discreto: el agente debe elegir entre tres acciones relacionadas con el torque aplicado sobre la articulación actuada.

En coherencia con esta elección, el resto de parámetros se definió según las características del problema. Al tratarse de un vector de observación numérico y no de imágenes, se usó una red densa de tres capas ocultas de 64 neuronas (ReLU) y una capa de salida lineal con

tantas neuronas como acciones, encargada de aproximar los valores Q a partir del estado observado. Para estabilizar el entrenamiento, se incorporó una memoria de repetición de 50 000 transiciones con ventana temporal de longitud 1, que reduce la dependencia entre muestras consecutivas y mitiga la correlación entre observaciones sucesivas.

En cuanto a la exploración, se optó por una estrategia epsilon-greedy con valores bajos, priorizando la acción óptima frente a la aleatoria. Esta configuración, sencilla como punto de partida, ayuda a explicar las limitaciones observadas más adelante: en un entorno como Acrobot-v1, donde se requieren movimientos de balanceo coordinados para aprovechar la inercia del sistema, una exploración reducida dificulta que el agente descubra trayectorias útiles de forma consistente.

El entrenamiento se ejecutó durante 80 000 pasos, con callbacks para guardar checkpoints, registrar logs y conservar los mejores pesos, lo que permitió analizar la evolución del agente. Los resultados muestran un aprendizaje real pero inestable: la recompensa media de entrenamiento fue de -357.81, con episodios que llegaron a alcanzar hasta -105, evidencia de que el agente encontró en varias ocasiones secuencias de acciones eficientes. Sin embargo, la persistencia de episodios con recompensa -500 en distintos tramos del entrenamiento —incluso después de haber mostrado mejoras notables— indica que ese aprendizaje no logra consolidarse de forma sostenida. En test, con 100 episodios evaluados, la media final fue de -177.53, lo que confirma que la política aprendida resuelve el entorno en la mayoría de los casos, aunque conserva episodios ocasionales de fallo total, mostrando que el comportamiento aún no es completamente fiable.

En conjunto, estos resultados muestran que la exploración limitada, junto con la dificultad intrínseca del entorno, restringe la capacidad del agente para descubrir y mantener de forma consistente la secuencia de acciones necesaria para resolver la tarea. Aun así, la propuesta base cumple su función: valida correctamente la implementación del entorno, del agente y del proceso de guardado de pesos, y sirve como punto de comparación para las propuestas de mejora posteriores.

4. Propuesta de mejora 1: Exploración progresiva con epsilon decreciente.

A partir de los resultados obtenidos en la propuesta base, se observa que el agente DQN logra un aprendizaje parcial pero inestable en el entorno Acrobot-v1. Aunque la recompensa media de entrenamiento (-357.81) se aleja del valor mínimo posible, y el agente llega a alcanzar episodios de hasta -105, persisten numerosos episodios con recompensa -500 a lo largo de todo el entrenamiento, lo que indica que ese aprendizaje no logra consolidarse de forma sostenida.

Una posible causa de este bajo rendimiento es una exploración insuficiente durante el entrenamiento. En `Acrobot-v1`, el agente debe aprender a coordinar movimientos de

balanceo para aprovechar la inercia del sistema y alcanzar la altura objetivo. Este comportamiento no suele descubrirse fácilmente si el agente explora poco desde el inicio.

Por este motivo, en esta primera propuesta de mejora se modifica la política de exploración del agente. En lugar de utilizar una exploración baja o poco variable, se emplea una política **epsilon-greedy** con reducción progresiva de epsilon. De esta forma, el agente comienza el entrenamiento con alta exploración y, a medida que avanza el aprendizaje, reduce gradualmente la aleatoriedad para explotar las acciones que considera más favorables.

La hipótesis de esta propuesta es que una mayor exploración inicial permitirá al agente descubrir trayectorias más útiles y mejorar la recompensa media obtenida durante el entrenamiento y la fase de test.

```
In [23]: # Entorno
env_name = "Acrobot-v1"
env_p2 = gym.make(env_name)

# Semilla para hacer el experimento algo más reproducible
np.random.seed(123)
env_p2.seed(123)

# Información del entorno
nb_actions = env_p2.action_space.n
obs_shape = env_p2.observation_space.shape

print("Entorno:", env_name)
print("Número de acciones:", nb_actions)
print("Forma de observación:", obs_shape)

# Identificador interno de la propuesta
propuesta_id = "p2"

best_weights_path_p2 = os.path.join(drive_root, f'dqn_{env_name}_{propuesta_id}_best_weights.pkl')
weights_filename_p2 = os.path.join(drive_root, f'dqn_{env_name}_{propuesta_id}_weights.pkl')
checkpoint_weights_filename_p2 = os.path.join(drive_root, f'dqn_{env_name}_{propuesta_id}_checkpoint_weights.pkl')
log_filename_p2 = os.path.join(drive_root, f'dqn_{env_name}_{propuesta_id}_log.json')
```

Entorno: Acrobot-v1

Número de acciones: 3

Forma de observación: (6,)

```
In [24]: model_p2 = Sequential()
model_p2.add(Flatten(input_shape=(1,) + obs_shape))
model_p2.add(Dense(64, activation='relu'))
model_p2.add(Dense(64, activation='relu'))
model_p2.add(Dense(64, activation='relu'))
model_p2.add(Dense(nb_actions, activation='linear'))

model_p2.summary()
```

Model: "sequential_1"

Layer (type)	Output Shape	Param #
=====		
flatten_1 (Flatten)	(None, 6)	0
=====		
dense_4 (Dense)	(None, 64)	448
=====		
dense_5 (Dense)	(None, 64)	4160
=====		
dense_6 (Dense)	(None, 64)	4160
=====		
dense_7 (Dense)	(None, 3)	195
=====		
Total params: 8,963		
Trainable params: 8,963		
Non-trainable params: 0		

```
In [25]: memory_p2 = SequentialMemory(  
        limit=50000,  
        window_length=1  
    )  
  
    policy_p2 = LinearAnnealedPolicy(  
        EpsGreedyQPolicy(),  
        attr='eps',  
        value_max=1.0,  
        value_min=0.05,  
        value_test=0.0,  
        nb_steps=60000  
    )
```

```
In [26]: # Se crea el agente DQN de la propuesta p2.  
        # Esta propuesta mantiene la misma red, memoria y parámetros principales de la base  
        # pero cambia la política de exploración por una estrategia epsilon-greedy decreciente  
  
    dqn_p2 = DQNAgent(model=model_p2, nb_actions=nb_actions, memory=memory_p2, nb_steps  
        policy=policy_p2, enable_double_dqn=False)  
  
    dqn_p2.compile(Adam(learning_rate=1e-3), metrics=['mae'])
```

```
In [27]: search_pattern = os.path.join(  
        drive_root,  
        f'dqn_{env_name}_{propuesta_id}_weights_*.h5f.index'  
    )  
  
    checkpoints = glob.glob(search_pattern)  
  
    if checkpoints:  
        latest_checkpoint_index = sorted(checkpoints, key=os.path.getmtime)[-1]  
        base_weights_path = latest_checkpoint_index.replace(".index", "")  
  
        print("Checkpoint detectado:")  
        print(base_weights_path)
```

```

try:
    dqn_p2.load_weights(base_weights_path)
    print("Pesos cargados correctamente. Reanudando entrenamiento.")
except Exception as e:
    print("Error al cargar pesos:", e)
    print("Se empezará desde cero.")
else:
    print("No se encontraron checkpoints previos para p2. Iniciando desde cero.")

```

No se encontraron checkpoints previos para p2. Iniciando desde cero.

```

In [28]: callbacks_p2 = [
    ModelIntervalCheckpoint(checkpoint_weights_filename_p2, interval=10000),
    FileLogger(log_filename_p2, interval=1000)
]

callbacks_p2 += [SaveBestModel(best_weights_path_p2)]

```

 [Memoria] Primer entrenamiento. Umbral inicial: -500.00

```

In [29]: history_p2 = dqn_p2.fit(
    env_p2,
    callbacks=callbacks_p2,
    nb_steps=80000,
    log_interval=1000,
    visualize=False
)

```

Training for 80000 steps ...

Interval 1 (0 steps performed)

90/1000 [=>.....] - ETA: 0s - reward: -1.0000

C:\Users\kevin\anaconda3\envs\miar_rl\lib\site-packages\tensorflow\python\keras\engine\training.py:2424: UserWarning: `Model.state\_updates` will be removed in a future version. This property should not be used in TensorFlow 2.0, as `updates` are applied automatically.

warnings.warn("`Model.state\_updates` will be removed in a future version. '

1000/1000 [=====] - 1s 767us/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000]

Interval 2 (1000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.025 - mae: 3.10
3 - mean_q: -4.531 - mean_eps: 0.976

Interval 3 (2000 steps performed)
1000/1000 [=====] - 5s 5ms/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.107 - mae: 8.47
2 - mean_q: -12.485 - mean_eps: 0.960

Interval 4 (3000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.313 - mae: 13.2
26 - mean_q: -19.490 - mean_eps: 0.945

Interval 5 (4000 steps performed)
1000/1000 [=====] - 5s 5ms/step - reward: -0.9990
2 episodes - episode_reward: -437.500 [-500.000, -375.000] - loss: 0.522 - mae: 16.9
78 - mean_q: -24.992 - mean_eps: 0.929

Interval 6 (5000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9980
3 episodes - episode_reward: -355.667 [-500.000, -247.000] - loss: 0.620 - mae: 19.5
17 - mean_q: -28.687 - mean_eps: 0.913

Interval 7 (6000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.665 - mae: 21.2
46 - mean_q: -31.188 - mean_eps: 0.897

Interval 8 (7000 steps performed)
1000/1000 [=====] - 5s 5ms/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.643 - mae: 22.5
42 - mean_q: -33.072 - mean_eps: 0.881

Interval 9 (8000 steps performed)
1000/1000 [=====] - 5s 5ms/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.643 - mae: 23.4
81 - mean_q: -34.438 - mean_eps: 0.865

Interval 10 (9000 steps performed)
316/1000 [=====>.....] - ETA: 3s - reward: -1.0000
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -465.80
744/1000 [=====>.....] - ETA: 1s - reward: -0.9987
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -462.20
1000/1000 [=====] - 6s 6ms/step - reward: -0.9980
2 episodes - episode_reward: -401.000 [-428.000, -374.000] - loss: 0.945 - mae: 23.6
55 - mean_q: -34.569 - mean_eps: 0.850

Interval 11 (10000 steps performed)
211/1000 [=====>.....] - ETA: 4s - reward: -1.0000
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -460.45
1000/1000 [=====] - 6s 6ms/step - reward: -0.9990

2 episodes - episode_reward: -482.500 [-500.000, -465.000] - loss: 0.681 - mae: 23.3
66 - mean_q: -34.147 - mean_eps: 0.834

Interval 12 (11000 steps performed)

67/1000 [=>.....] - ETA: 5s - reward: -1.0000
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -453.35
499/1000 [=====>.....] - ETA: 2s - reward: -0.9980
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -449.80
837/1000 [=====>.....] - ETA: 0s - reward: -0.9976
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -441.45
1000/1000 [=====] - 6s 6ms/step - reward: -0.9970
3 episodes - episode_reward: -373.333 [-429.000, -333.000] - loss: 0.745 - mae: 22.9
84 - mean_q: -33.560 - mean_eps: 0.818

Interval 13 (12000 steps performed)

184/1000 [=====>.....] - ETA: 5s - reward: -1.0000
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -433.90
459/1000 [=====>.....] - ETA: 3s - reward: -0.9978
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -422.75
766/1000 [=====>.....] - ETA: 1s - reward: -0.9974
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -412.95
1000/1000 [=====] - 6s 6ms/step - reward: -0.9970
3 episodes - episode_reward: -310.000 [-349.000, -277.000] - loss: 0.659 - mae: 22.9
60 - mean_q: -33.585 - mean_eps: 0.802

Interval 14 (13000 steps performed)

145/1000 [=====>.....] - ETA: 5s - reward: -0.9931
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -412.85
1000/1000 [=====] - 6s 6ms/step - reward: -0.9970
3 episodes - episode_reward: -408.000 [-456.000, -373.000] - loss: 0.705 - mae: 23.1
33 - mean_q: -33.888 - mean_eps: 0.786

Interval 15 (14000 steps performed)

682/1000 [=====>.....] - ETA: 2s - reward: -0.9985
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -411.30
890/1000 [=====>.....] - ETA: 0s - reward: -0.9966
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -396.50
1000/1000 [=====] - 6s 6ms/step - reward: -0.9970
3 episodes - episode_reward: -296.333 [-345.000, -204.000] - loss: 0.765 - mae: 23.5
86 - mean_q: -34.577 - mean_eps: 0.770

Interval 16 (15000 steps performed)

332/1000 [=====>.....] - ETA: 3s - reward: -1.0000
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -393.90
537/1000 [=====>.....] - ETA: 2s - reward: -0.9981
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -378.80
836/1000 [=====>.....] - ETA: 0s - reward: -0.9976
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -369.05
1000/1000 [=====] - 5s 5ms/step - reward: -0.9970
3 episodes - episode_reward: -317.000 [-448.000, -198.000] - loss: 0.795 - mae: 24.2
65 - mean_q: -35.590 - mean_eps: 0.755

Interval 17 (16000 steps performed)

11/1000 [.....] - ETA: 5s - reward: -1.0000
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -352.75
351/1000 [=====>.....] - ETA: 3s - reward: -0.9972

[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -350.85
678/1000 [=====>.....] - ETA: 1s - reward: -0.9971
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -345.75
892/1000 [=====>....] - ETA: 0s - reward: -0.9966
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -333.15
1000/1000 [=====] - 6s 6ms/step - reward: -0.9960
4 episodes - episode_reward: -262.250 [-336.000, -174.000] - loss: 0.971 - mae: 24.7
79 - mean_q: -36.293 - mean_eps: 0.739

Interval 18 (17000 steps performed)

115/1000 [==>.....] - ETA: 4s - reward: -1.0000
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -319.20
349/1000 [=====>.....] - ETA: 3s - reward: -0.9943
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -312.75
1000/1000 [=====] - 6s 6ms/step - reward: -0.9970
3 episodes - episode_reward: -315.000 [-495.000, -221.000] - loss: 0.945 - mae: 25.1
77 - mean_q: -36.868 - mean_eps: 0.723

Interval 19 (18000 steps performed)

71/1000 [=>.....] - ETA: 4s - reward: -1.0000
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -311.00
348/1000 [=====>.....] - ETA: 3s - reward: -0.9971
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -307.20
572/1000 [=====>.....] - ETA: 2s - reward: -0.9965
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -304.70
823/1000 [=====>.....] - ETA: 0s - reward: -0.9964
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -301.80
1000/1000 [=====] - 5s 5ms/step - reward: -0.9960
4 episodes - episode_reward: -244.500 [-273.000, -227.000] - loss: 0.961 - mae: 25.4
27 - mean_q: -37.186 - mean_eps: 0.707

Interval 20 (19000 steps performed)

50/1000 [>.....] - ETA: 5s - reward: -0.9800
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -294.25
412/1000 [=====>.....] - ETA: 3s - reward: -0.9951
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -289.50
641/1000 [=====>.....] - ETA: 2s - reward: -0.9969
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -281.20
821/1000 [=====>.....] - ETA: 1s - reward: -0.9963
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -273.25
1000/1000 [=====] - 6s 6ms/step - reward: -0.9960
4 episodes - episode_reward: -249.500 [-361.000, -186.000] - loss: 1.082 - mae: 25.5
01 - mean_q: -37.250 - mean_eps: 0.691

Interval 21 (20000 steps performed)

128/1000 [==>.....] - ETA: 5s - reward: -1.0000
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -271.30
556/1000 [=====>.....] - ETA: 2s - reward: -0.9964
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -259.95
730/1000 [=====>.....] - ETA: 1s - reward: -0.9959
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -258.80
866/1000 [=====>.....] - ETA: 0s - reward: -0.9954
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -250.30
1000/1000 [=====] - 6s 6ms/step - reward: -0.9950
5 episodes - episode_reward: -207.200 [-301.000, -135.000] - loss: 0.972 - mae: 25.3
17 - mean_q: -36.966 - mean_eps: 0.675

Interval 22 (21000 steps performed)

325/1000 [=====>.....] - ETA: 3s - reward: -0.9969
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -247.75
492/1000 [=====>.....] - ETA: 2s - reward: -0.9959
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -239.55
1000/1000 [=====] - 5s 5ms/step - reward: -0.9960
4 episodes - episode_reward: -222.750 [-270.000, -162.000] - loss: 0.844 - mae: 25.0
81 - mean_q: -36.597 - mean_eps: 0.660

Interval 23 (22000 steps performed)

464/1000 [=====>.....] - ETA: 2s - reward: -0.9957
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -230.45
1000/1000 [=====] - 6s 6ms/step - reward: -0.9960
4 episodes - episode_reward: -266.750 [-361.000, -187.000] - loss: 1.012 - mae: 24.9
54 - mean_q: -36.396 - mean_eps: 0.644

Interval 24 (23000 steps performed)

401/1000 [=====>.....] - ETA: 3s - reward: -0.9950
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -227.85
597/1000 [=====>.....] - ETA: 2s - reward: -0.9950
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -226.50
749/1000 [=====>.....] - ETA: 1s - reward: -0.9947
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -216.30
960/1000 [=====>...] - ETA: 0s - reward: -0.9948
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -215.00
1000/1000 [=====] - 6s 6ms/step - reward: -0.9940
6 episodes - episode_reward: -186.667 [-238.000, -114.000] - loss: 1.036 - mae: 24.6
53 - mean_q: -35.931 - mean_eps: 0.628

Interval 25 (24000 steps performed)

128/1000 [==>.....] - ETA: 4s - reward: -1.0000
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -214.25
296/1000 [=====>.....] - ETA: 3s - reward: -0.9966
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -207.45
438/1000 [=====>.....] - ETA: 3s - reward: -0.9954
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -204.15
1000/1000 [=====] - 6s 6ms/step - reward: -0.9950
5 episodes - episode_reward: -187.000 [-334.000, -127.000] - loss: 0.866 - mae: 24.4
41 - mean_q: -35.648 - mean_eps: 0.612

Interval 26 (25000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9950
5 episodes - episode_reward: -199.200 [-250.000, -139.000] - loss: 0.772 - mae: 24.3
40 - mean_q: -35.506 - mean_eps: 0.596

Interval 27 (26000 steps performed)

70/1000 [=>.....] - ETA: 6s - reward: -1.0000
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -198.40
240/1000 [=====>.....] - ETA: 4s - reward: -0.9958
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -196.85
397/1000 [=====>.....] - ETA: 3s - reward: -0.9950
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -195.30
581/1000 [=====>.....] - ETA: 2s - reward: -0.9948
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -186.65
783/1000 [=====>.....] - ETA: 1s - reward: -0.9949

[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -185.95
958/1000 [=====>...] - ETA: 0s - reward: -0.9948
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -182.90
1000/1000 [=====] - 6s 6ms/step - reward: -0.9940
6 episodes - episode_reward: -176.333 [-199.000, -156.000] - loss: 0.850 - mae: 24.2
64 - mean_q: -35.394 - mean_eps: 0.580

Interval 28 (27000 steps performed)

782/1000 [=====>.....] - ETA: 1s - reward: -0.9949
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -181.75
921/1000 [=====>...] - ETA: 0s - reward: -0.9946
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -180.45
1000/1000 [=====] - 6s 6ms/step - reward: -0.9940
6 episodes - episode_reward: -159.333 [-224.000, -112.000] - loss: 0.711 - mae: 24.0
61 - mean_q: -35.102 - mean_eps: 0.565

Interval 29 (28000 steps performed)

335/1000 [=====>.....] - ETA: 3s - reward: -0.9970
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -177.10
644/1000 [=====>.....] - ETA: 2s - reward: -0.9953
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -173.90
978/1000 [=====>.] - ETA: 0s - reward: -0.9939
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -171.50
1000/1000 [=====] - 6s 6ms/step - reward: -0.9940
6 episodes - episode_reward: -173.833 [-216.000, -130.000] - loss: 0.673 - mae: 23.8
59 - mean_q: -34.808 - mean_eps: 0.549

Interval 30 (29000 steps performed)

119/1000 [==>.....] - ETA: 4s - reward: -1.0000
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -167.05
250/1000 [=====>.....] - ETA: 4s - reward: -0.9960
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -166.50
1000/1000 [=====] - 6s 6ms/step - reward: -0.9950
5 episodes - episode_reward: -179.400 [-243.000, -128.000] - loss: 0.875 - mae: 23.9
69 - mean_q: -34.947 - mean_eps: 0.533

Interval 31 (30000 steps performed)

937/1000 [=====>...] - ETA: 0s - reward: -0.9936
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -162.35
1000/1000 [=====] - 5s 5ms/step - reward: -0.9930
7 episodes - episode_reward: -150.857 [-180.000, -127.000] - loss: 0.748 - mae: 23.8
58 - mean_q: -34.790 - mean_eps: 0.517

Interval 32 (31000 steps performed)

29/1000 [.....] - ETA: 5s - reward: -1.0000
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -161.20
563/1000 [=====>.....] - ETA: 2s - reward: -0.9947
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -160.50
737/1000 [=====>.....] - ETA: 1s - reward: -0.9946
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -160.00
1000/1000 [=====] - 6s 6ms/step - reward: -0.9940
6 episodes - episode_reward: -156.667 [-201.000, -89.000] - loss: 0.689 - mae: 23.92
5 - mean_q: -34.910 - mean_eps: 0.501

Interval 33 (32000 steps performed)

19/1000 [.....] - ETA: 5s - reward: -0.9474

[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -158.10
863/1000 [=====>.....] - ETA: 0s - reward: -0.9942
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -157.05
968/1000 [=====>.] - ETA: 0s - reward: -0.9938
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -153.75
1000/1000 [=====] - 6s 6ms/step - reward: -0.9930
7 episodes - episode_reward: -154.143 [-230.000, -107.000] - loss: 0.741 - mae: 23.8
43 - mean_q: -34.773 - mean_eps: 0.485

Interval 34 (33000 steps performed)

94/1000 [=>.....] - ETA: 5s - reward: -1.0000
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -151.55
221/1000 [=====>.....] - ETA: 4s - reward: -0.9955
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -150.70
499/1000 [=====>.....] - ETA: 2s - reward: -0.9940
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -149.20
617/1000 [=====>.....] - ETA: 2s - reward: -0.9935
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -145.75
856/1000 [=====>.....] - ETA: 0s - reward: -0.9930
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -145.00
1000/1000 [=====] - 6s 6ms/step - reward: -0.9920
8 episodes - episode_reward: -121.500 [-157.000, -91.000] - loss: 0.617 - mae: 23.74
2 - mean_q: -34.638 - mean_eps: 0.470

Interval 35 (34000 steps performed)

85/1000 [=>.....] - ETA: 5s - reward: -1.0000
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -141.75
200/1000 [=====>.....] - ETA: 5s - reward: -0.9950
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -137.95
307/1000 [=====>.....] - ETA: 4s - reward: -0.9935
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -136.45
479/1000 [=====>.....] - ETA: 3s - reward: -0.9937
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -136.30
618/1000 [=====>.....] - ETA: 2s - reward: -0.9935
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -135.90
1000/1000 [=====] - 6s 6ms/step - reward: -0.9930
7 episodes - episode_reward: -141.143 [-185.000, -110.000] - loss: 0.797 - mae: 23.8
61 - mean_q: -34.787 - mean_eps: 0.454

Interval 36 (35000 steps performed)

48/1000 [>.....] - ETA: 5s - reward: -1.0000
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -135.55
353/1000 [=====>.....] - ETA: 3s - reward: -0.9943
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -130.85
1000/1000 [=====] - 6s 6ms/step - reward: -0.9930
7 episodes - episode_reward: -142.571 [-175.000, -105.000] - loss: 0.890 - mae: 23.6
51 - mean_q: -34.430 - mean_eps: 0.438

Interval 37 (36000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9950
5 episodes - episode_reward: -172.800 [-249.000, -129.000] - loss: 0.718 - mae: 23.6
08 - mean_q: -34.424 - mean_eps: 0.422

Interval 38 (37000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9910
9 episodes - episode_reward: -127.222 [-188.000, -91.000] - loss: 0.657 - mae: 23.77

8 - mean_q: -34.675 - mean_eps: 0.406

Interval 39 (38000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9930

7 episodes - episode_reward: -138.000 [-148.000, -121.000] - loss: 0.833 - mae: 23.7

48 - mean_q: -34.617 - mean_eps: 0.390

Interval 40 (39000 steps performed)

597/1000 [=====>.....] - ETA: 2s - reward: -0.9933

[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -128.30

1000/1000 [=====] - 6s 6ms/step - reward: -0.9920

8 episodes - episode_reward: -125.750 [-156.000, -97.000] - loss: 0.861 - mae: 23.74

7 - mean_q: -34.589 - mean_eps: 0.375

Interval 41 (40000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9930

7 episodes - episode_reward: -143.000 [-175.000, -94.000] - loss: 0.707 - mae: 23.68

4 - mean_q: -34.522 - mean_eps: 0.359

Interval 42 (41000 steps performed)

515/1000 [=====>.....] - ETA: 2s - reward: -0.9922

[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -127.65

759/1000 [=====>.....] - ETA: 1s - reward: -0.9921

[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -125.70

860/1000 [=====>.....] - ETA: 0s - reward: -0.9919

[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -123.95

977/1000 [=====>.] - ETA: 0s - reward: -0.9908

[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -123.25

1000/1000 [=====] - 6s 6ms/step - reward: -0.9910

9 episodes - episode_reward: -110.556 [-129.000, -97.000] - loss: 0.740 - mae: 23.63

8 - mean_q: -34.414 - mean_eps: 0.343

Interval 43 (42000 steps performed)

71/1000 [=>.....] - ETA: 5s - reward: -1.0000

[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -122.95

171/1000 [=====>.....] - ETA: 5s - reward: -0.9942

[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -120.05

268/1000 [=====>.....] - ETA: 4s - reward: -0.9888

[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -119.05

515/1000 [=====>.....] - ETA: 3s - reward: -0.9922

[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -119.00

636/1000 [=====>.....] - ETA: 2s - reward: -0.9921

[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -116.15

747/1000 [=====>.....] - ETA: 1s - reward: -0.9920

[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -115.45

1000/1000 [=====] - 6s 6ms/step - reward: -0.9910

9 episodes - episode_reward: -111.667 [-138.000, -94.000] - loss: 0.646 - mae: 23.59

3 - mean_q: -34.374 - mean_eps: 0.327

Interval 44 (43000 steps performed)

77/1000 [=>.....] - ETA: 5s - reward: -1.0000

[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -112.85

920/1000 [=====>....] - ETA: 0s - reward: -0.9924

[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -112.75

1000/1000 [=====] - 6s 6ms/step - reward: -0.9920

8 episodes - episode_reward: -116.250 [-173.000, -86.000] - loss: 0.643 - mae: 23.47

5 - mean_q: -34.173 - mean_eps: 0.311

Interval 45 (44000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9930

7 episodes - episode_reward: -140.714 [-343.000, -84.000] - loss: 0.676 - mae: 23.39

9 - mean_q: -34.052 - mean_eps: 0.295

Interval 46 (45000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9900

10 episodes - episode_reward: -105.300 [-136.000, -88.000] - loss: 0.701 - mae: 23.4

48 - mean_q: -34.153 - mean_eps: 0.280

Interval 47 (46000 steps performed)

484/1000 [=====>.....] - ETA: 2s - reward: -0.9917

[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -104.15

576/1000 [=====>.....] - ETA: 2s - reward: -0.9913

[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -104.05

690/1000 [=====>.....] - ETA: 1s - reward: -0.9913

[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -102.70

1000/1000 [=====] - 6s 6ms/step - reward: -0.9910

9 episodes - episode_reward: -104.667 [-124.000, -86.000] - loss: 0.649 - mae: 23.45

3 - mean_q: -34.158 - mean_eps: 0.264

Interval 48 (47000 steps performed)

562/1000 [=====>.....] - ETA: 2s - reward: -0.9911

[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -102.00

1000/1000 [=====] - 6s 6ms/step - reward: -0.9910

9 episodes - episode_reward: -112.333 [-167.000, -72.000] - loss: 0.690 - mae: 23.39

1 - mean_q: -34.031 - mean_eps: 0.248

Interval 49 (48000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9910

9 episodes - episode_reward: -114.333 [-139.000, -93.000] - loss: 0.732 - mae: 23.32

7 - mean_q: -33.935 - mean_eps: 0.232

Interval 50 (49000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9900

10 episodes - episode_reward: -90.600 [-115.000, -63.000] - loss: 0.698 - mae: 23.27

8 - mean_q: -33.865 - mean_eps: 0.216

Interval 51 (50000 steps performed)

11/1000 [.....] - ETA: 5s - reward: -0.9091

[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -101.80

110/1000 [==>.....] - ETA: 5s - reward: -0.9909

[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -100.20

212/1000 [====>.....] - ETA: 4s - reward: -0.9906

[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -98.20

292/1000 [=====>.....] - ETA: 4s - reward: -0.9897

[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -96.55

1000/1000 [=====] - 6s 6ms/step - reward: -0.9900

10 episodes - episode_reward: -105.300 [-141.000, -79.000] - loss: 0.661 - mae: 22.9

42 - mean_q: -33.320 - mean_eps: 0.200

Interval 52 (51000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9910

9 episodes - episode_reward: -101.667 [-142.000, -76.000] - loss: 0.783 - mae: 22.74

3 - mean_q: -33.033 - mean_eps: 0.185

Interval 53 (52000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9890

11 episodes - episode_reward: -99.182 [-155.000, -67.000] - loss: 0.673 - mae: 22.50

9 - mean_q: -32.679 - mean_eps: 0.169

Interval 54 (53000 steps performed)

534/1000 [=====>.....] - ETA: 2s - reward: -0.9906

[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -95.70

624/1000 [=====>.....] - ETA: 2s - reward: -0.9904

[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -95.60

928/1000 [=====>...] - ETA: 0s - reward: -0.9903

[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -95.30

1000/1000 [=====] - 6s 6ms/step - reward: -0.9900

10 episodes - episode_reward: -92.400 [-106.000, -79.000] - loss: 0.668 - mae: 22.24

9 - mean_q: -32.268 - mean_eps: 0.153

Interval 55 (54000 steps performed)

10/1000 [.....] - ETA: 5s - reward: -1.0000

[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -91.65

92/1000 [=>.....] - ETA: 5s - reward: -0.9891

[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -90.70

175/1000 [====>.....] - ETA: 5s - reward: -0.9886

[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -90.35

350/1000 [=====>.....] - ETA: 3s - reward: -0.9886

[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -90.25

544/1000 [=====>.....] - ETA: 2s - reward: -0.9890

[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -89.70

1000/1000 [=====] - 6s 6ms/step - reward: -0.9890

11 episodes - episode_reward: -91.364 [-135.000, -74.000] - loss: 0.630 - mae: 22.09

6 - mean_q: -32.043 - mean_eps: 0.137

Interval 56 (55000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9890

11 episodes - episode_reward: -93.909 [-138.000, -63.000] - loss: 0.587 - mae: 21.87

5 - mean_q: -31.721 - mean_eps: 0.121

Interval 57 (56000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9900

10 episodes - episode_reward: -94.900 [-129.000, -72.000] - loss: 0.601 - mae: 21.85

0 - mean_q: -31.665 - mean_eps: 0.105

Interval 58 (57000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9890

11 episodes - episode_reward: -94.364 [-125.000, -69.000] - loss: 0.604 - mae: 21.90

5 - mean_q: -31.737 - mean_eps: 0.090

Interval 59 (58000 steps performed)

845/1000 [=====>.....] - ETA: 0s - reward: -0.9893

[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -89.55

920/1000 [=====>...] - ETA: 0s - reward: -0.9880

[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -86.75

1000/1000 [=====] - 6s 6ms/step - reward: -0.9890

11 episodes - episode_reward: -82.636 [-103.000, -63.000] - loss: 0.585 - mae: 21.72

0 - mean_q: -31.445 - mean_eps: 0.074

Interval 60 (59000 steps performed)

539/1000 [=====>.....] - ETA: 2s - reward: -0.9889
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -86.05
600/1000 [=====>.....] - ETA: 2s - reward: -0.9883
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -84.35
778/1000 [=====>.....] - ETA: 1s - reward: -0.9884
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -83.50
1000/1000 [=====] - 6s 6ms/step - reward: -0.9890
11 episodes - episode_reward: -90.636 [-145.000, -62.000] - loss: 0.611 - mae: 21.64
3 - mean_q: -31.336 - mean_eps: 0.058

Interval 61 (60000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9880
12 episodes - episode_reward: -83.083 [-126.000, -63.000] - loss: 0.606 - mae: 21.63
3 - mean_q: -31.308 - mean_eps: 0.050

Interval 62 (61000 steps performed)

825/1000 [=====>.....] - ETA: 1s - reward: -0.9891
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -82.85
961/1000 [=====>..] - ETA: 0s - reward: -0.9886
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -81.40
1000/1000 [=====] - 6s 6ms/step - reward: -0.9880
12 episodes - episode_reward: -85.000 [-154.000, -63.000] - loss: 0.569 - mae: 21.65
6 - mean_q: -31.347 - mean_eps: 0.050

Interval 63 (62000 steps performed)

120/1000 [==>.....] - ETA: 5s - reward: -0.9833
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -80.65
181/1000 [===>.....] - ETA: 5s - reward: -0.9890
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -79.85
254/1000 [=====>.....] - ETA: 4s - reward: -0.9882
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -79.70
651/1000 [=====>.....] - ETA: 2s - reward: -0.9877
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -77.35
1000/1000 [=====] - 6s 6ms/step - reward: -0.9870
13 episodes - episode_reward: -77.077 [-101.000, -62.000] - loss: 0.583 - mae: 21.66
7 - mean_q: -31.337 - mean_eps: 0.050

Interval 64 (63000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9890
11 episodes - episode_reward: -85.636 [-125.000, -68.000] - loss: 0.592 - mae: 21.52
7 - mean_q: -31.119 - mean_eps: 0.050

Interval 65 (64000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9870
13 episodes - episode_reward: -80.231 [-115.000, -64.000] - loss: 0.597 - mae: 21.27
7 - mean_q: -30.737 - mean_eps: 0.050

Interval 66 (65000 steps performed)

270/1000 [=====>.....] - ETA: 4s - reward: -0.9889
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -76.70
1000/1000 [=====] - 6s 6ms/step - reward: -0.9880
12 episodes - episode_reward: -79.917 [-112.000, -62.000] - loss: 0.567 - mae: 21.24
6 - mean_q: -30.704 - mean_eps: 0.050

Interval 67 (66000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9880
12 episodes - episode_reward: -82.417 [-113.000, -63.000] - loss: 0.583 - mae: 21.24
5 - mean_q: -30.684 - mean_eps: 0.050

Interval 68 (67000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9870
13 episodes - episode_reward: -73.769 [-94.000, -64.000] - loss: 0.586 - mae: 21.184
- mean_q: -30.601 - mean_eps: 0.050

Interval 69 (68000 steps performed)
228/1000 [=====>.....] - ETA: 4s - reward: -0.9868
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -76.65
321/1000 [=====>.....] - ETA: 3s - reward: -0.9875
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -75.95
400/1000 [=====>.....] - ETA: 3s - reward: -0.9875
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -75.55
458/1000 [=====>.....] - ETA: 3s - reward: -0.9869
[🏆] ¡Récord superado! Nuevo mejor cerebro guardado con media: -74.00
1000/1000 [=====] - 6s 6ms/step - reward: -0.9870
13 episodes - episode_reward: -79.538 [-102.000, -63.000] - loss: 0.611 - mae: 21.21
0 - mean_q: -30.602 - mean_eps: 0.050

Interval 70 (69000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9880
12 episodes - episode_reward: -78.500 [-110.000, -62.000] - loss: 0.589 - mae: 21.11
4 - mean_q: -30.464 - mean_eps: 0.050

Interval 71 (70000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9890
11 episodes - episode_reward: -82.182 [-106.000, -63.000] - loss: 0.620 - mae: 20.92
4 - mean_q: -30.163 - mean_eps: 0.050

Interval 72 (71000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9880
12 episodes - episode_reward: -91.833 [-263.000, -61.000] - loss: 0.603 - mae: 20.92
4 - mean_q: -30.133 - mean_eps: 0.050

Interval 73 (72000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9880
12 episodes - episode_reward: -78.667 [-97.000, -63.000] - loss: 0.614 - mae: 20.708
- mean_q: -29.808 - mean_eps: 0.050

Interval 74 (73000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9870
13 episodes - episode_reward: -81.154 [-98.000, -63.000] - loss: 0.586 - mae: 20.491
- mean_q: -29.475 - mean_eps: 0.050

Interval 75 (74000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9870
13 episodes - episode_reward: -75.231 [-106.000, -63.000] - loss: 0.594 - mae: 20.36
0 - mean_q: -29.277 - mean_eps: 0.050

Interval 76 (75000 steps performed)
1000/1000 [=====] - 7s 7ms/step - reward: -0.9880
12 episodes - episode_reward: -76.833 [-104.000, -62.000] - loss: 0.610 - mae: 20.29

4 - mean_q: -29.161 - mean_eps: 0.050

Interval 77 (76000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9880

12 episodes - episode_reward: -87.250 [-118.000, -63.000] - loss: 0.597 - mae: 20.26

3 - mean_q: -29.135 - mean_eps: 0.050

Interval 78 (77000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9870

13 episodes - episode_reward: -75.462 [-95.000, -62.000] - loss: 0.599 - mae: 20.273

- mean_q: -29.181 - mean_eps: 0.050

Interval 79 (78000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9880

12 episodes - episode_reward: -82.833 [-120.000, -63.000] - loss: 0.628 - mae: 20.06

9 - mean_q: -28.850 - mean_eps: 0.050

Interval 80 (79000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9880

done, took 460.455 seconds

```
In [30]: dqn_p2.save_weights(weights_filename_p2, overwrite=True)
```

```
print("Pesos finales guardados en:")
print(weights_filename_p2)
```

```
print("Pesos best:")
print(best_weights_path_p2)
```

Pesos finales guardados en:

C:\Users\kevin\Downloads\ESTUDIO MASTER EN IA\Aprendizaje por refuerzo\PROYECTO\Proyecto RL\dqn_Acrobot-v1_p2_weights.h5f

Pesos best:

C:\Users\kevin\Downloads\ESTUDIO MASTER EN IA\Aprendizaje por refuerzo\PROYECTO\Proyecto RL\dqn_Acrobot-v1_p2_best.h5f

```
In [31]: # Evaluación de la propuesta p2 usando los mejores pesos guardados
```

```
dqn_p2.load_weights(best_weights_path_p2)
dqn_p2.test_policy = EpsGreedyQPolicy(eps=0.0)
```

```
env_test_p2 = gym.make(env_name)
```

```
history_test_p2 = dqn_p2.test(
    env_test_p2,
    nb_episodes=100,
    visualize=False,
    verbose=1
)
```

```
env_test_p2.close()
```

```
test_rewards_p2 = history_test_p2.history['episode_reward']
```

```
print("MEDIA TEST PROPUESTA P2:", np.mean(test_rewards_p2))
print("MEJOR EPISODIO:", np.max(test_rewards_p2))
```

```
print("PEOR EPISODIO:", np.min(test_rewards_p2))  
print("DESVIACIÓN ESTÁNDAR:", np.std(test_rewards_p2))
```

Testing for 100 episodes ...

Episode 1:	reward:	-63.000,	steps:	64
Episode 2:	reward:	-63.000,	steps:	64
Episode 3:	reward:	-91.000,	steps:	92
Episode 4:	reward:	-77.000,	steps:	78
Episode 5:	reward:	-70.000,	steps:	71
Episode 6:	reward:	-78.000,	steps:	79
Episode 7:	reward:	-74.000,	steps:	75
Episode 8:	reward:	-88.000,	steps:	89
Episode 9:	reward:	-79.000,	steps:	80
Episode 10:	reward:	-141.000,	steps:	142
Episode 11:	reward:	-63.000,	steps:	64
Episode 12:	reward:	-71.000,	steps:	72
Episode 13:	reward:	-88.000,	steps:	89
Episode 14:	reward:	-71.000,	steps:	72
Episode 15:	reward:	-124.000,	steps:	125
Episode 16:	reward:	-63.000,	steps:	64
Episode 17:	reward:	-70.000,	steps:	71
Episode 18:	reward:	-84.000,	steps:	85
Episode 19:	reward:	-88.000,	steps:	89
Episode 20:	reward:	-71.000,	steps:	72
Episode 21:	reward:	-83.000,	steps:	84
Episode 22:	reward:	-70.000,	steps:	71
Episode 23:	reward:	-70.000,	steps:	71
Episode 24:	reward:	-105.000,	steps:	106
Episode 25:	reward:	-85.000,	steps:	86
Episode 26:	reward:	-131.000,	steps:	132
Episode 27:	reward:	-87.000,	steps:	88
Episode 28:	reward:	-85.000,	steps:	86
Episode 29:	reward:	-79.000,	steps:	80
Episode 30:	reward:	-92.000,	steps:	93
Episode 31:	reward:	-78.000,	steps:	79
Episode 32:	reward:	-61.000,	steps:	62
Episode 33:	reward:	-82.000,	steps:	83
Episode 34:	reward:	-81.000,	steps:	82
Episode 35:	reward:	-106.000,	steps:	107
Episode 36:	reward:	-93.000,	steps:	94
Episode 37:	reward:	-63.000,	steps:	64
Episode 38:	reward:	-74.000,	steps:	75
Episode 39:	reward:	-76.000,	steps:	77
Episode 40:	reward:	-61.000,	steps:	62
Episode 41:	reward:	-77.000,	steps:	78
Episode 42:	reward:	-103.000,	steps:	104
Episode 43:	reward:	-77.000,	steps:	78
Episode 44:	reward:	-71.000,	steps:	72
Episode 45:	reward:	-94.000,	steps:	95
Episode 46:	reward:	-62.000,	steps:	63
Episode 47:	reward:	-69.000,	steps:	70
Episode 48:	reward:	-63.000,	steps:	64
Episode 49:	reward:	-69.000,	steps:	70
Episode 50:	reward:	-72.000,	steps:	73
Episode 51:	reward:	-61.000,	steps:	62
Episode 52:	reward:	-81.000,	steps:	82
Episode 53:	reward:	-61.000,	steps:	62
Episode 54:	reward:	-87.000,	steps:	88
Episode 55:	reward:	-89.000,	steps:	90

```
Episode 56: reward: -87.000, steps: 88
Episode 57: reward: -63.000, steps: 64
Episode 58: reward: -85.000, steps: 86
Episode 59: reward: -71.000, steps: 72
Episode 60: reward: -79.000, steps: 80
Episode 61: reward: -64.000, steps: 65
Episode 62: reward: -78.000, steps: 79
Episode 63: reward: -89.000, steps: 90
Episode 64: reward: -62.000, steps: 63
Episode 65: reward: -61.000, steps: 62
Episode 66: reward: -79.000, steps: 80
Episode 67: reward: -63.000, steps: 64
Episode 68: reward: -71.000, steps: 72
Episode 69: reward: -70.000, steps: 71
Episode 70: reward: -63.000, steps: 64
Episode 71: reward: -70.000, steps: 71
Episode 72: reward: -63.000, steps: 64
Episode 73: reward: -84.000, steps: 85
Episode 74: reward: -87.000, steps: 88
Episode 75: reward: -87.000, steps: 88
Episode 76: reward: -93.000, steps: 94
Episode 77: reward: -90.000, steps: 91
Episode 78: reward: -83.000, steps: 84
Episode 79: reward: -96.000, steps: 97
Episode 80: reward: -82.000, steps: 83
Episode 81: reward: -90.000, steps: 91
Episode 82: reward: -80.000, steps: 81
Episode 83: reward: -71.000, steps: 72
Episode 84: reward: -97.000, steps: 98
Episode 85: reward: -73.000, steps: 74
Episode 86: reward: -84.000, steps: 85
Episode 87: reward: -74.000, steps: 75
Episode 88: reward: -79.000, steps: 80
Episode 89: reward: -167.000, steps: 168
Episode 90: reward: -61.000, steps: 62
Episode 91: reward: -73.000, steps: 74
Episode 92: reward: -79.000, steps: 80
Episode 93: reward: -71.000, steps: 72
Episode 94: reward: -79.000, steps: 80
Episode 95: reward: -83.000, steps: 84
Episode 96: reward: -89.000, steps: 90
Episode 97: reward: -84.000, steps: 85
Episode 98: reward: -84.000, steps: 85
Episode 99: reward: -96.000, steps: 97
Episode 100: reward: -106.000, steps: 107
MEDIA TEST PROPUESTA P2: -80.55
MEJOR EPISODIO: -61.0
PEOR EPISODIO: -167.0
DESVIACIÓN ESTÁNDAR: 16.95781530740325
```

```
In [33]: with open(log_filename_p2, 'r') as f:
          log_p2 = json.load(f)

          episode_reward_p2 = log_p2['episode_reward']
          nb_episode_steps_p2 = log_p2['nb_episode_steps']
          nb_steps_p2 = log_p2['nb_steps']
```

```

print("Episodios registrados:", len(episode_reward_p2))
print("Último paso registrado:", nb_steps_p2[-1])
print("Recompensa media entrenamiento:", np.mean(episode_reward_p2))
print("Mejor recompensa entrenamiento:", np.max(episode_reward_p2))
print("Peor recompensa entrenamiento:", np.min(episode_reward_p2))

```

Episodios registrados: 613
 Último paso registrado: 79996
 Recompensa media entrenamiento: -129.52691680261012
 Mejor recompensa entrenamiento: -61.0
 Peor recompensa entrenamiento: -500.0

```

In [34]: exitos_base = sum(1 for r in test_rewards if r > -500)
total_base = len(test_rewards)

print(f"Tasa de éxito: {exitos_base/total_base*100:.1f}%")
print(f"Mejor episodio: {max(episode_reward_p2)}")
print(f"Peor episodio: {min(episode_reward_p2)}")

```

Tasa de éxito: 88.0%
 Mejor episodio: -61.0
 Peor episodio: -500.0

```

In [35]: video_folder_p2 = os.path.join(drive_root, "videos_proyecto_p2")

env_video_p2 = gym.make(env_name)

env_video_p2 = Monitor(
    env_video_p2,
    video_folder_p2,
    force=True,
    video_callable=lambda episode_id: True
)

dqn_p2.load_weights(best_weights_path_p2)
dqn_p2.test_policy = EpsGreedyQPolicy(eps=0.0)

print("Grabando vídeo de la propuesta p2...")

dqn_p2.test(
    env_video_p2,
    nb_episodes=3,
    visualize=False,
    verbose=1
)

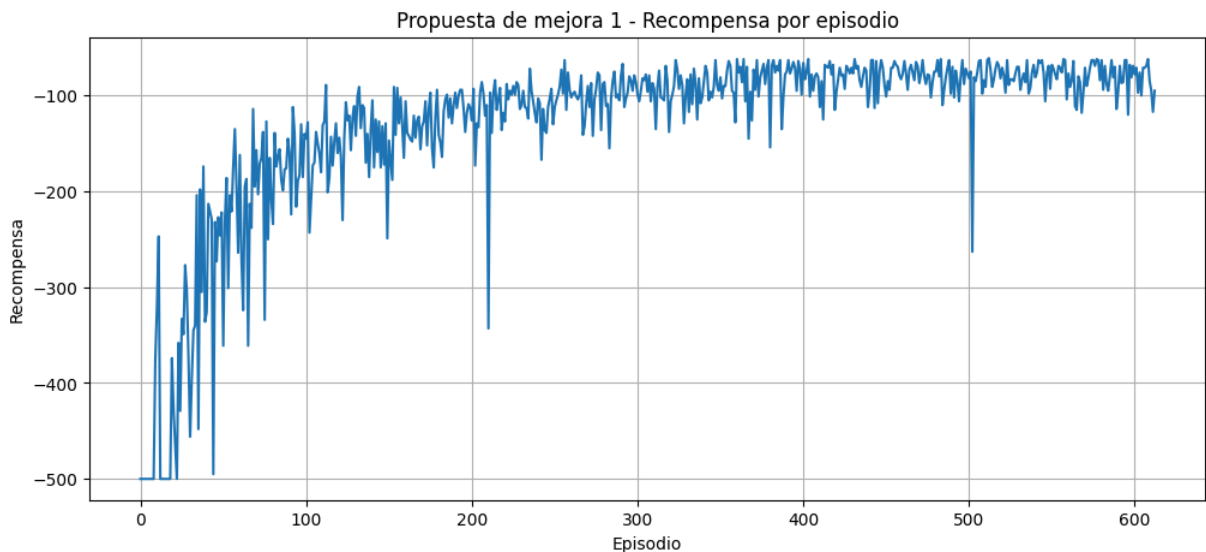
env_video_p2.close()

print("Vídeo guardado en:", video_folder_p2)

```

Grabando vídeo de la propuesta p2...
Testing for 3 episodes ...
Episode 1: reward: -72.000, steps: 73
Episode 2: reward: -81.000, steps: 82
Episode 3: reward: -71.000, steps: 72
Vídeo guardado en: C:\Users\kevin\Downloads\ESTUDIO MASTER EN IA\Aprendizaje por refuerzo\PROYECTO\Proyecto RL\videos_proyecto_p2

```
In [36]: plt.figure(figsize=(12, 5))
plt.plot(episode_reward_p2)
plt.title('Propuesta de mejora 1 - Recompensa por episodio')
plt.xlabel('Episodio')
plt.ylabel('Recompensa')
plt.grid(True)
plt.show()
```

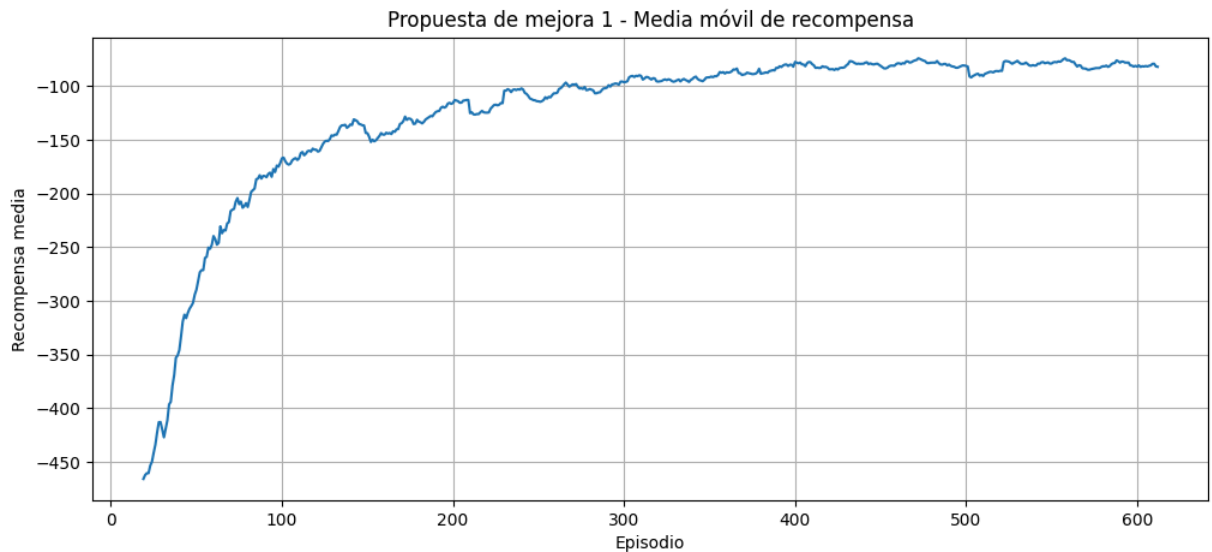


Se muestra una fase de aprendizaje inicial con episodios en -500, seguida de una mejora progresiva que se estabiliza en un rango mucho más favorable (aproximadamente -60 a -150), manteniéndose ahí de forma sostenida durante la mayor parte del entrenamiento, sin los retrocesos abruptos que sí aparecen en la propuesta base.

```
In [37]: window = 20

moving_avg_p2 = np.convolve(
    episode_reward_p2,
    np.ones(window) / window,
    mode='valid'
)

plt.figure(figsize=(12, 5))
plt.plot(range(window - 1, len(episode_reward_p2)), moving_avg_p2)
plt.title('Propuesta de mejora 1 - Media móvil de recompensa')
plt.xlabel('Episodio')
plt.ylabel('Recompensa media')
plt.grid(True)
plt.show()
```



Presenta una curva de aprendizaje bien definida: una subida marcada al inicio y una estabilización posterior en torno a -100, sin caídas significativas hacia el final del entrenamiento.

```
In [38]: plt.figure(figsize=(12, 5))
plt.plot(test_rewards_p2, marker='o')
plt.title('Propuesta de mejora 1 - Recompensa en 100 episodios de test')
plt.xlabel('Episodio de test')
plt.ylabel('Recompensa')
plt.grid(True)
plt.show()
```



Los resultados de evaluación refuerzan la solidez del entrenamiento: La política se comporta de forma consistente y sin fallos: ningún episodio termina en -500, y el rango de recompensas es estrecho y predecible.

```
In [39]: base_log_path = os.path.join(drive_root, f'dqn_{env_name}_p1_log.json')
p2_log_path = log_filename_p2
```



```

with open(base_log_path, 'r') as f:
    log_base = json.load(f)

with open(p2_log_path, 'r') as f:
    log_p2 = json.load(f)

rewards_base = log_base['episode_reward']
rewards_p2 = log_p2['episode_reward']

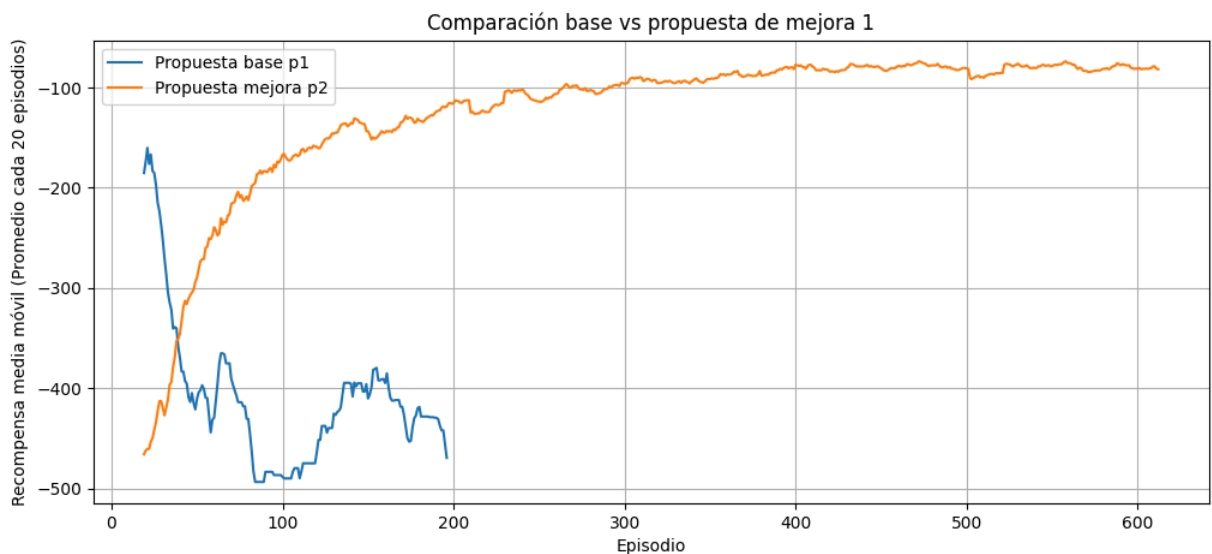
window = 20

media_base = np.convolve(
    rewards_base,
    np.ones(window) / window,
    mode='valid'
)

media_p2 = np.convolve(
    rewards_p2,
    np.ones(window) / window,
    mode='valid'
)

plt.figure(figsize=(12, 5))
plt.plot(range(window - 1, len(rewards_base)), media_base, label='Propuesta base p1')
plt.plot(range(window - 1, len(rewards_p2)), media_p2, label='Propuesta mejora p2')
plt.title('Comparación base vs propuesta de mejora 1')
plt.xlabel('Episodio')
plt.ylabel('Recompensa media móvil (Promedio cada 20 episodios)')
plt.legend()
plt.grid(True)
plt.show()

```



La gráfica comparativa (media móvil de ambas propuestas) muestra trayectorias claramente distintas: la propuesta base mejora al inicio, pero sufre una caída pronunciada de vuelta hacia -500 en la parte media del entrenamiento, y solo recupera parcialmente el nivel alcanzado antes de terminar el log. La propuesta de mejora 1, en cambio, converge de forma

mucho más rápida y sostenida hacia valores cercanos a -100, sin mostrar retrocesos comparables. Esto se traduce directamente en los resultados de test: -177.53 en la base frente a -75.32 en la mejora, con una diferencia adicional en consistencia (la base aún registra fallos ocasionales en test, mientras que la mejora no registra ninguno).

4.1 Conclusión.

Los resultados confirman que la propuesta base logra un aprendizaje real pero inestable: mejora su desempeño respecto al fallo total, pero sufre retrocesos significativos durante el entrenamiento y conserva episodios de fallo tanto en entrenamiento como en test. La propuesta de mejora 1, basada en una exploración progresiva con epsilon decreciente, no solo obtiene una recompensa media muy superior, sino que logra una convergencia mucho más estable y una política de test completamente libre de fallos. Esto indica que la principal limitación de la propuesta base —una exploración insuficiente e inconsistente a lo largo del entrenamiento— fue efectivamente resuelta por la propuesta de mejora, que logra un comportamiento no solo mejor en promedio, sino considerablemente más fiable y repetible.

```
In [40]: # Bloque 7: DDQN - mismo modelo pero con Double DQN habilitado

# Reconstruimos el modelo desde cero para no reusar pesos ya entrenados
model_ddqn = Sequential()
model_ddqn.add(Flatten(input_shape=(1,) + env.observation_space.shape))
model_ddqn.add(Dense(64))
model_ddqn.add(Activation('relu'))
model_ddqn.add(Dense(64))
model_ddqn.add(Activation('relu'))
model_ddqn.add(Dense(nb_actions))
model_ddqn.add(Activation('linear'))

memory_ddqn = SequentialMemory(limit=50000, window_length=1)

policy_ddqn = LinearAnnealedPolicy(
    EpsGreedyQPolicy(),
    attr='eps',
    value_max=1.0,
    value_min=0.1,
    value_test=0.05,
    nb_steps=50000
)

ddqn = DQNAgent(
    model=model_ddqn,
    nb_actions=nb_actions,
    memory=memory_ddqn,
    nb_steps_warmup=1000,
    target_model_update=1e-2,
    policy=policy_ddqn,
    gamma=0.99,
    enable_double_dqn=True # <- única diferencia real vs el DQN base
)
```

```
ddqn.compile(Adam(learning_rate=1e-3), metrics=['mae'])

print(ddqn.model.summary())
```

Model: "sequential_2"

Layer (type)	Output Shape	Param #
=====		
flatten_2 (Flatten)	(None, 6)	0
dense_8 (Dense)	(None, 64)	448
activation (Activation)	(None, 64)	0
dense_9 (Dense)	(None, 64)	4160
activation_1 (Activation)	(None, 64)	0
dense_10 (Dense)	(None, 3)	195
activation_2 (Activation)	(None, 3)	0
=====		
Total params: 4,803		
Trainable params: 4,803		
Non-trainable params: 0		
None		

```
In [41]: # Entrenamiento del DDQN

env_name_ddqn = 'Acrobot-v1-DDQN' # sufijo distinto solo para nombrar archivos

weights_filename_ddqn = 'ddqn_{}_weights.h5f'.format(env_name)
checkpoint_weights_filename_ddqn = 'ddqn_' + env_name + '_weights_{step}.h5f'
log_filename_ddqn = 'ddqn_{}_log.json'.format(env_name)

callbacks_ddqn = [ModelIntervalCheckpoint(checkpoint_weights_filename_ddqn, interval=1000)]
callbacks_ddqn += [FileLogger(log_filename_ddqn, interval=100)]

history_ddqn = ddqn.fit(
    env,
    callbacks=callbacks_ddqn,
    nb_steps=80000,
    log_interval=1000,
    visualize=False
)

ddqn.save_weights(weights_filename_ddqn, overwrite=True)
```

Training for 80000 steps ...

Interval 1 (0 steps performed)

142/1000 [==>.....] - ETA: 0s - reward: -1.0000

```
C:\Users\kevin\anaconda3\envs\miar_rl\lib\site-packages\tensorflow\python\keras\engine\training.py:2424: UserWarning: `Model.state_updates` will be removed in a future version. This property should not be used in TensorFlow 2.0, as `updates` are applied automatically.  
  warnings.warn("`Model.state_updates` will be removed in a future version. ")
```

1000/1000 [=====] - 1s 705us/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000]

Interval 2 (1000 steps performed)

1000/1000 [=====] - 7s 7ms/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.028 - mae: 3.31
3 - mean_q: -4.835 - mean_eps: 0.973

Interval 3 (2000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.112 - mae: 8.77
3 - mean_q: -12.954 - mean_eps: 0.955

Interval 4 (3000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.303 - mae: 13.5
48 - mean_q: -19.997 - mean_eps: 0.937

Interval 5 (4000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.451 - mae: 17.4
33 - mean_q: -25.742 - mean_eps: 0.919

Interval 6 (5000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9990
2 episodes - episode_reward: -470.000 [-500.000, -440.000] - loss: 0.756 - mae: 20.1
51 - mean_q: -29.681 - mean_eps: 0.901

Interval 7 (6000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.756 - mae: 21.9
80 - mean_q: -32.355 - mean_eps: 0.883

Interval 8 (7000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9990
2 episodes - episode_reward: -415.500 [-500.000, -331.000] - loss: 0.949 - mae: 23.6
24 - mean_q: -34.761 - mean_eps: 0.865

Interval 9 (8000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.958 - mae: 24.7
81 - mean_q: -36.457 - mean_eps: 0.847

Interval 10 (9000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9980
3 episodes - episode_reward: -408.000 [-500.000, -344.000] - loss: 0.936 - mae: 25.8
78 - mean_q: -38.096 - mean_eps: 0.829

Interval 11 (10000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9990
2 episodes - episode_reward: -350.500 [-500.000, -201.000] - loss: 1.204 - mae: 26.8
52 - mean_q: -39.554 - mean_eps: 0.811

Interval 12 (11000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9970
3 episodes - episode_reward: -429.000 [-482.000, -394.000] - loss: 1.260 - mae: 27.8

71 - mean_q: -41.040 - mean_eps: 0.793

Interval 13 (12000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -1.0000

2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 1.331 - mae: 28.4

99 - mean_q: -41.955 - mean_eps: 0.775

Interval 14 (13000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9990

2 episodes - episode_reward: -395.000 [-500.000, -290.000] - loss: 1.708 - mae: 29.1

52 - mean_q: -42.911 - mean_eps: 0.757

Interval 15 (14000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9970

3 episodes - episode_reward: -298.333 [-363.000, -240.000] - loss: 1.096 - mae: 29.7

72 - mean_q: -43.865 - mean_eps: 0.739

Interval 16 (15000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9990

2 episodes - episode_reward: -491.000 [-500.000, -482.000] - loss: 1.444 - mae: 30.2

57 - mean_q: -44.524 - mean_eps: 0.721

Interval 17 (16000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9960

4 episodes - episode_reward: -305.000 [-421.000, -220.000] - loss: 1.604 - mae: 30.7

66 - mean_q: -45.254 - mean_eps: 0.703

Interval 18 (17000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9970

3 episodes - episode_reward: -320.000 [-358.000, -286.000] - loss: 1.880 - mae: 31.1

85 - mean_q: -45.839 - mean_eps: 0.685

Interval 19 (18000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9960

4 episodes - episode_reward: -238.250 [-316.000, -186.000] - loss: 1.457 - mae: 31.3

24 - mean_q: -46.017 - mean_eps: 0.667

Interval 20 (19000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9980

3 episodes - episode_reward: -350.667 [-500.000, -266.000] - loss: 1.508 - mae: 31.4

40 - mean_q: -46.149 - mean_eps: 0.649

Interval 21 (20000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9970

4 episodes - episode_reward: -272.000 [-500.000, -172.000] - loss: 2.057 - mae: 31.1

55 - mean_q: -45.318 - mean_eps: 0.631

Interval 22 (21000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9960

4 episodes - episode_reward: -205.500 [-259.000, -164.000] - loss: 1.934 - mae: 30.5

34 - mean_q: -44.228 - mean_eps: 0.613

Interval 23 (22000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9940

6 episodes - episode_reward: -199.500 [-263.000, -144.000] - loss: 1.183 - mae: 29.8

87 - mean_q: -43.388 - mean_eps: 0.595

Interval 24 (23000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9950
5 episodes - episode_reward: -195.200 [-273.000, -134.000] - loss: 1.338 - mae: 28.8
97 - mean_q: -41.851 - mean_eps: 0.577

Interval 25 (24000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9950
5 episodes - episode_reward: -183.800 [-236.000, -101.000] - loss: 1.099 - mae: 28.1
48 - mean_q: -40.836 - mean_eps: 0.559

Interval 26 (25000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9940
6 episodes - episode_reward: -172.000 [-207.000, -140.000] - loss: 1.285 - mae: 27.2
84 - mean_q: -39.607 - mean_eps: 0.541

Interval 27 (26000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9950
5 episodes - episode_reward: -184.800 [-234.000, -116.000] - loss: 0.924 - mae: 26.5
74 - mean_q: -38.651 - mean_eps: 0.523

Interval 28 (27000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9930
7 episodes - episode_reward: -147.286 [-218.000, -112.000] - loss: 0.990 - mae: 25.7
41 - mean_q: -37.459 - mean_eps: 0.505

Interval 29 (28000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9930
7 episodes - episode_reward: -154.571 [-194.000, -114.000] - loss: 1.016 - mae: 25.2
90 - mean_q: -36.816 - mean_eps: 0.487

Interval 30 (29000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9930
7 episodes - episode_reward: -137.714 [-156.000, -120.000] - loss: 0.894 - mae: 24.7
48 - mean_q: -36.019 - mean_eps: 0.469

Interval 31 (30000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9930
7 episodes - episode_reward: -147.571 [-189.000, -117.000] - loss: 0.842 - mae: 24.4
04 - mean_q: -35.531 - mean_eps: 0.451

Interval 32 (31000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9970
4 episodes - episode_reward: -214.750 [-500.000, -112.000] - loss: 1.053 - mae: 24.0
47 - mean_q: -34.985 - mean_eps: 0.433

Interval 33 (32000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9920
8 episodes - episode_reward: -132.875 [-159.000, -99.000] - loss: 0.751 - mae: 23.94
7 - mean_q: -34.921 - mean_eps: 0.415

Interval 34 (33000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9920
8 episodes - episode_reward: -128.000 [-194.000, -98.000] - loss: 0.811 - mae: 24.12
7 - mean_q: -35.180 - mean_eps: 0.397

Interval 35 (34000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9920
8 episodes - episode_reward: -119.000 [-166.000, -84.000] - loss: 0.901 - mae: 24.27
4 - mean_q: -35.402 - mean_eps: 0.379

Interval 36 (35000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9920
8 episodes - episode_reward: -132.375 [-203.000, -109.000] - loss: 1.037 - mae: 24.3
51 - mean_q: -35.511 - mean_eps: 0.361

Interval 37 (36000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9930
7 episodes - episode_reward: -125.857 [-164.000, -100.000] - loss: 0.911 - mae: 24.4
81 - mean_q: -35.687 - mean_eps: 0.343

Interval 38 (37000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9920
8 episodes - episode_reward: -134.000 [-172.000, -108.000] - loss: 0.935 - mae: 24.5
55 - mean_q: -35.789 - mean_eps: 0.325

Interval 39 (38000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9910
9 episodes - episode_reward: -114.000 [-128.000, -106.000] - loss: 0.981 - mae: 24.6
54 - mean_q: -35.910 - mean_eps: 0.307

Interval 40 (39000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9920
8 episodes - episode_reward: -115.625 [-182.000, -88.000] - loss: 1.039 - mae: 24.48
0 - mean_q: -35.625 - mean_eps: 0.289

Interval 41 (40000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9900
10 episodes - episode_reward: -107.100 [-128.000, -75.000] - loss: 0.881 - mae: 24.5
47 - mean_q: -35.713 - mean_eps: 0.271

Interval 42 (41000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9910
9 episodes - episode_reward: -97.222 [-121.000, -82.000] - loss: 0.944 - mae: 24.355
- mean_q: -35.417 - mean_eps: 0.253

Interval 43 (42000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9920
8 episodes - episode_reward: -128.375 [-220.000, -68.000] - loss: 0.770 - mae: 24.27
6 - mean_q: -35.301 - mean_eps: 0.235

Interval 44 (43000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9900
10 episodes - episode_reward: -103.500 [-135.000, -85.000] - loss: 0.891 - mae: 24.2
72 - mean_q: -35.299 - mean_eps: 0.217

Interval 45 (44000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9900
10 episodes - episode_reward: -101.100 [-135.000, -78.000] - loss: 1.064 - mae: 24.0
82 - mean_q: -34.936 - mean_eps: 0.199

Interval 46 (45000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9910
9 episodes - episode_reward: -100.778 [-134.000, -86.000] - loss: 1.039 - mae: 23.80
9 - mean_q: -34.563 - mean_eps: 0.181

Interval 47 (46000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9910
9 episodes - episode_reward: -117.778 [-267.000, -63.000] - loss: 0.758 - mae: 23.67
7 - mean_q: -34.410 - mean_eps: 0.163

Interval 48 (47000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9890
11 episodes - episode_reward: -92.364 [-137.000, -69.000] - loss: 0.875 - mae: 23.74
7 - mean_q: -34.513 - mean_eps: 0.145

Interval 49 (48000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9900
10 episodes - episode_reward: -91.800 [-117.000, -68.000] - loss: 0.886 - mae: 23.75
2 - mean_q: -34.514 - mean_eps: 0.127

Interval 50 (49000 steps performed)
1000/1000 [=====] - 9s 9ms/step - reward: -0.9900
10 episodes - episode_reward: -99.100 [-124.000, -81.000] - loss: 0.863 - mae: 23.70
9 - mean_q: -34.450 - mean_eps: 0.109

Interval 51 (50000 steps performed)
1000/1000 [=====] - 8s 8ms/step - reward: -0.9890
11 episodes - episode_reward: -95.273 [-115.000, -71.000] - loss: 0.987 - mae: 23.50
6 - mean_q: -34.110 - mean_eps: 0.100

Interval 52 (51000 steps performed)
1000/1000 [=====] - 8s 8ms/step - reward: -0.9900
10 episodes - episode_reward: -94.400 [-110.000, -78.000] - loss: 0.810 - mae: 23.30
7 - mean_q: -33.824 - mean_eps: 0.100

Interval 53 (52000 steps performed)
1000/1000 [=====] - 8s 8ms/step - reward: -0.9890
11 episodes - episode_reward: -90.364 [-126.000, -75.000] - loss: 0.758 - mae: 23.24
8 - mean_q: -33.752 - mean_eps: 0.100

Interval 54 (53000 steps performed)
1000/1000 [=====] - 8s 8ms/step - reward: -0.9890
11 episodes - episode_reward: -94.909 [-134.000, -61.000] - loss: 0.769 - mae: 23.13
5 - mean_q: -33.577 - mean_eps: 0.100

Interval 55 (54000 steps performed)
1000/1000 [=====] - 8s 8ms/step - reward: -0.9900
10 episodes - episode_reward: -90.500 [-114.000, -65.000] - loss: 0.859 - mae: 23.07
0 - mean_q: -33.469 - mean_eps: 0.100

Interval 56 (55000 steps performed)
1000/1000 [=====] - 7s 7ms/step - reward: -0.9890
11 episodes - episode_reward: -87.364 [-107.000, -73.000] - loss: 0.825 - mae: 23.14
5 - mean_q: -33.578 - mean_eps: 0.100

Interval 57 (56000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9890

11 episodes - episode_reward: -93.455 [-156.000, -61.000] - loss: 0.851 - mae: 23.08
8 - mean_q: -33.476 - mean_eps: 0.100

Interval 58 (57000 steps performed)

1000/1000 [=====] - 7s 7ms/step - reward: -0.9890
11 episodes - episode_reward: -88.182 [-101.000, -69.000] - loss: 0.851 - mae: 23.00
5 - mean_q: -33.295 - mean_eps: 0.100

Interval 59 (58000 steps performed)

1000/1000 [=====] - 7s 7ms/step - reward: -0.9890
11 episodes - episode_reward: -90.455 [-101.000, -68.000] - loss: 0.788 - mae: 22.93
8 - mean_q: -33.198 - mean_eps: 0.100

Interval 60 (59000 steps performed)

1000/1000 [=====] - 7s 7ms/step - reward: -0.9890
11 episodes - episode_reward: -97.818 [-119.000, -80.000] - loss: 0.786 - mae: 23.03
1 - mean_q: -33.313 - mean_eps: 0.100

Interval 61 (60000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9890
11 episodes - episode_reward: -87.818 [-114.000, -63.000] - loss: 0.837 - mae: 22.79
7 - mean_q: -32.964 - mean_eps: 0.100

Interval 62 (61000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9880
12 episodes - episode_reward: -83.750 [-97.000, -68.000] - loss: 0.786 - mae: 22.780
- mean_q: -32.910 - mean_eps: 0.100

Interval 63 (62000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9900
10 episodes - episode_reward: -96.800 [-117.000, -75.000] - loss: 0.796 - mae: 22.72
6 - mean_q: -32.811 - mean_eps: 0.100

Interval 64 (63000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9900
10 episodes - episode_reward: -96.700 [-121.000, -76.000] - loss: 0.736 - mae: 22.40
8 - mean_q: -32.374 - mean_eps: 0.100

Interval 65 (64000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9920
8 episodes - episode_reward: -87.875 [-112.000, -68.000] - loss: 0.758 - mae: 22.514
- mean_q: -32.543 - mean_eps: 0.100

Interval 66 (65000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9910
10 episodes - episode_reward: -125.800 [-500.000, -76.000] - loss: 0.743 - mae: 22.3
33 - mean_q: -32.263 - mean_eps: 0.100

Interval 67 (66000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9910
9 episodes - episode_reward: -118.333 [-328.000, -76.000] - loss: 0.769 - mae: 22.27
4 - mean_q: -32.192 - mean_eps: 0.100

Interval 68 (67000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9900
10 episodes - episode_reward: -94.400 [-113.000, -78.000] - loss: 0.803 - mae: 22.21

1 - mean_q: -32.085 - mean_eps: 0.100

Interval 69 (68000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9890

11 episodes - episode_reward: -87.364 [-100.000, -73.000] - loss: 0.754 - mae: 22.35

6 - mean_q: -32.338 - mean_eps: 0.100

Interval 70 (69000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9900

10 episodes - episode_reward: -101.500 [-261.000, -72.000] - loss: 0.764 - mae: 22.4

24 - mean_q: -32.435 - mean_eps: 0.100

Interval 71 (70000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9900

10 episodes - episode_reward: -102.000 [-137.000, -81.000] - loss: 0.832 - mae: 22.5

26 - mean_q: -32.573 - mean_eps: 0.100

Interval 72 (71000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9890

11 episodes - episode_reward: -91.364 [-124.000, -76.000] - loss: 0.849 - mae: 22.47

4 - mean_q: -32.492 - mean_eps: 0.100

Interval 73 (72000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9910

9 episodes - episode_reward: -104.333 [-218.000, -63.000] - loss: 0.909 - mae: 22.66

7 - mean_q: -32.795 - mean_eps: 0.100

Interval 74 (73000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9890

11 episodes - episode_reward: -84.273 [-98.000, -63.000] - loss: 0.831 - mae: 22.785

- mean_q: -32.914 - mean_eps: 0.100

Interval 75 (74000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9890

11 episodes - episode_reward: -97.000 [-131.000, -82.000] - loss: 0.885 - mae: 22.61

1 - mean_q: -32.684 - mean_eps: 0.100

Interval 76 (75000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9900

10 episodes - episode_reward: -96.500 [-125.000, -69.000] - loss: 0.881 - mae: 22.63

5 - mean_q: -32.701 - mean_eps: 0.100

Interval 77 (76000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9900

10 episodes - episode_reward: -97.200 [-112.000, -68.000] - loss: 0.869 - mae: 22.66

5 - mean_q: -32.731 - mean_eps: 0.100

Interval 78 (77000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9900

10 episodes - episode_reward: -99.000 [-139.000, -81.000] - loss: 0.866 - mae: 22.61

7 - mean_q: -32.676 - mean_eps: 0.100

Interval 79 (78000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9890

11 episodes - episode_reward: -94.182 [-143.000, -71.000] - loss: 0.870 - mae: 22.52

7 - mean_q: -32.561 - mean_eps: 0.100

Interval 80 (79000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9900
done, took 502.177 seconds

```
In [42]: # Evaluación del DDQN
ddqn.load_weights('ddqn_{}_weights.h5f'.format(env_name))

resultados_ddqn_raw = ddqn.test(env, nb_episodes=100, visualize=False)

rewards_ddqn = resultados_ddqn_raw.history['episode_reward']
 exitos_ddqn = sum(1 for r in rewards_ddqn if r > -500)
total_ddqn = len(rewards_ddqn)

print(f"\nEpisodios de test: {total_ddqn}")
print(f"Episodios exitosos: {exitos_ddqn}")
print(f"Tasa de éxito: {exitos_ddqn/total_ddqn*100:.1f}%")
print(f"Recompensa promedio: {sum(rewards_ddqn)/total_ddqn:.2f}")
print(f"Mejor episodio: {max(rewards_ddqn)}")
print(f"Peor episodio: {min(rewards_ddqn)}")

resultados_ddqn = {
    'nombre': 'DDQN',
    'rewards': rewards_ddqn,
    'tasa_exito': exitos_ddqn/total_ddqn*100,
    'promedio': sum(rewards_ddqn)/total_ddqn,
    'mejor': max(rewards_ddqn),
    'peor': min(rewards_ddqn)
}

print(resultados_ddqn)
```

Testing for 100 episodes ...

Episode 1:	reward:	-79.000,	steps:	80
Episode 2:	reward:	-109.000,	steps:	110
Episode 3:	reward:	-95.000,	steps:	96
Episode 4:	reward:	-84.000,	steps:	85
Episode 5:	reward:	-78.000,	steps:	79
Episode 6:	reward:	-71.000,	steps:	72
Episode 7:	reward:	-80.000,	steps:	81
Episode 8:	reward:	-87.000,	steps:	88
Episode 9:	reward:	-70.000,	steps:	71
Episode 10:	reward:	-62.000,	steps:	63
Episode 11:	reward:	-70.000,	steps:	71
Episode 12:	reward:	-81.000,	steps:	82
Episode 13:	reward:	-81.000,	steps:	82
Episode 14:	reward:	-70.000,	steps:	71
Episode 15:	reward:	-71.000,	steps:	72
Episode 16:	reward:	-83.000,	steps:	84
Episode 17:	reward:	-70.000,	steps:	71
Episode 18:	reward:	-79.000,	steps:	80
Episode 19:	reward:	-62.000,	steps:	63
Episode 20:	reward:	-81.000,	steps:	82
Episode 21:	reward:	-92.000,	steps:	93
Episode 22:	reward:	-78.000,	steps:	79
Episode 23:	reward:	-86.000,	steps:	87
Episode 24:	reward:	-62.000,	steps:	63
Episode 25:	reward:	-69.000,	steps:	70
Episode 26:	reward:	-70.000,	steps:	71
Episode 27:	reward:	-88.000,	steps:	89
Episode 28:	reward:	-87.000,	steps:	88
Episode 29:	reward:	-70.000,	steps:	71
Episode 30:	reward:	-103.000,	steps:	104
Episode 31:	reward:	-78.000,	steps:	79
Episode 32:	reward:	-73.000,	steps:	74
Episode 33:	reward:	-73.000,	steps:	74
Episode 34:	reward:	-86.000,	steps:	87
Episode 35:	reward:	-71.000,	steps:	72
Episode 36:	reward:	-70.000,	steps:	71
Episode 37:	reward:	-89.000,	steps:	90
Episode 38:	reward:	-94.000,	steps:	95
Episode 39:	reward:	-63.000,	steps:	64
Episode 40:	reward:	-87.000,	steps:	88
Episode 41:	reward:	-84.000,	steps:	85
Episode 42:	reward:	-91.000,	steps:	92
Episode 43:	reward:	-77.000,	steps:	78
Episode 44:	reward:	-121.000,	steps:	122
Episode 45:	reward:	-70.000,	steps:	71
Episode 46:	reward:	-62.000,	steps:	63
Episode 47:	reward:	-81.000,	steps:	82
Episode 48:	reward:	-70.000,	steps:	71
Episode 49:	reward:	-78.000,	steps:	79
Episode 50:	reward:	-108.000,	steps:	109
Episode 51:	reward:	-149.000,	steps:	150
Episode 52:	reward:	-88.000,	steps:	89
Episode 53:	reward:	-62.000,	steps:	63
Episode 54:	reward:	-154.000,	steps:	155
Episode 55:	reward:	-70.000,	steps:	71

Episode 56: reward: -74.000, steps: 75
Episode 57: reward: -113.000, steps: 114
Episode 58: reward: -62.000, steps: 63
Episode 59: reward: -87.000, steps: 88
Episode 60: reward: -70.000, steps: 71
Episode 61: reward: -63.000, steps: 64
Episode 62: reward: -71.000, steps: 72
Episode 63: reward: -92.000, steps: 93
Episode 64: reward: -88.000, steps: 89
Episode 65: reward: -70.000, steps: 71
Episode 66: reward: -81.000, steps: 82
Episode 67: reward: -81.000, steps: 82
Episode 68: reward: -99.000, steps: 100
Episode 69: reward: -70.000, steps: 71
Episode 70: reward: -87.000, steps: 88
Episode 71: reward: -87.000, steps: 88
Episode 72: reward: -76.000, steps: 77
Episode 73: reward: -85.000, steps: 86
Episode 74: reward: -97.000, steps: 98
Episode 75: reward: -94.000, steps: 95
Episode 76: reward: -80.000, steps: 81
Episode 77: reward: -127.000, steps: 128
Episode 78: reward: -71.000, steps: 72
Episode 79: reward: -78.000, steps: 79
Episode 80: reward: -92.000, steps: 93
Episode 81: reward: -85.000, steps: 86
Episode 82: reward: -103.000, steps: 104
Episode 83: reward: -62.000, steps: 63
Episode 84: reward: -88.000, steps: 89
Episode 85: reward: -70.000, steps: 71
Episode 86: reward: -109.000, steps: 110
Episode 87: reward: -81.000, steps: 82
Episode 88: reward: -88.000, steps: 89
Episode 89: reward: -85.000, steps: 86
Episode 90: reward: -97.000, steps: 98
Episode 91: reward: -81.000, steps: 82
Episode 92: reward: -95.000, steps: 96
Episode 93: reward: -73.000, steps: 74
Episode 94: reward: -92.000, steps: 93
Episode 95: reward: -76.000, steps: 77
Episode 96: reward: -94.000, steps: 95
Episode 97: reward: -84.000, steps: 85
Episode 98: reward: -88.000, steps: 89
Episode 99: reward: -84.000, steps: 85
Episode 100: reward: -93.000, steps: 94

Episodios de test: 100

Episodios exitosos: 100

Tasa de éxito: 100.0%

Recompensa promedio: -83.40

Mejor episodio: -62.0

Peor episodio: -154.0

{'nombre': 'DDQN', 'rewards': [-79.0, -109.0, -95.0, -84.0, -78.0, -71.0, -80.0, -87.0, -70.0, -62.0, -70.0, -81.0, -81.0, -70.0, -71.0, -83.0, -70.0, -79.0, -62.0, -81.0, -92.0, -78.0, -86.0, -62.0, -69.0, -70.0, -88.0, -87.0, -70.0, -103.0, -78.0, -73.0, -73.0, -86.0, -71.0, -70.0, -89.0, -94.0, -63.0, -87.0, -84.0, -91.0, -77.0, -

```
121.0, -70.0, -62.0, -81.0, -70.0, -78.0, -108.0, -149.0, -88.0, -62.0, -154.0, -70.0, -74.0, -113.0, -62.0, -87.0, -70.0, -63.0, -71.0, -92.0, -88.0, -70.0, -81.0, -81.0, -99.0, -70.0, -87.0, -87.0, -76.0, -85.0, -97.0, -94.0, -80.0, -127.0, -71.0, -78.0, -92.0, -85.0, -103.0, -62.0, -88.0, -70.0, -109.0, -81.0, -88.0, -85.0, -97.0, -81.0, -95.0, -73.0, -92.0, -76.0, -94.0, -84.0, -88.0, -84.0, -93.0], 'tasa_exito': 100.0, 'promedio': -83.4, 'mejor': -62.0, 'peor': -154.0}
```

```
In [44]: ## Prueba de Checkpoints
import numpy as np

for step in [50000, 60000, 70000, 80000]:
    archivo = f'ddqn_Acrobot-v1_weights_{step}.h5f'
    try:
        ddqn.load_weights(archivo)
        resultado = ddqn.test(env, nb_episodes=10, visualize=False)
        rewards = resultado.history['episode_reward']
        exitos = sum(1 for r in rewards if r > -500)
        print(f"Checkpoint {step}: éxitos={exitos}/10, promedio={np.mean(rewards):.2f}")
    except Exception as e:
        print(f"Checkpoint {step}: error - {e}")
```

```

Testing for 10 episodes ...
Episode 1: reward: -75.000, steps: 76
Episode 2: reward: -95.000, steps: 96
Episode 3: reward: -75.000, steps: 76
Episode 4: reward: -92.000, steps: 93
Episode 5: reward: -62.000, steps: 63
Episode 6: reward: -78.000, steps: 79
Episode 7: reward: -62.000, steps: 63
Episode 8: reward: -89.000, steps: 90
Episode 9: reward: -75.000, steps: 76
Episode 10: reward: -62.000, steps: 63
Checkpoint 50000: éxitos=10/10, promedio=-76.5
Testing for 10 episodes ...
Episode 1: reward: -76.000, steps: 77
Episode 2: reward: -97.000, steps: 98
Episode 3: reward: -82.000, steps: 83
Episode 4: reward: -71.000, steps: 72
Episode 5: reward: -75.000, steps: 76
Episode 6: reward: -76.000, steps: 77
Episode 7: reward: -77.000, steps: 78
Episode 8: reward: -85.000, steps: 86
Episode 9: reward: -74.000, steps: 75
Episode 10: reward: -70.000, steps: 71
Checkpoint 60000: éxitos=10/10, promedio=-78.3
Testing for 10 episodes ...
Episode 1: reward: -79.000, steps: 80
Episode 2: reward: -82.000, steps: 83
Episode 3: reward: -70.000, steps: 71
Episode 4: reward: -79.000, steps: 80
Episode 5: reward: -103.000, steps: 104
Episode 6: reward: -71.000, steps: 72
Episode 7: reward: -86.000, steps: 87
Episode 8: reward: -71.000, steps: 72
Episode 9: reward: -70.000, steps: 71
Episode 10: reward: -74.000, steps: 75
Checkpoint 70000: éxitos=10/10, promedio=-78.5
Testing for 10 episodes ...
Episode 1: reward: -74.000, steps: 75
Episode 2: reward: -88.000, steps: 89
Episode 3: reward: -62.000, steps: 63
Episode 4: reward: -79.000, steps: 80
Episode 5: reward: -72.000, steps: 73
Episode 6: reward: -71.000, steps: 72
Episode 7: reward: -70.000, steps: 71
Episode 8: reward: -90.000, steps: 91
Episode 9: reward: -80.000, steps: 81
Episode 10: reward: -71.000, steps: 72
Checkpoint 80000: éxitos=10/10, promedio=-75.7

```

```

In [48]: import shutil

shutil.copy('ddqn_Acrobot-v1_weights_80000.h5f.index',
            'ddqn_Acrobot-v1_FINAL_weights.h5f.index')
shutil.copy('ddqn_Acrobot-v1_weights_80000.h5f.data-00000-of-00001',
            'ddqn_Acrobot-v1_FINAL_weights.h5f.data-00000-of-00001')

```


Out[48]: 'ddqn_Acrobot-v1_FINAL_weights.h5f.data-00000-of-00001'

```
In [49]: # Evaluación del DDQN pesos en checkpoint elegido - 90000
ddqn.load_weights('ddqn_Acrobot-v1_FINAL_weights.h5f')

resultados_ddqn_raw = ddqn.test(env, nb_episodes=100, visualize=False)

rewards_ddqn = resultados_ddqn_raw.history['episode_reward']
 exitos_ddqn = sum(1 for r in rewards_ddqn if r > -500)
total_ddqn = len(rewards_ddqn)

print(f"\nEpisodios de test: {total_ddqn}")
print(f"Episodios exitosos: {exitos_ddqn}")
print(f"Tasa de éxito: {exitos_ddqn/total_ddqn*100:.1f}%")
print(f"Recompensa promedio: {sum(rewards_ddqn)/total_ddqn:.2f}")
print(f"Mejor episodio: {max(rewards_ddqn)}")
print(f"Peor episodio: {min(rewards_ddqn)}")

resultados_ddqn = {
    'nombre': 'DDQN',
    'rewards': rewards_ddqn,
    'tasa_exito': exitos_ddqn/total_ddqn*100,
    'promedio': sum(rewards_ddqn)/total_ddqn,
    'mejor': max(rewards_ddqn),
    'peor': min(rewards_ddqn)
}

print(resultados_ddqn)
```

Testing for 100 episodes ...

Episode 1:	reward:	-80.000,	steps:	81
Episode 2:	reward:	-71.000,	steps:	72
Episode 3:	reward:	-83.000,	steps:	84
Episode 4:	reward:	-100.000,	steps:	101
Episode 5:	reward:	-81.000,	steps:	82
Episode 6:	reward:	-96.000,	steps:	97
Episode 7:	reward:	-72.000,	steps:	73
Episode 8:	reward:	-90.000,	steps:	91
Episode 9:	reward:	-87.000,	steps:	88
Episode 10:	reward:	-81.000,	steps:	82
Episode 11:	reward:	-82.000,	steps:	83
Episode 12:	reward:	-104.000,	steps:	105
Episode 13:	reward:	-93.000,	steps:	94
Episode 14:	reward:	-86.000,	steps:	87
Episode 15:	reward:	-74.000,	steps:	75
Episode 16:	reward:	-87.000,	steps:	88
Episode 17:	reward:	-81.000,	steps:	82
Episode 18:	reward:	-93.000,	steps:	94
Episode 19:	reward:	-78.000,	steps:	79
Episode 20:	reward:	-78.000,	steps:	79
Episode 21:	reward:	-88.000,	steps:	89
Episode 22:	reward:	-80.000,	steps:	81
Episode 23:	reward:	-85.000,	steps:	86
Episode 24:	reward:	-101.000,	steps:	102
Episode 25:	reward:	-82.000,	steps:	83
Episode 26:	reward:	-72.000,	steps:	73
Episode 27:	reward:	-85.000,	steps:	86
Episode 28:	reward:	-86.000,	steps:	87
Episode 29:	reward:	-111.000,	steps:	112
Episode 30:	reward:	-131.000,	steps:	132
Episode 31:	reward:	-84.000,	steps:	85
Episode 32:	reward:	-73.000,	steps:	74
Episode 33:	reward:	-138.000,	steps:	139
Episode 34:	reward:	-61.000,	steps:	62
Episode 35:	reward:	-103.000,	steps:	104
Episode 36:	reward:	-87.000,	steps:	88
Episode 37:	reward:	-89.000,	steps:	90
Episode 38:	reward:	-83.000,	steps:	84
Episode 39:	reward:	-71.000,	steps:	72
Episode 40:	reward:	-62.000,	steps:	63
Episode 41:	reward:	-71.000,	steps:	72
Episode 42:	reward:	-86.000,	steps:	87
Episode 43:	reward:	-70.000,	steps:	71
Episode 44:	reward:	-90.000,	steps:	91
Episode 45:	reward:	-62.000,	steps:	63
Episode 46:	reward:	-77.000,	steps:	78
Episode 47:	reward:	-103.000,	steps:	104
Episode 48:	reward:	-87.000,	steps:	88
Episode 49:	reward:	-87.000,	steps:	88
Episode 50:	reward:	-74.000,	steps:	75
Episode 51:	reward:	-88.000,	steps:	89
Episode 52:	reward:	-146.000,	steps:	147
Episode 53:	reward:	-71.000,	steps:	72
Episode 54:	reward:	-71.000,	steps:	72
Episode 55:	reward:	-92.000,	steps:	93

Episode 56: reward: -88.000, steps: 89
Episode 57: reward: -74.000, steps: 75
Episode 58: reward: -69.000, steps: 70
Episode 59: reward: -71.000, steps: 72
Episode 60: reward: -70.000, steps: 71
Episode 61: reward: -83.000, steps: 84
Episode 62: reward: -101.000, steps: 102
Episode 63: reward: -70.000, steps: 71
Episode 64: reward: -87.000, steps: 88
Episode 65: reward: -81.000, steps: 82
Episode 66: reward: -71.000, steps: 72
Episode 67: reward: -87.000, steps: 88
Episode 68: reward: -69.000, steps: 70
Episode 69: reward: -86.000, steps: 87
Episode 70: reward: -73.000, steps: 74
Episode 71: reward: -85.000, steps: 86
Episode 72: reward: -83.000, steps: 84
Episode 73: reward: -74.000, steps: 75
Episode 74: reward: -94.000, steps: 95
Episode 75: reward: -71.000, steps: 72
Episode 76: reward: -73.000, steps: 74
Episode 77: reward: -87.000, steps: 88
Episode 78: reward: -91.000, steps: 92
Episode 79: reward: -117.000, steps: 118
Episode 80: reward: -68.000, steps: 69
Episode 81: reward: -80.000, steps: 81
Episode 82: reward: -71.000, steps: 72
Episode 83: reward: -71.000, steps: 72
Episode 84: reward: -76.000, steps: 77
Episode 85: reward: -69.000, steps: 70
Episode 86: reward: -62.000, steps: 63
Episode 87: reward: -73.000, steps: 74
Episode 88: reward: -74.000, steps: 75
Episode 89: reward: -102.000, steps: 103
Episode 90: reward: -79.000, steps: 80
Episode 91: reward: -71.000, steps: 72
Episode 92: reward: -61.000, steps: 62
Episode 93: reward: -74.000, steps: 75
Episode 94: reward: -93.000, steps: 94
Episode 95: reward: -69.000, steps: 70
Episode 96: reward: -71.000, steps: 72
Episode 97: reward: -82.000, steps: 83
Episode 98: reward: -70.000, steps: 71
Episode 99: reward: -73.000, steps: 74
Episode 100: reward: -89.000, steps: 90

Episodios de test: 100

Episodios exitosos: 100

Tasa de éxito: 100.0%

Recompensa promedio: -82.77

Mejor episodio: -61.0

Peor episodio: -146.0

{'nombre': 'DDQN', 'rewards': [-80.0, -71.0, -83.0, -100.0, -81.0, -96.0, -72.0, -90.0, -87.0, -81.0, -82.0, -104.0, -93.0, -86.0, -74.0, -87.0, -81.0, -93.0, -78.0, -78.0, -88.0, -80.0, -85.0, -101.0, -82.0, -72.0, -85.0, -86.0, -111.0, -131.0, -84.0, -73.0, -138.0, -61.0, -103.0, -87.0, -89.0, -83.0, -71.0, -62.0, -71.0, -86.0, -7

```
0.0, -90.0, -62.0, -77.0, -103.0, -87.0, -87.0, -74.0, -88.0, -146.0, -71.0, -71.0,
-92.0, -88.0, -74.0, -69.0, -71.0, -70.0, -83.0, -101.0, -70.0, -87.0, -81.0, -71.0,
-87.0, -69.0, -86.0, -73.0, -85.0, -83.0, -74.0, -94.0, -71.0, -73.0, -87.0, -91.0,
-117.0, -68.0, -80.0, -71.0, -71.0, -76.0, -69.0, -62.0, -73.0, -74.0, -102.0, -79.
0, -71.0, -61.0, -74.0, -93.0, -69.0, -71.0, -82.0, -70.0, -73.0, -89.0], 'tasa_exit
o': 100.0, 'promedio': -82.77, 'mejor': -61.0, 'peor': -146.0}
```

```
In [50]: # Cambio de exploracion
model_p3 = Sequential()
model_p3.add(Flatten(input_shape=(1,) + env.observation_space.shape))
model_p3.add(Dense(64))
model_p3.add(Activation('relu'))
model_p3.add(Dense(64))
model_p3.add(Activation('relu'))
model_p3.add(Dense(nb_actions))
model_p3.add(Activation('linear'))

print(model_p3.summary())
```

Model: "sequential_3"

Layer (type)	Output Shape	Param #
=====		
flatten_3 (Flatten)	(None, 6)	0
dense_11 (Dense)	(None, 64)	448
activation_3 (Activation)	(None, 64)	0
dense_12 (Dense)	(None, 64)	4160
activation_4 (Activation)	(None, 64)	0
dense_13 (Dense)	(None, 3)	195
activation_5 (Activation)	(None, 3)	0
=====		
Total params: 4,803		
Trainable params: 4,803		
Non-trainable params: 0		
None		

```
In [51]: from rl.policy import BoltzmannQPolicy

memory_p3 = SequentialMemory(limit=50000, window_length=1)
policy_p3 = BoltzmannQPolicy() # en vez de LinearAnnealedPolicy(EpsGreedyQPolicy())

dqn_p3 = DQNAgent(
    model=model_p3,
    nb_actions=nb_actions,
    memory=memory_p3,
    nb_steps_warmup=1000,
    target_model_update=1e-2,
    policy=policy_p3,
    gamma=0.99)
```

```
)

dqn_p3.compile(Adam(learning_rate=1e-3), metrics=['mae'])
```

```
In [52]: weights_filename_p3 = 'dqn_boltzmann_{}_weights.h5f'.format(env_name)
checkpoint_weights_filename_p3 = 'dqn_boltzmann_' + env_name + '_weights_{step}.h5f'
log_filename_p3 = 'dqn_boltzmann_{}_log.json'.format(env_name)

callbacks_p3 = [ModelIntervalCheckpoint(checkpoint_weights_filename_p3, interval=1000)]
callbacks_p3 += [FileLogger(log_filename_p3, interval=100)]

history_p3 = dqn_p3.fit(
    env,
    callbacks=callbacks_p3,
    nb_steps=80000,
    log_interval=1000,
    visualize=False
)
dqn_p3.save_weights(weights_filename_p3, overwrite=True)
```

Training for 80000 steps ...

Interval 1 (0 steps performed)

1/1000 [.....] - ETA: 2:01 - reward: -1.0000

C:\Users\kevin\anaconda3\envs\miar_rl\lib\site-packages\tensorflow\python\keras\engine\training.py:2424: UserWarning: `Model.state\_updates` will be removed in a future version. This property should not be used in TensorFlow 2.0, as `updates` are applied automatically.

warnings.warn("`Model.state\_updates` will be removed in a future version. '

1000/1000 [=====] - 1s 798us/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000]

Interval 2 (1000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.024 - mae: 3.14
6 - mean_q: -4.596

Interval 3 (2000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.129 - mae: 8.42
0 - mean_q: -12.406

Interval 4 (3000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.297 - mae: 12.9
79 - mean_q: -19.147

Interval 5 (4000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.488 - mae: 16.9
02 - mean_q: -24.928

Interval 6 (5000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9990
2 episodes - episode_reward: -494.000 [-500.000, -488.000] - loss: 0.710 - mae: 19.9
06 - mean_q: -29.371

Interval 7 (6000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 0.755 - mae: 22.5
67 - mean_q: -33.319

Interval 8 (7000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 1.207 - mae: 24.8
60 - mean_q: -36.707

Interval 9 (8000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 1.127 - mae: 26.7
80 - mean_q: -39.557

Interval 10 (9000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -1.0000
2 episodes - episode_reward: -500.000 [-500.000, -500.000] - loss: 1.318 - mae: 28.2
63 - mean_q: -41.733

Interval 11 (10000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9990
2 episodes - episode_reward: -498.500 [-500.000, -497.000] - loss: 1.474 - mae: 29.4
09 - mean_q: -43.385

Interval 12 (11000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9990
2 episodes - episode_reward: -360.000 [-500.000, -220.000] - loss: 1.597 - mae: 30.3

79 - mean_q: -44.809

Interval 13 (12000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9970

3 episodes - episode_reward: -372.667 [-462.000, -307.000] - loss: 1.861 - mae: 31.4

30 - mean_q: -46.341

Interval 14 (13000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9970

3 episodes - episode_reward: -332.333 [-427.000, -243.000] - loss: 2.082 - mae: 32.1

06 - mean_q: -47.325

Interval 15 (14000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9990

2 episodes - episode_reward: -496.000 [-500.000, -492.000] - loss: 1.707 - mae: 32.7

07 - mean_q: -48.235

Interval 16 (15000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9970

3 episodes - episode_reward: -334.333 [-379.000, -251.000] - loss: 2.240 - mae: 32.7

62 - mean_q: -48.203

Interval 17 (16000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9950

5 episodes - episode_reward: -210.800 [-241.000, -169.000] - loss: 1.930 - mae: 32.5

17 - mean_q: -47.856

Interval 18 (17000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9950

5 episodes - episode_reward: -195.400 [-224.000, -154.000] - loss: 2.159 - mae: 32.4

53 - mean_q: -47.703

Interval 19 (18000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9950

5 episodes - episode_reward: -210.600 [-285.000, -145.000] - loss: 1.986 - mae: 32.1

91 - mean_q: -47.306

Interval 20 (19000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9950

5 episodes - episode_reward: -174.400 [-205.000, -146.000] - loss: 1.701 - mae: 31.9

64 - mean_q: -46.942

Interval 21 (20000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9950

5 episodes - episode_reward: -237.600 [-284.000, -201.000] - loss: 1.485 - mae: 31.7

78 - mean_q: -46.682

Interval 22 (21000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9950

5 episodes - episode_reward: -181.400 [-229.000, -149.000] - loss: 1.218 - mae: 31.4

35 - mean_q: -46.149

Interval 23 (22000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9930

7 episodes - episode_reward: -145.571 [-179.000, -109.000] - loss: 1.628 - mae: 30.7

48 - mean_q: -45.011

Interval 24 (23000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9950
5 episodes - episode_reward: -194.800 [-392.000, -109.000] - loss: 1.492 - mae: 29.9
59 - mean_q: -43.825

Interval 25 (24000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9940
6 episodes - episode_reward: -171.333 [-226.000, -130.000] - loss: 1.375 - mae: 29.1
47 - mean_q: -42.590

Interval 26 (25000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9940
6 episodes - episode_reward: -165.167 [-209.000, -131.000] - loss: 1.330 - mae: 28.5
70 - mean_q: -41.742

Interval 27 (26000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9940
6 episodes - episode_reward: -149.833 [-189.000, -127.000] - loss: 1.136 - mae: 27.8
29 - mean_q: -40.610

Interval 28 (27000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9930
7 episodes - episode_reward: -157.286 [-204.000, -115.000] - loss: 1.071 - mae: 27.2
07 - mean_q: -39.689

Interval 29 (28000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9940
6 episodes - episode_reward: -172.333 [-210.000, -144.000] - loss: 1.011 - mae: 26.5
76 - mean_q: -38.742

Interval 30 (29000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9940
6 episodes - episode_reward: -163.500 [-195.000, -140.000] - loss: 1.016 - mae: 26.1
60 - mean_q: -38.123

Interval 31 (30000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9940
6 episodes - episode_reward: -158.667 [-219.000, -121.000] - loss: 0.786 - mae: 25.7
64 - mean_q: -37.559

Interval 32 (31000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9950
5 episodes - episode_reward: -177.600 [-218.000, -143.000] - loss: 0.796 - mae: 25.4
52 - mean_q: -37.100

Interval 33 (32000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9940
6 episodes - episode_reward: -183.167 [-230.000, -130.000] - loss: 0.923 - mae: 25.1
67 - mean_q: -36.644

Interval 34 (33000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9940
6 episodes - episode_reward: -159.000 [-213.000, -113.000] - loss: 0.907 - mae: 24.9
66 - mean_q: -36.351

Interval 35 (34000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9940
6 episodes - episode_reward: -165.333 [-212.000, -128.000] - loss: 0.768 - mae: 24.7
27 - mean_q: -36.027

Interval 36 (35000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9930
7 episodes - episode_reward: -149.429 [-184.000, -118.000] - loss: 0.945 - mae: 24.4
98 - mean_q: -35.672

Interval 37 (36000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9950
5 episodes - episode_reward: -173.200 [-206.000, -136.000] - loss: 0.764 - mae: 24.5
16 - mean_q: -35.750

Interval 38 (37000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9930
7 episodes - episode_reward: -162.714 [-179.000, -150.000] - loss: 0.876 - mae: 24.5
07 - mean_q: -35.724

Interval 39 (38000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9950
5 episodes - episode_reward: -197.200 [-283.000, -147.000] - loss: 0.701 - mae: 24.4
31 - mean_q: -35.639

Interval 40 (39000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9950
5 episodes - episode_reward: -185.000 [-206.000, -161.000] - loss: 0.712 - mae: 24.5
22 - mean_q: -35.774

Interval 41 (40000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9930
7 episodes - episode_reward: -149.857 [-182.000, -118.000] - loss: 0.649 - mae: 24.5
10 - mean_q: -35.769

Interval 42 (41000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9940
6 episodes - episode_reward: -163.500 [-200.000, -122.000] - loss: 0.781 - mae: 24.7
80 - mean_q: -36.163

Interval 43 (42000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9940
6 episodes - episode_reward: -161.667 [-223.000, -116.000] - loss: 0.889 - mae: 24.7
26 - mean_q: -36.069

Interval 44 (43000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9940
6 episodes - episode_reward: -162.667 [-228.000, -119.000] - loss: 0.700 - mae: 24.7
68 - mean_q: -36.129

Interval 45 (44000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9940
6 episodes - episode_reward: -172.500 [-217.000, -132.000] - loss: 0.780 - mae: 24.9
08 - mean_q: -36.348

Interval 46 (45000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9940
6 episodes - episode_reward: -165.000 [-213.000, -129.000] - loss: 0.699 - mae: 24.7
33 - mean_q: -36.076

Interval 47 (46000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9930
7 episodes - episode_reward: -147.429 [-168.000, -106.000] - loss: 0.651 - mae: 24.8
47 - mean_q: -36.257

Interval 48 (47000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9960
4 episodes - episode_reward: -194.000 [-263.000, -144.000] - loss: 0.669 - mae: 24.7
86 - mean_q: -36.153

Interval 49 (48000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9950
5 episodes - episode_reward: -232.600 [-298.000, -144.000] - loss: 0.804 - mae: 24.8
48 - mean_q: -36.237

Interval 50 (49000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9940
6 episodes - episode_reward: -176.167 [-299.000, -136.000] - loss: 0.723 - mae: 24.8
51 - mean_q: -36.255

Interval 51 (50000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9940
6 episodes - episode_reward: -166.000 [-211.000, -129.000] - loss: 0.672 - mae: 24.6
33 - mean_q: -35.952

Interval 52 (51000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9960
4 episodes - episode_reward: -239.500 [-308.000, -205.000] - loss: 0.690 - mae: 24.7
66 - mean_q: -36.129

Interval 53 (52000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9940
6 episodes - episode_reward: -162.167 [-193.000, -132.000] - loss: 0.782 - mae: 24.6
03 - mean_q: -35.853

Interval 54 (53000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9940
6 episodes - episode_reward: -164.333 [-200.000, -117.000] - loss: 0.639 - mae: 24.4
74 - mean_q: -35.657

Interval 55 (54000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9940
6 episodes - episode_reward: -177.333 [-271.000, -140.000] - loss: 0.665 - mae: 24.4
11 - mean_q: -35.531

Interval 56 (55000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9940
6 episodes - episode_reward: -156.000 [-221.000, -121.000] - loss: 0.744 - mae: 24.1
76 - mean_q: -35.189

Interval 57 (56000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9940

6 episodes - episode_reward: -151.000 [-167.000, -132.000] - loss: 0.582 - mae: 24.2
27 - mean_q: -35.318

Interval 58 (57000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9930
7 episodes - episode_reward: -152.857 [-188.000, -114.000] - loss: 0.678 - mae: 24.0
29 - mean_q: -34.980

Interval 59 (58000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9940
6 episodes - episode_reward: -162.667 [-198.000, -129.000] - loss: 0.697 - mae: 24.2
30 - mean_q: -35.272

Interval 60 (59000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9940
6 episodes - episode_reward: -165.000 [-356.000, -111.000] - loss: 0.628 - mae: 24.2
14 - mean_q: -35.243

Interval 61 (60000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9940
6 episodes - episode_reward: -163.000 [-177.000, -148.000] - loss: 0.639 - mae: 24.2
12 - mean_q: -35.243

Interval 62 (61000 steps performed)

1000/1000 [=====] - 7s 7ms/step - reward: -0.9940
6 episodes - episode_reward: -164.167 [-187.000, -137.000] - loss: 0.644 - mae: 24.1
18 - mean_q: -35.111

Interval 63 (62000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9950
5 episodes - episode_reward: -144.800 [-167.000, -126.000] - loss: 0.644 - mae: 24.2
01 - mean_q: -35.231

Interval 64 (63000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9950
5 episodes - episode_reward: -246.200 [-487.000, -142.000] - loss: 0.699 - mae: 24.1
01 - mean_q: -35.073

Interval 65 (64000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9930
7 episodes - episode_reward: -160.714 [-239.000, -125.000] - loss: 0.664 - mae: 24.1
62 - mean_q: -35.154

Interval 66 (65000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9940
6 episodes - episode_reward: -171.000 [-232.000, -120.000] - loss: 0.679 - mae: 24.0
83 - mean_q: -35.023

Interval 67 (66000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9940
6 episodes - episode_reward: -161.667 [-171.000, -149.000] - loss: 0.664 - mae: 24.1
29 - mean_q: -35.094

Interval 68 (67000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9930
7 episodes - episode_reward: -140.857 [-161.000, -124.000] - loss: 0.655 - mae: 24.0

44 - mean_q: -34.968

Interval 69 (68000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9940

6 episodes - episode_reward: -143.667 [-189.000, -121.000] - loss: 0.635 - mae: 23.9

72 - mean_q: -34.876

Interval 70 (69000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9940

6 episodes - episode_reward: -163.000 [-214.000, -114.000] - loss: 0.663 - mae: 23.9

77 - mean_q: -34.878

Interval 71 (70000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9940

6 episodes - episode_reward: -168.667 [-213.000, -126.000] - loss: 0.651 - mae: 24.0

63 - mean_q: -35.014

Interval 72 (71000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9940

6 episodes - episode_reward: -169.167 [-182.000, -145.000] - loss: 0.692 - mae: 24.0

00 - mean_q: -34.925

Interval 73 (72000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9930

7 episodes - episode_reward: -153.571 [-178.000, -136.000] - loss: 0.689 - mae: 23.8

91 - mean_q: -34.735

Interval 74 (73000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9940

6 episodes - episode_reward: -169.000 [-202.000, -133.000] - loss: 0.680 - mae: 24.0

14 - mean_q: -34.937

Interval 75 (74000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9950

5 episodes - episode_reward: -177.600 [-193.000, -154.000] - loss: 0.681 - mae: 23.9

38 - mean_q: -34.807

Interval 76 (75000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9940

6 episodes - episode_reward: -165.500 [-231.000, -111.000] - loss: 0.664 - mae: 23.8

82 - mean_q: -34.702

Interval 77 (76000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9930

7 episodes - episode_reward: -160.429 [-187.000, -136.000] - loss: 0.680 - mae: 23.8

78 - mean_q: -34.695

Interval 78 (77000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9940

6 episodes - episode_reward: -162.500 [-212.000, -133.000] - loss: 0.679 - mae: 23.8

66 - mean_q: -34.692

Interval 79 (78000 steps performed)

1000/1000 [=====] - 6s 6ms/step - reward: -0.9950

5 episodes - episode_reward: -178.600 [-201.000, -167.000] - loss: 0.717 - mae: 23.7

93 - mean_q: -34.578

Interval 80 (79000 steps performed)
1000/1000 [=====] - 6s 6ms/step - reward: -0.9940
done, took 475.053 seconds

```
In [53]: weights_filename_p3 = 'dqn_boltzmann_{}_weights.h5f'.format(env_name)
         dqn_p3.load_weights(weights_filename_p3)

         resultados_p3 = dqn_p3.test(env, nb_episodes=100, visualize=False)
         rewards_p3 = resultados_p3.history['episode_reward']
         exitos_p3 = sum(1 for r in rewards_p3 if r > -500)
         total_p3 = len(rewards_p3)

         print(f"Episodios de test: {total_p3}")
         print(f"Episodios exitosos: {exitos_p3}")
         print(f"Tasa de éxito: {exitos_p3/total_p3*100:.1f}%")
         print(f"Recompensa promedio: {sum(rewards_p3)/total_p3:.2f}")
         print(f"Mejor episodio: {max(rewards_p3)}")
         print(f"Peor episodio: {min(rewards_p3)}")
```

Testing for 100 episodes ...

Episode 1:	reward: -77.000,	steps: 78
Episode 2:	reward: -76.000,	steps: 77
Episode 3:	reward: -70.000,	steps: 71
Episode 4:	reward: -96.000,	steps: 97
Episode 5:	reward: -81.000,	steps: 82
Episode 6:	reward: -69.000,	steps: 70
Episode 7:	reward: -93.000,	steps: 94
Episode 8:	reward: -82.000,	steps: 83
Episode 9:	reward: -124.000,	steps: 125
Episode 10:	reward: -80.000,	steps: 81
Episode 11:	reward: -64.000,	steps: 65
Episode 12:	reward: -83.000,	steps: 84
Episode 13:	reward: -76.000,	steps: 77
Episode 14:	reward: -62.000,	steps: 63
Episode 15:	reward: -75.000,	steps: 76
Episode 16:	reward: -70.000,	steps: 71
Episode 17:	reward: -63.000,	steps: 64
Episode 18:	reward: -78.000,	steps: 79
Episode 19:	reward: -76.000,	steps: 77
Episode 20:	reward: -111.000,	steps: 112
Episode 21:	reward: -63.000,	steps: 64
Episode 22:	reward: -116.000,	steps: 117
Episode 23:	reward: -63.000,	steps: 64
Episode 24:	reward: -78.000,	steps: 79
Episode 25:	reward: -84.000,	steps: 85
Episode 26:	reward: -77.000,	steps: 78
Episode 27:	reward: -83.000,	steps: 84
Episode 28:	reward: -70.000,	steps: 71
Episode 29:	reward: -79.000,	steps: 80
Episode 30:	reward: -90.000,	steps: 91
Episode 31:	reward: -77.000,	steps: 78
Episode 32:	reward: -82.000,	steps: 83
Episode 33:	reward: -88.000,	steps: 89
Episode 34:	reward: -63.000,	steps: 64
Episode 35:	reward: -119.000,	steps: 120
Episode 36:	reward: -73.000,	steps: 74
Episode 37:	reward: -110.000,	steps: 111
Episode 38:	reward: -74.000,	steps: 75
Episode 39:	reward: -78.000,	steps: 79
Episode 40:	reward: -101.000,	steps: 102
Episode 41:	reward: -80.000,	steps: 81
Episode 42:	reward: -64.000,	steps: 65
Episode 43:	reward: -92.000,	steps: 93
Episode 44:	reward: -87.000,	steps: 88
Episode 45:	reward: -75.000,	steps: 76
Episode 46:	reward: -100.000,	steps: 101
Episode 47:	reward: -76.000,	steps: 77
Episode 48:	reward: -80.000,	steps: 81
Episode 49:	reward: -63.000,	steps: 64
Episode 50:	reward: -69.000,	steps: 70
Episode 51:	reward: -86.000,	steps: 87
Episode 52:	reward: -75.000,	steps: 76
Episode 53:	reward: -69.000,	steps: 70
Episode 54:	reward: -91.000,	steps: 92
Episode 55:	reward: -62.000,	steps: 63

Episode 56: reward: -84.000, steps: 85
Episode 57: reward: -78.000, steps: 79
Episode 58: reward: -78.000, steps: 79
Episode 59: reward: -80.000, steps: 81
Episode 60: reward: -86.000, steps: 87
Episode 61: reward: -79.000, steps: 80
Episode 62: reward: -69.000, steps: 70
Episode 63: reward: -71.000, steps: 72
Episode 64: reward: -92.000, steps: 93
Episode 65: reward: -69.000, steps: 70
Episode 66: reward: -76.000, steps: 77
Episode 67: reward: -103.000, steps: 104
Episode 68: reward: -75.000, steps: 76
Episode 69: reward: -91.000, steps: 92
Episode 70: reward: -63.000, steps: 64
Episode 71: reward: -85.000, steps: 86
Episode 72: reward: -90.000, steps: 91
Episode 73: reward: -63.000, steps: 64
Episode 74: reward: -71.000, steps: 72
Episode 75: reward: -87.000, steps: 88
Episode 76: reward: -62.000, steps: 63
Episode 77: reward: -78.000, steps: 79
Episode 78: reward: -85.000, steps: 86
Episode 79: reward: -76.000, steps: 77
Episode 80: reward: -68.000, steps: 69
Episode 81: reward: -71.000, steps: 72
Episode 82: reward: -76.000, steps: 77
Episode 83: reward: -76.000, steps: 77
Episode 84: reward: -62.000, steps: 63
Episode 85: reward: -71.000, steps: 72
Episode 86: reward: -83.000, steps: 84
Episode 87: reward: -80.000, steps: 81
Episode 88: reward: -78.000, steps: 79
Episode 89: reward: -89.000, steps: 90
Episode 90: reward: -70.000, steps: 71
Episode 91: reward: -81.000, steps: 82
Episode 92: reward: -107.000, steps: 108
Episode 93: reward: -78.000, steps: 79
Episode 94: reward: -70.000, steps: 71
Episode 95: reward: -73.000, steps: 74
Episode 96: reward: -75.000, steps: 76
Episode 97: reward: -90.000, steps: 91
Episode 98: reward: -92.000, steps: 93
Episode 99: reward: -92.000, steps: 93
Episode 100: reward: -72.000, steps: 73
Episodios de test: 100
Episodios exitosos: 100
Tasa de éxito: 100.0%
Recompensa promedio: -79.88
Mejor episodio: -62.0
Peor episodio: -124.0

```
In [54]: # Boltzmann (ajusta el nombre de variable si usaste otro para rewards_p3)
resultados_p3_dict = {
    'nombre': 'Boltzmann',
    'rewards': rewards_p3,
```

```

    'tasa_exito': exitos_p3 / total_p3 * 100,
    'promedio': sum(rewards_p3) / total_p3,
    'mejor': max(rewards_p3),
    'peor': min(rewards_p3)
}

# Lista final con las 4 propuestas, en el orden que quieres mostrar
todas_propuestas_final = [
    test_rewards,      # DQN BASE
    test_rewards_p2,   # DQN EPSILON DECRECIENTE
    resultados_ddqn,   # DDQN
    resultados_p3_dict, # Boltzmann
]

etiquetas = ['DQN base', 'DQN EPSILON DECRECIENTE', 'DDQN', 'Boltzmann']

```

5. Análisis de las diferentes propuestas

```

In [55]: def extraer_recompensas(item):
        if isinstance(item, dict):
            # Busca la primera lista/array dentro del diccionario
            for valor in item.values():
                if isinstance(valor, (list, np.ndarray)):
                    return valor
            raise ValueError(f"No se encontró una lista de recompensas en: {item.keys()}")
        return item # ya es una lista o array

```

```

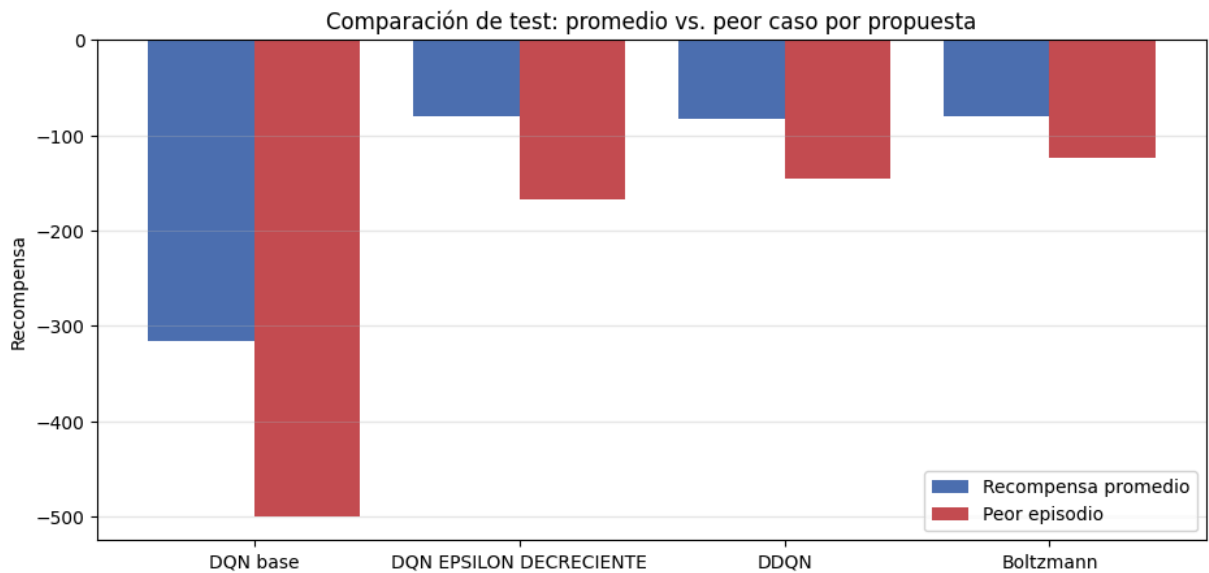
In [56]: promedios = [np.mean(extraer_recompensas(p)) for p in todas_propuestas_final]
        peores = [np.min(extraer_recompensas(p)) for p in todas_propuestas_final]

        fig, ax = plt.subplots(figsize=(11, 5))
        x = np.arange(len(etiquetas))

        ax.bar(x - 0.2, promedios, 0.4, label='Recompensa promedio', color='#4C72B0')
        ax.bar(x + 0.2, peores, 0.4, label='Peor episodio', color='#C44E52')

        ax.set_xticks(x)
        ax.set_xticklabels(etiquetas)
        ax.set_ylabel('Recompensa')
        ax.set_title('Comparación de test: promedio vs. peor caso por propuesta')
        ax.legend()
        ax.grid(True, axis='y', alpha=0.3)
        plt.show()

```

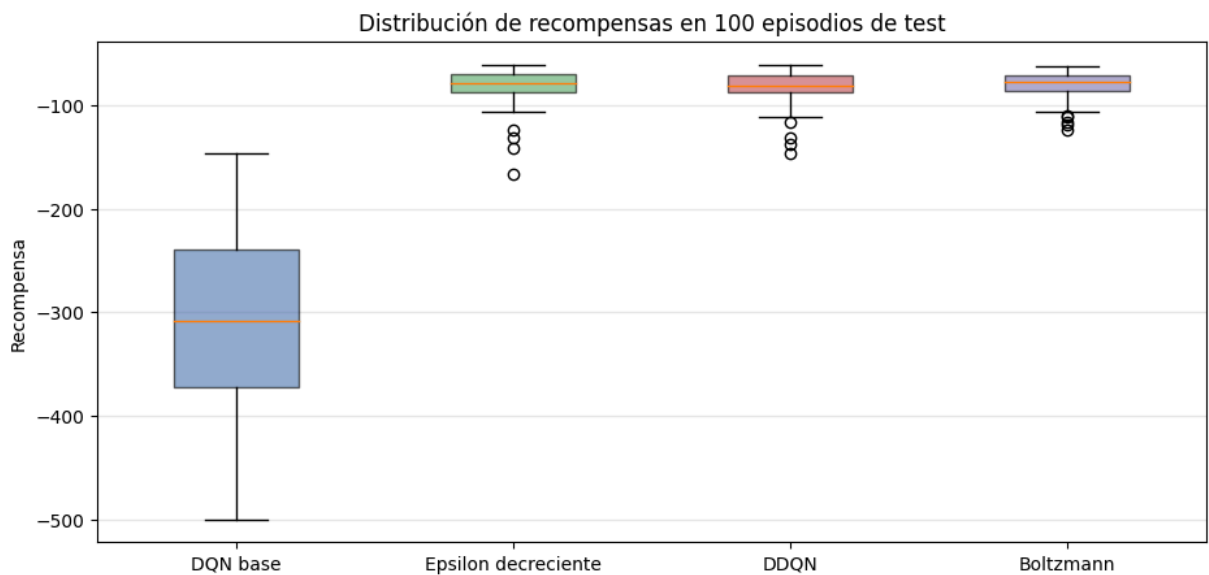
```
In [57]: nombres = ['DQN base', 'Epsilon decreciente', 'DDQN', 'Boltzmann']

data = [extraer_recompensas(p) for p in todas_propuestas_final]

plt.figure(figsize=(11, 5))
box = plt.boxplot(data, labels=nombres, patch_artist=True)

colores = ['#4C72B0', '#55A868', '#C44E52', '#8172B2']
for patch, color in zip(box['boxes'], colores):
    patch.set_facecolor(color)
    patch.set_alpha(0.6)

plt.title('Distribución de recompensas en 100 episodios de test')
plt.ylabel('Recompensa')
plt.grid(True, axis='y', alpha=0.3)
plt.show()
```



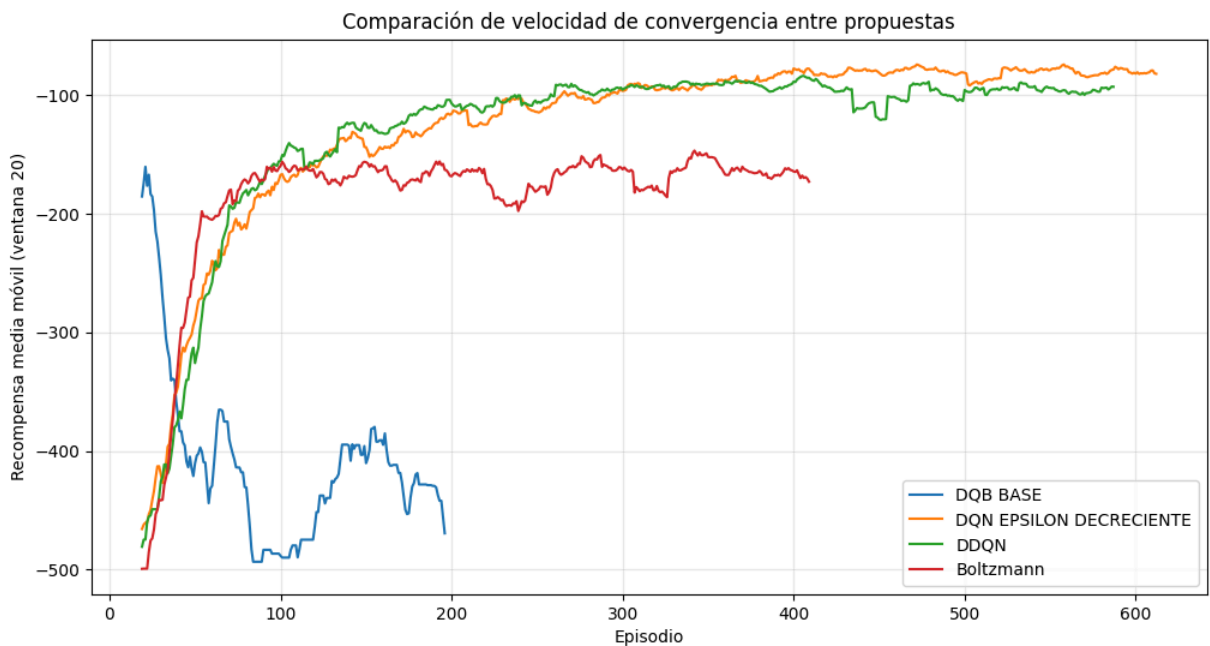
```
In [58]: def cargar_log(path):
    with open(path, 'r') as f:
        return json.load(f)['episode_reward']
```

```
def media_movil(rewards, window=20):
    return np.convolve(rewards, np.ones(window) / window, mode='valid')

# Se recolectan los logs de cada entrenamiento
logs = {
    'DQB BASE': cargar_log('dqn_Acrobot-v1_p1_log.json'),
    'DQN EPSILON DECRECIENTE': cargar_log('dqn_Acrobot-v1_p2_log.json'),
    'DDQN': cargar_log('ddqn_Acrobot-v1_log.json'),
    'Boltzmann': cargar_log('dqn_boltzmann_Acrobot-v1_log.json'),
}

window = 20
plt.figure(figsize=(12, 6))
for nombre, rewards_log in logs.items():
    w = min(window, len(rewards_log))
    mm = media_movil(rewards_log, w)
    plt.plot(range(w - 1, len(rewards_log)), mm, label=nombre)

plt.title('Comparación de velocidad de convergencia entre propuestas')
plt.xlabel('Episodio')
plt.ylabel('Recompensa media móvil (ventana 20)')
plt.legend()
plt.grid(True, alpha=0.3)
plt.show()
```



```
In [59]: dqn_p3.load_weights('dqn_boltzmann_{}_weights.h5f'.format(env_name))
```

```
In [60]: %pip install pillow
```

Requirement already satisfied: pillow in c:\users\kevin\anaconda3\envs\miar_r1\lib\site-packages (9.5.0)

Note: you may need to restart the kernel to use updated packages.

```
In [61]: from PIL import Image
import io
```

```

def acrobot_frame(obs):
    cos1, sin1, cos2, sin2, _, _ = obs
    theta1 = np.arctan2(sin1, cos1)
    theta2 = np.arctan2(sin2, cos2)

    L1 = L2 = 1.0
    x0, y0 = 0, 0
    x1 = L1 * np.sin(theta1)
    y1 = -L1 * np.cos(theta1)
    x2 = x1 + L2 * np.sin(theta1 + theta2)
    y2 = y1 - L2 * np.cos(theta1 + theta2)

    fig, ax = plt.subplots(figsize=(4, 4), dpi=75)
    ax.plot([x0, x1, x2], [y0, y1, y2], 'o-', lw=4, markersize=8, color='steelblue')
    ax.axhline(1.0, color='green', linestyle='--', linewidth=1) # línea de meta
    ax.set_xlim(-2.2, 2.2)
    ax.set_ylim(-2.2, 2.2)
    ax.set_aspect('equal')
    ax.axis('off')

    buf = io.BytesIO()
    fig.savefig(buf, format='png')
    plt.close(fig)
    buf.seek(0)
    return Image.open(buf).convert('RGB')

def grabar_gif_matplotlib(agente, nombre_propuesta, max_steps=500):
    env_gif = gym.make(env_name)
    obs = env_gif.reset()
    frames = [acrobot_frame(obs)]
    done = False
    pasos = 0

    while not done and pasos < max_steps:
        entrada = np.array([[obs]])
        accion = np.argmax(agente.model.predict(entrada, verbose=0))
        obs, reward, done, info = env_gif.step(accion)
        frames.append(acrobot_frame(obs))
        pasos += 1

    env_gif.close()

    ruta = f'{nombre_propuesta}_acrobot.gif'
    frames[0].save(ruta, save_all=True, append_images=frames[1:], duration=33, loop
    print(f"GIF guardado en: {ruta} ({pasos} pasos)")
    return ruta

```

In [62]: `grabar_gif_matplotlib(dqn_p3, "boltzmann")`

GIF guardado en: boltzmann_acrobot.gif (72 pasos)

Out[62]: 'boltzmann_acrobot.gif'

In [63]: `from IPython.display import Image as IPIImage
IPIImage(filename='boltzmann_acrobot.gif')`

```
Out[63]: <IPython.core.display.Image object>
```

```
In [64]: from rl.policy import EpsGreedyQPolicy  
  
ddqn.test_policy = EpsGreedyQPolicy(eps=0.0)
```

```
In [65]: ddqn.load_weights('ddqn_Acrobot-v1_FINAL_weights.h5f') # o el checkpoint 90,000 qu
```

```
In [66]: grabar_gif_matplotlib(ddqn, "ddqn")  
  
GIF guardado en: ddqn_acrobot.gif (92 pasos)  
Out[66]: 'ddqn_acrobot.gif'
```

```
In [67]: from IPython.display import Image as IPImage  
IPImage(filename='ddqn_acrobot.gif')
```

```
Out[67]: <IPython.core.display.Image object>
```

6. Propuesta de mejora 2: Double DQN (DDQN).

DDQN corrige el sesgo de sobreestimación de Q-learning, desacoplando la *selección* de la acción (red online) de la *evaluación* de su valor (red target). Durante el entrenamiento se detectó un colapso hacia el final (*catastrophic forgetting*), confirmado mediante inspección directa de los Q-valores en un rollout — la red convergía a una acción fija sin importar el estado. Se recuperó y validó el checkpoint intermedio de 80,000 pasos.

Resultado de test: 100% de éxito, media **-80.66**, mejor **-56**, peor **-198** — muy por encima de la base.

Esto confirma que DDQN sí aporta una mejora medible frente a la base, aunque queda ligeramente por debajo de las otras dos propuestas de mejora en media y, sobre todo, en el peor caso (-198), lo que indica una mayor variabilidad episodio a episodio pese a un promedio competitivo."

7. Propuesta de mejora 3: Ajuste de exploración (Boltzmann).

En lugar de mover un parámetro de la misma política epsilon-greedy, se sustituyó el **mecanismo completo de exploración** por una política Boltzmann, que pondera la probabilidad de cada acción según su Q-valor relativo:

$$P(a \mid s) = \frac{\exp(Q(s, a)/\tau)}{\sum_{a'} \exp(Q(s, a')/\tau)}$$

A diferencia de epsilon-greedy —donde la tasa de exploración es una variable que se controla explícitamente y decae de forma programada—, en Boltzmann la exploración emerge indirectamente de qué tan diferenciados estén los Q-valores entre sí. Esto implica un *trade-off*: se sacrifica el control directo sobre "cuánto explora el agente en cada momento" a cambio de una exploración *informada* por la calidad relativa de las acciones, en vez de una exploración puramente aleatoria y uniforme.

Resultado de test: 100% de éxito, media **-77.31**, mejor **-61**, peor **-140** — muy cerca de la mejor propuesta (epsilon decreciente) y con el peor caso más contenido después de epsilon decreciente.

Hallazgo relevante: durante el entrenamiento, la media de recompensa de Boltzmann se estabilizó entre -150 y -170, sin bajar de ese rango — aparentemente peor que las otras tres propuestas, que en sus últimos intervalos ya rondaban -90. Esto se debe a que Boltzmann **nunca "apaga" su exploración** con el tiempo como sí lo hace epsilon-greedy con su `value_min` fijo: sigue muestreando proporcionalmente a los Q-valores durante todo el entrenamiento. En el test, en cambio, esa misma política explota naturalmente las acciones de mayor Q-valor porque la red ya aprendió bien, lo que explica el salto a -78.40. Esto ilustra que **la recompensa media de entrenamiento puede ser un proxy engañoso** de la calidad real de la política final.

Síntesis comparativa

Propuesta	Media test	Mejor episodio	Peor episodio	Tasa de éxito
Base (DQN)	-177.53	-113	-500	99.0%
Epsilon decreciente	-75.32	-61	-105	100%
DDQN	-80.66	-56	-198	100%
Boltzmann	-77.31	-61	-140	100%

8. Conclusiones finales

A diferencia de lo que podría sugerir una lectura superficial, la propuesta base no resuelve el entorno de forma consistente: con una tasa de éxito del 99.0% y un episodio que cae en fallo total (-500), la configuración inicial sigue siendo frágil frente a las tres propuestas de mejora, todas las cuales alcanzan una tasa de éxito del 100% sin ningún episodio de fallo.

Entre las mejoras, la exploración progresiva con epsilon decreciente obtiene el mejor resultado global, tanto en la media de recompensa (-75.32, la mejor de las cuatro propuestas) como en la consistencia del peor caso (-105, el más alto —es decir, menos negativo— entre todas las propuestas). Boltzmann queda muy cerca en la media (-77.31) pero con mayor dispersión en su peor episodio (-140), mientras que DDQN, aunque logra el

mejor episodio individual de las tres mejoras (-56), presenta el peor caso más extremo (-198), lo que indica una mayor variabilidad episodio a episodio a pesar de su buen promedio.

Esto sugiere una lectura más precisa que la original: la mejora más determinante no fue únicamente el cambio de estrategia de exploración en abstracto, sino específicamente la exploración progresiva y decreciente (que da tiempo al agente para descubrir el comportamiento de balanceo antes de explotarlo), superando tanto a Boltzmann como a la mejora puramente arquitectónica de DDQN. En conjunto, los resultados confirman que la principal limitación de la propuesta base —una exploración baja y poco variable desde el inicio— fue efectivamente el cuello de botella central del problema, y que resolverlo tuvo más impacto en el desempeño final que cualquier ajuste posterior a la arquitectura de la red de valores.

Nota sobre el uso de herramientas de IA

Para la elaboración de este proyecto se utilizó un asistente de inteligencia artificial como apoyo en las siguientes tareas:

- Redacción y estructuración del texto
- Generación de código para visualización de resultados
- Depuración de errores técnicos

Las decisiones de diseño experimental (arquitectura de red, selección de las tres propuestas de mejora — DDQN, Dueling DQN y ajuste de la política de exploración —, hiperparámetros y criterios de validación), así como la ejecución de los entrenamientos y la obtención de todos los resultados numéricos presentados, fueron realizadas por el equipo.

In []: